

ION-ATOM CHARGE TRANSFER REACTIONS  
AT COLD AND ULTRA-COLD  
TEMPERATURES

By

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## Abstract

We study radiative charge-transfer reactions and the radiative association of  $Yb$  with  $Ca^+$  in the cold-to-ultracold temperature range. Our findings confirm the validity of Langevin behavior at temperatures in the single Kelvin range. However, we also show that this behavior breaks down at ultracold temperatures, resulting in a dramatic isotope effect.

We explore resonant charge transfer (RCT) reactions in two-dimensions for the  $H^+ + H$  system. We introduce a toy model in which the collision partners are effective anyons and investigate how their introduction affects RCT rates.

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## Chapter 1

### Introduction

## 1.1 Overview

Charge transfer (CT) refers to the movement of electrons or ions from one species (atom, molecule, or ion) to another. This transfer can occur through various mechanisms, including chemical reactions, electrochemical processes, and interactions with electromagnetic fields. Charge transfer can occur through several mechanisms, including chemical reactions, Electrochemical processes, Photoinduced charge transfer, and ionization.

- Chemical reactions

Charge transfer occurs when electrons are exchanged between the reactants.

- Electrochemical processes

Electrochemical reactions involve the transfer of electrons between electrodes and electrolytes. This is the main process in batteries, where the transfer of electrons creates an electric current.

- Photo-induced charge transfer

In this scenario, the charge transfer is induced by light. It is the process in photosynthesis, photovoltaic cells, and optical devices.

- Ionization processes

These processes often occur in plasmas or atmospheric chemistry. The charge transfer in this environment leads to the formation of ions. A large part of the visible uni-

verse is plasma, including stars, nebulae, interstellar space, and intergalactic space. Understanding charge transfer leads to a better understanding of the formation of these cosmic bodies.

### 1.1.1 CT in the ultra-cold regime

Over the past several decades, cold and ultracold atom physics has evolved into a cornerstone of atomic, molecular, and optical (AMO) science. Advances in laser cooling, trapping, and coherent control have enabled the preparation of atomic systems at temperatures at which quantum mechanics governs both internal and external degrees of freedom. Within the AMO framework, cold-atom research is distinguished by its exceptional experimental tunability and isolation from environmental noise, enabling quantitative tests of quantum theory and the controlled exploration of interacting many-body systems. These capabilities place cold atoms at the intersection of traditional atomic physics, quantum optics, and modern quantum many-body theory.

A central motivation for cold atom research is the ability to study atomic interactions in the ultralow-energy regime, where scattering processes are governed by universal principles. At microkelvin and nanokelvin temperatures, collisions are dominated by a small number of partial waves, allowing interactions to be characterized by a few experimentally accessible parameters. The development of magnetically tunable Feshbach resonances has provided a powerful tool for controlling these interactions, enabling continuous tuning from weak to strong coupling. This level of control has facilitated precision studies of few-body physics, including universal bound states and resonant scattering phenomena, as well as systematic investigations of strongly correlated quantum gases. From an AMO perspective, these advances represent a natural extension of collision and scattering physics into a fully quantum-controlled regime.

Cold-atom platforms also play a vital role in precision measurements, a long-standing strength of AMO physics. Laser-cooled atoms underpin the most accurate frequency standards currently available, with optical lattice clocks achieving extraordinary stability and accuracy. These systems provide sensitive probes of fundamental symmetries and potential variations of fundamental constants, while also enabling practical advances in timekeeping and navigation. In parallel, matter-wave interferometry with cold atoms exploits long coherence times and precise control over atomic wave packets to achieve high sensitivity to inertial and gravitational effects. Such experiments exemplify the dual intellectual and practical impact of cold atom research within the AMO tradition.

Beyond precision measurement, cold atoms serve as model quantum systems in which complex many-body phenomena can be studied from first principles. Optical lattices formed by interfering laser fields allow neutral atoms to be arranged in highly controllable periodic potentials, providing clean realizations of lattice Hamiltonians relevant to condensed matter physics. Unlike solid-state systems, these atomic analogues are largely free of disorder and can be dynamically tuned, enabling controlled exploration of quantum magnetism, superfluidity, and correlated fermionic behavior. For AMO researchers, this represents an opportunity to investigate emergent quantum phenomena while retaining the microscopic control and measurement capabilities characteristic of atomic physics.

Recent years have also seen growing interest in hybrid systems that combine trapped ions with ultracold neutral atoms. Cold ion-atom platforms merge two mature areas of AMO physics ion trapping and ultracold neutral gases into a single experimental setting. The ion-atom interaction is dominated by a long-range polarization potential, leading to scattering and collision dynamics that differ qualitatively from those of neutral atoms alone. These systems, therefore, open a distinct regime of cold collision physics, offering new opportunities to test theoretical models of quantum scattering, charge exchange, and controlled chemical processes at ultralow energies.

From a fundamental standpoint, cold ion–atom systems provide a versatile platform for studying impurity physics and nonequilibrium dynamics in quantum environments. A single trapped ion immersed in a cold atomic gas can act as a well-localized probe of many-body behavior, enabling studies of decoherence, energy exchange, and transport at the microscopic level. These questions connect directly to core AMO themes, including open quantum systems and the interplay between coherent control and dissipation. While hybrid ion–atom experiments pose unique technical challenges, continued progress in this area is rapidly expanding the range of experimentally accessible phenomena.

In summary, research in cold atom physics is motivated by its unique capacity to unify precision measurement, controlled interactions, and quantum many-body dynamics within a single AMO framework. Over the past few decades, cold atoms have transformed the scope of atomic physics, enabling quantitative studies of complex quantum phenomena with unprecedented clarity. The emergence of cold ion–atom systems represents a natural and complementary extension of this progress, broadening the experimental and theoretical landscape of AMO physics. Together, these platforms continue to provide fertile ground for fundamental discovery while supporting the development of future quantum technologies.

In this thesis, we focus on two distinct applications of CT. In a recent study, published in the authoritative Physical Review Letters [68] Rellergert et al reported an enhanced radiative association rate of the  $Yb^+$  ion in collisions with  $Ca$  at temperatures in the sub-Kelvin regime. The reported rate was about five orders of magnitude larger than previous measurements of similar species at those temperatures, and the authors posited that a new mechanism beyond Langevin behavior was responsible for the enhanced rate.

Here, we report on our independent investigation of this collision process. Our results, published in [93], find that Langevin behavior persists into the sub-mK region, and predict a CT association rate for the  $YbCa^+$  molecule that is orders of magnitude smaller than

that reported in the cited measurements. The results of our investigation have prompted the authors of the laboratory study to reconsider the interpretation of their measurements. However, in carrying out calculations into the ultracold regime in which only a few partial waves contribute, we find dramatic isotopic behavior not reported, to the best of our knowledge, in the current literature. This discovery was highlighted in a Europhysics feature article [27] subsequent to our publications. Our calculations are the first of their kind for this collision system, and in the first part of this thesis, we review and elaborate on our calculations.

In the second part of the thesis, we explore charge transfer (CT) in a two-dimensional (2D) environment, in which the collision partners are confined to trajectories on a 2D plane. This investigation is motivated by two primary factors.

First, advances in laboratory techniques and device fabrication over the past few decades have significantly improved the development of 2D materials. Therefore, it is relevant to examine a fundamental process like CT through theoretical investigations.

Second, it is recognized that quantum exchange symmetry behaves quite differently in 2D compared to three-dimensional (3D) systems. In 3D, particles are classified into two types: fermions and bosons. The probability amplitude for fermions changes sign under particle interchange, whereas for bosons it remains symmetric. These characteristics have profound implications for the structure of matter.

In the early 1980s, it became widely accepted that the conventional distinctions between fermionic and bosonic behavior no longer applied to 2D systems. Indeed, in 2D, exchange symmetry allows for a variety of phase dependencies in the scattering amplitudes of identical particles. Particles that exhibit these properties are referred to as anyons.

In this thesis, we will focus on the  $H - H^+$  system as the subject of our investigations. In this context, the ground states for  $H_2^+$  are degenerate, and the reaction, in which an

electron from a neutral hydrogen atom is captured by a proton, is referred to as a resonant charge transfer (CT). Since collisions in the ultra-cold regime require a molecular-based description, we will need to generate potential energy curves in a two-dimensional (2D) domain. Most quantum chemistry codes use three-dimensional (3D) orbitals, developed over the past 70 years for 3D dynamics; therefore, we must develop techniques specifically for the 2D system. Additionally, we will explore, for the first time, how anyonic behavior affects resonant charge transfer in the ultralow energy regime.

The outline of this thesis is structured as follows: Below, we provide a brief review of the general theoretical approaches to calculating charge transfer rates in ion-atom processes in 3D. We will then focus, in chapter 2 an review the theory and methodology employed to calculate radiative charge transfer. A detailed discussion is provided in the later chapters, describing the methodologies used and the results obtained. In the last chapter we briefly address the theory behind anyons, which is relevant for the 2 dimensional case.

## 1.2 Overview of CT Theories

### Classical descriptions: COB model

The classical 'over the barrier model' [17] treats an electron as being inside a potential well. The well's potential is described by the function

$$V(r) = -\frac{q}{r} - \frac{1}{|R - r|} \quad (1.1)$$

where  $r$  is the distance of the bound electron from the incoming nucleus.  $R$  is the distance between the nuclei, and  $q$  is the charge of the ion.

Potential has its maximum as  $V_{max} = -(\sqrt{q} + 1)^2 / R$ . The electron attaches itself to an incoming ion if its binding energy, perturbed by the electronion Coulomb energy  $-1/2 - q/R$  becomes greater than  $V_{max}$ .

Assuming that the electron's binding energy is  $E_n = -q^2/2n^2$  where  $n$  is a principal quantum number, energy conservation requires that

$$-\frac{1}{2} - \frac{q}{R} = E_n - \frac{1}{R} \quad (1.2)$$

Together these relations predict the largest value of  $R$  where the electron is transferred to an incoming ion with an energy  $E_n$  to be:

$$R_n = \frac{2(q-1)}{q^2/n^2 - 1} \quad (1.3)$$

Assuming the transfer probability of 1/2 this model predicts the total charge transfer cross section to be:

$$\sigma_{CT} = \frac{\pi}{2} R_{n^*}^2 \quad (1.4)$$

where  $n^*$  is the value of  $n$  in equation (1.3) where the equality holds. We shall compare this value with the value obtained by our method. But generally speaking this method fails to give the correct values for the very low energy processes [37].

## Quantum descriptions I: Langevin Orbiting

Since the preceding model fails to give the correct prediction a new, still classical model was suggested by P. Langevin. In this model, the incoming ion introduces the energy shift of the electron's ground state by

$$\Delta E = -\frac{1}{2}\alpha|E|^2 \quad (1.5)$$

where  $\alpha$  is called the polarizability of the atom and  $E = q/R^2$ . The mutually attractive potential has the form :

$$V_{pol}(R) = -\frac{C_4}{R^4} \quad C_4 = \frac{\alpha}{2}q^2 \quad (1.6)$$

in addition to the centripetal potential

$$V_L(R) = \frac{L(L+1)}{2\mu R^2} \quad (1.7)$$

where  $L = \mu vb$  is an angular momentum and  $b$  is an impact parameter. For a given  $b$ , the effective potential, the sum of  $V_{pol}$  and  $V_L$  possesses a local maximum  $V_{max}(b)$  at critical value  $R_c$ . At the end this model predicts the total charge transfer cross section to be:

$$\sigma_{CT} = 2\pi \int_0^{b_c} db b = \pi b_c^2 = \frac{\pi}{v} \sqrt{\frac{8C_4}{\mu}} = \frac{2\pi q}{v} \sqrt{\frac{\alpha}{\mu}} \quad (1.8)$$

## Quantum descriptions II: LandauZenerRosen Theories

This is a semi-classical method that offers a quick analytic formula, which takes into account the quantum nature of the motion, to estimate CT rates in systems in which the molecular potentials share an avoided crossing. We shall skip the detailed derivation, just to note that this model predicts the cross section to be:

$$\sigma_{CT} = 2\pi \int_0^{R_c} b P(b) db \quad (1.9)$$

$$P(b) = \frac{1}{2} \text{sech}^2 \left( \frac{\pi \Delta E(R_c)}{2\lambda v(b)} \right).$$

where  $R$  is the internuclear distance,  $R_c$  is the value where the energy defect  $\Delta E = E_2(R) - E_1(R)$  has a local minimum. The  $E_{1,2}$  are the energies of the two states,  $b$  is the impact parameter as defined in [94] and  $v$  is the relative velocity of the collision partners.

### 1.2.1 Quantum descriptions III, MOCC Approach

The Molecular Orbital Approach (MOCC) assumes that the total system amplitude is expressed as a sum of electronic eigenstates, whose expansion coefficients describe the motion of the nuclear coordinate  $R$ . There are two versions of this theory. In the first, the nuclear motion is treated classically, but the electronic degrees of freedom are treated quantum mechanically. This approach, the method of perturbed stationary states (PSS), was first advocated by Massey and Smith in 1932. The quantum version of this theory treats both the nuclear motion and the electronic degrees of freedom quantum mechanically. It was

developed by David Bates and coworkers in the 1950s. It was not much used since it required a sizable computing power. So only in the 80s the fully quantum MOCC computation becomes available. [94].

In the fully quantum expressions of the PSS equations we get for the wavefunction

$$\Psi(\mathbf{r}, \mathbf{R}) = \sum_n^N \phi_n(\mathbf{r}, \mathbf{R}) F_n(R) \quad (1.10)$$

where  $F_n(R)$  are quantal amplitudes that predict the probability for the system to occupy the  $n$ -th electronic states at  $\mathbf{R}$ . Index  $N$  basically includes all the states that are energetically open. The states  $\phi_n$  form a complete set

$$\sum_n \phi_n^\dagger(\mathbf{r}, \mathbf{R}) F_n(R) \phi_n(\mathbf{r}', \mathbf{R}) F_n(R) = \delta(\mathbf{r} - \mathbf{r}') \quad (1.11)$$

and we can integrate out the degrees of freedom associated with the electrons and derive an effective Schrödinger description, for the amplitudes  $F_n(\mathbf{R})$  that describe the motion of the nuclei. In Chapter 2, we will apply this method to calculate the CT and CT-induced molecular association rates for the  $Yb^+ - Ca$  system.

## Chapter 2

### Radiative Charge Transfer and Association for the $Yb^+ + Ca$ System

## 2.1 Description

The analysis of the radiative and charge transfer processes is related to the analysis of the general scattering process. Typically, in scattering processes, we can distinguish between radiative and non-radiative processes. We can consider these charge transfers from an energetic perspective and observe that they are endothermic or exothermic.

In the endothermic case, the binding energy of the electron on the ion is less than that of the atom. In this case, to satisfy the energy conservation requirement, an incoming ion must supply the additional energy. However, if the ion approaches slowly, no additional energy is supplied. So when additional energy is not available, charge transfer can only occur if the final electron binding energy is greater than or equal to its binding energy on the atom.

If the energy is equal, the reaction is called resonant charge transfer.

If not, the reaction will be processed with the possibility of the emission of a photon. If radiation is emitted, the reaction is termed radiative charge transfer/association; if not, it is termed direct charge transfer.

## 2.2 Radiative Charge Transfer and Association

Radiative charge transfer (abbreviated RCT) is an inelastic, irreversible quantum collision process in which an electron is transferred and a photon is emitted.

In general the process looks like:  $A(n+1\ ^1S) + B \longleftrightarrow A(n\ ^1S) + B + \hbar\omega$

It is governed by the dipole matrix element  $D(R) = \langle \psi_f | r | \psi_i \rangle$  where  $|\psi_f\rangle$  and  $|\psi_i\rangle$  are the electronic wavefunctions associated with the initial and final states, respectively.

In our case process is  $Yb^+ + Ca \rightarrow Ca^+ + Yb + \gamma$ , for radiative charge transfer.

In this reaction, the electron jumps from the upper adiabatic curve to the lower one by emitting a photon. The nuclei continue moving while the electron radiates and drops to the lower curve, after which the electron localizes on one proton. This is a true inelastic process, controlled by the dipole elements between initial and final states.

## 2.3 Radiative Association

Related radiative process is Radiative association:

$$A(n+1\ ^1S) + B \longleftrightarrow A(n\ ^1S)B + \hbar\omega .$$

There is also emission of a photon, but as a result, a molecule is formed. In this case, a photon removes excess electronic energy and the relative kinetic energy. So in this case, from a wavefunction point of view, there is a transition from a continuum state to a bound vibrationalrotational state.

This process is dominant at very low collision energies. This is a key mechanism for molecule formation in the interstellar medium.

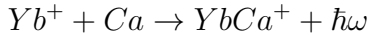
There are several theoretical approaches and treatments of these classes of problems. [24], [25], [91].

Fairly recently, there have been advances in trapping, cooling, and manipulation of ultra-cold atoms [77]. The development of hybrid ion-atom traps has enabled researchers to explore competing pathways, including non-radiative and radiative charge transfer, as well as radiative association, in which ions and atoms recombine and rearrange at low tempera-

tures.

Some possible applications include astrophysics [10], single-ion clocks, and, more recently, quantum computing. Ultracold atoms, cooled to nanokelvin temperatures near absolute zero  $\approx \mu K$  using laser cooling and evaporative techniques, serve as a premier platform for quantum simulation and computing. By trapping atoms in optical lattices or tweezers, they act as stable qubits for quantum information processing and simulate complex, many-body systems.

There has been a recent [77] laboratory study that used a hybrid trap to investigate the interactions among cold  $^{174}Yb^+$  ions with  $^{40}Ca$  atoms. That study reported a large chemical reaction rate coefficient of  $2 \times 10^{-10} cm^3 s^{-1}$  and, based on their calculations, argued that radiative charge transfer is the dominant process contributing to this rate. This is several orders of magnitude larger than similar reactions for the lighter species. The paper by Zygelman and Hunt [92] repeated the calculation using the optical potential method [91] for the total of two reactions.



We will show that our results are several orders of magnitude smaller than those given in [77]. Based on these results, there are some doubts about the conclusion given in that paper concerning the role of radiative relaxation. Following the approach in the [93], we apply a semi-classical approach to the system of  $YbCa$  molecules. This approach at ultra-low energies (temperatures) has not been tested before. In addition, we used molecular data obtained from the illustrations in [77]. At these energies, small details in the potential surfaces and dipole moments can be important. For these reasons, we have repeated the calculations using the original data provided in [77]. We compare the predictions of the local-optical potential with the results obtained using the Fermi-Golden-Rule prescription [91]. The result obtained by our calculation in the subsequent chapter is several orders of

magnitude smaller than that given in [77].

The local optical potential method has its roots in the semi-classical theory of radiative association first developed by Kramers and Ter-Haar [46]. In the subsequent discussion, based on that theory, we assume that a photon is emitted with energy equal to the energy difference between the two Born-Oppenheimer potential surfaces at the transition distance. The efficiency of a transition is determined by an Einstein-A coefficient at that internuclear distance. This leads to the Local Optical Potential Method, and following the semi-classical approach, the total rate is estimated as a classical integral over all localized transitions.

In our discussion below, we briefly summarize the various theoretical approaches and use them to calculate the rates for processes.

We provide a rigorous upper bound for the sum of rates given in (2.1).

### 2.3.1 Theory

Here we follow the discussion in [93], and apply the theory to the  $YbCa^+$  system. The cross section of the spontaneous radiative association is given by [10]. This can also be derived using Fermi's Golden Rule (FGR)(which was actually published by Dirac 20 years before Fermi)[80]: Considering the radiative association process:

$$Yb^+ + Ca \rightarrow YbCa^+ + \hbar\omega \quad (2.2)$$

where  $\hbar\omega$  is the energy of the emitted photon. The cross-section for the radiative association process is given by:

$$\sigma_{RA} = \sum_J \sum_n \frac{8}{3} \frac{\pi^2 \omega_{nJ}^2}{c^3 k^2} \left[ (J+1) M_{J+1,J}^2(k, k') + J M_{J-1,J}^2(k, k') \right] \quad (2.3)$$

$$M_{J,J'}(k, n) = \int_0^\infty dR f_J(kR) D(R) \phi_{J'n}(R)$$

where  $D(R)$  is the transition dipole moment between  $X^2\Sigma^+$  and  $A^2\Sigma^+$  states of the  $YbCa^+$  molecular ion.  $\phi_{J'n}(R)$  is a rho-vibrational eigenstate of the  $X^2\Sigma^+$  ground state, with energy eigenvalue  $\epsilon_{nJ}$ . The  $J$  and  $n$  represent the angular and vibrational momentum quantum numbers, respectively.  $f_J(kR)$  is the wavefunction that satisfies the radial Schrodinger equation

$$f_J''(kR) - \frac{J(J+1)}{R^2}f_J(kR) + 2\mu V_A(R)f_J(kR) + k^2 f_J(kR) = 0 \quad (2.4)$$

where  $V_A(R)$  is the Born-Openheimer energy of the excited  $A^2\Sigma^+$  state,  $\mu$  is the reduced mass of the collision system, and  $k$  is the wavenumber of the incident collision partners in that channel. The asymptotic solution of the  $f_J(kR)$  has the form:

$$f_J(kR) \leftarrow \sqrt{\frac{2\mu}{\pi k}} \sin(kR - \frac{J\pi}{2} + \delta_J) \quad (2.5)$$

where  $\delta_J$  is a phaseshift, as  $R \rightarrow \infty$ . The energy of the emitted photon is given by

$$\hbar\omega = \frac{\hbar^2 k^2}{2\mu} + V_A(\infty) - \epsilon_{nJ} - V_X(\infty) \quad (2.6)$$

This process is governed by the selection rules,

- Rotational:  $\Delta J = \pm 1$
- Electric Dipole:  $\langle f|D|i\rangle$
- Electronic angular momentum:  $\Delta\Lambda = 0, \pm 1$

– where  $\Lambda = 0, 1, 2$  corresponds to  $\Sigma, \Pi, \Delta$  states

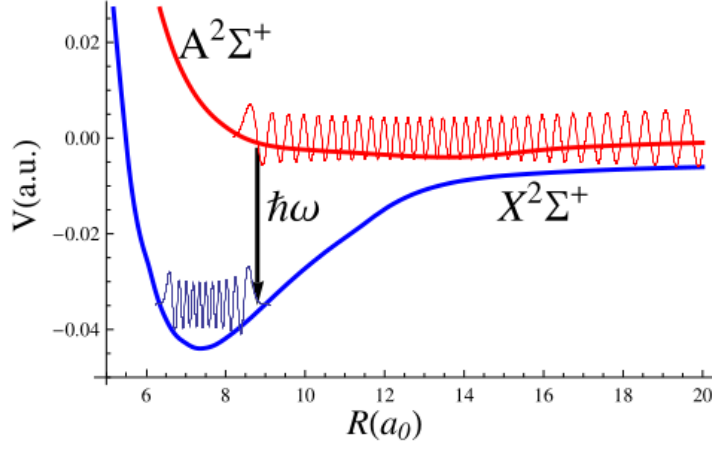


Figure 2.1: Illustration of the BO molecular potential curves (solid thick lines) participating in the radiative association process. In the  $A^2\Sigma^+$  entrance channel, the wave function is shown by the light undulating line. The oscillations are due to the strong polarization force in the entrance channel, leading to a potential minimum at  $R \approx 14a_0$ . Association is precipitated by the emission of a photon of energy  $\omega$  near the classical turning point. The final bound rho-vibrational state, in the  $X^2\Sigma^+$  channel, is shown by the thin line.

## 2.4 Radiative Charge Transfer Theory

Here, we consider the radiative charge transfer process

$$Yb^+ + Ca = Yb + Ca^+ + \hbar\omega \quad (2.7)$$

where  $\hbar\omega$  is the energy of the emitted photon given by the expression (2.10). The cross-section for the radiative charge transfer process is given by

$$\sigma_{CT} = \int_0^{\omega_{max}} d\omega \frac{d\sigma}{d\omega}, \quad (2.8)$$

$$\frac{d\sigma}{d\omega} = \sum_J \frac{8}{3} \frac{\pi^2 \omega_{nJ}^3}{c^3 k^2} \left[ (J+1) M_{J+1,J}^2(k, k') + J M_{J-1,J}^2(k, k') \right]$$

where

$$M_{J,J'}(k, n) = \int_0^\infty dR f_J(kR) D(R) f'_{J'}(k'R) \quad (2.9)$$

Here  $f_J(kR)$  is a solution to (2.4) and  $f'_{J'}(k'R)$  obeys the corresponding equation for the  $X^2\Sigma^+$  exit channel with wavenumber and partial wave  $k'$ ,  $J'$  respectively. The radial wavefunctions are normalized as in (2.5) and

$$\hbar\omega = \frac{\hbar^2 k^2}{2\mu} - \frac{\hbar k'^2}{2\mu} + \Delta E \quad (2.10)$$

$$\Delta E = V_A(\infty) - V_X(\infty)$$

From the (2.10) the maximum angular frequency  $\omega_{max}$  is

$$\hbar\omega_{max} = \frac{\hbar^2 k^2}{2\mu} + \Delta E \quad (2.11)$$

We avoid evaluating the sum (2.8).

Instead, we will compute an upper bound on the charge transfer cross section. From the equation above, we get for the frequency of the emitted photon, during an RCT transition, to be

$$\hbar\omega = \frac{\hbar^2 k^2}{2\mu} - \frac{\hbar^2 k'^2}{2\mu} + V_A(\infty) - V_X(\infty) = \frac{\hbar^2 k^2}{2\mu} - E' + V_A(\infty) - V_X(\infty) \quad (2.12)$$

Now, from the equation above, we can see that to achieve the maximum energy of the emitted photon,  $\hbar\omega_{max}$ , we need to have  $E' = 0$ . The maximum value of  $E'$  is achieved when  $\hbar\omega = 0$  therefore:

$$E'_{max} = \frac{\hbar^2 k^2}{2\mu} + V_A(\infty) - V_X(\infty) \quad (2.13)$$

Now the (2.8) can be written as:

$$\sigma = \frac{8}{3} \frac{\pi^2}{c^3 k^2} \int_0^{E'_{max}} dE' \omega^3(E') \left[ (J+1) M_{J,J+1}^2(k, E') + J M_{J,J-1}^2(k, E') \right] \quad (2.14)$$

where we have the inequality

$$\sigma_{CT} < \frac{8}{3} \frac{\pi^2 \omega^3(E')_{max}}{c^3 k^2} \int_0^{E'_{max}} dE' \left[ (J+1) M_{J,J+1}^2(k, E') + J M_{J,J-1}^2(k, E') \right] \quad (2.15)$$

Now we can replace the expression for  $M_{J+1,J}^2(k, E')$  in (2.15) with the expression in (2.9)

and squaring it, we get the integral

$$\int_0^\infty dE' JM_{J,J-1}^2(k, E') = \int_0^\infty dE' \int_0^\infty dR f_J(kR) D(R) f_{J-1}(k'R) \int_0^\infty dR' f_J(kR') D(R) f_{J-1}(k'R') \quad (2.16)$$

so the inequality becomes

$$\int_0^\infty dE' JM_{J,J-1}^2(k, E') < \sum_{E'} JM_{J,J-1}^2(k, E') = J \int_0^\infty dR f_J^2(kR) D^2(R) \quad (2.17)$$

and we get for the upper bound

$$\sigma_{CT} < \tilde{\sigma}_{CT} \frac{8\pi^2 \omega^3(E')_{max}}{c^3 k^2} \sum_J (2J+1) \int_0^\infty dR f_J^2(kR) D^2(R) \quad (2.18)$$

### 2.4.1 Optical Potential Approach

The alternative way to compute the cross section is to use the local optical potential method [91]. The loss of flux from the entrance channel equals the radiative probability, which is converted to a radiative cross section. In this method, the collision system in the incoming  $A^2\Sigma^+$  state experiences an imaginary (absorptive) potential  $iA(R)/2$  added to the entrance-channel BO potential. So we define a complex potential

$$V(R) = V_{BO}(R) + V_{OPT}(R) = V_{BO}(R) + i \frac{A(R)}{2} \quad (2.19)$$

$$A(R) = \frac{4}{3c^2} D(R) [V_A(R) - V_X(R)]$$

$A(R)$  is the Einstein-A coefficient. We shall use a 2D version of the optical theorem to compute the radiative charge transfer in the 2D case in the next chapter. The next step is to solve the usual radial scattering equation, but with  $V(R)$  complex. The resulting S matrix becomes non-unitary,  $|S| < 1$ , and the missing probability is the radiative loss.

The cross-section for radiative quenching is given by

$$\sigma = \frac{\pi}{k^2} \sum_J (2J+1) (1 - e^{-4\eta_j}) \quad (2.20)$$

where  $\eta_j$  is the imaginary part of the J'th partial wave phase shift  $\delta_j$  for the radial wave

$f_J(kR)$  that satisfies

$$f_J''(kR) - \frac{J(J+1)}{R^2} f_J(kR) + 2\mu[V_A(R) + V_{opt}(R)]f_J(kR) + k^2 f_J(kR) = 0 \quad (2.21)$$

The Local Optical Potential is generally applicable when the following conditions are satisfied, which aligns well with our case.

- Transition happens over a region small compared to relevant wavelengths
- Emission is relatively weak compared to the collision energies
- It is accurate enough for the computation of total probability

## 2.4.2 Ultra Cold Limit

In the limit of ultra-cold temperatures, only the s-wave state participates in the process, so the total cross section takes a simpler form.

$$\sigma = \sum_n \frac{16\mu\pi\omega_n^3}{3c^3k} \left| \int_0^{R_0} dR \phi(R) D(R) \phi_{J=1}^n(R) \right|^2 \quad (2.22)$$

where  $\phi(R)$  is the s-wave solution to (2.4). The boundary condition for  $\phi(R)$

$$\frac{d\phi(R)}{dR} \Big|_{R_0} = 1 \quad (2.23)$$

is valid for a sufficiently large radius  $R$  and  $\phi_{J=1}^n(R)$  are  $J = 0$  rho-vibrational states of the  $X^2\Sigma^+$  potential. Because there is only an s-wave, the integral (2.22) does not depend on the incoming wave number  $k$ . Therefore, the equation (2.22) predicts that the association cross section, in the ultra cold regime, scales as the inverse of the incoming velocity, i.e., the rate tends to a constant.

We shall show in the subsequent chapter the method used to compute the phase shift. The idea is to match the numerical solution for  $f_J(kR)$  to its asymptotic form for large  $R$ .

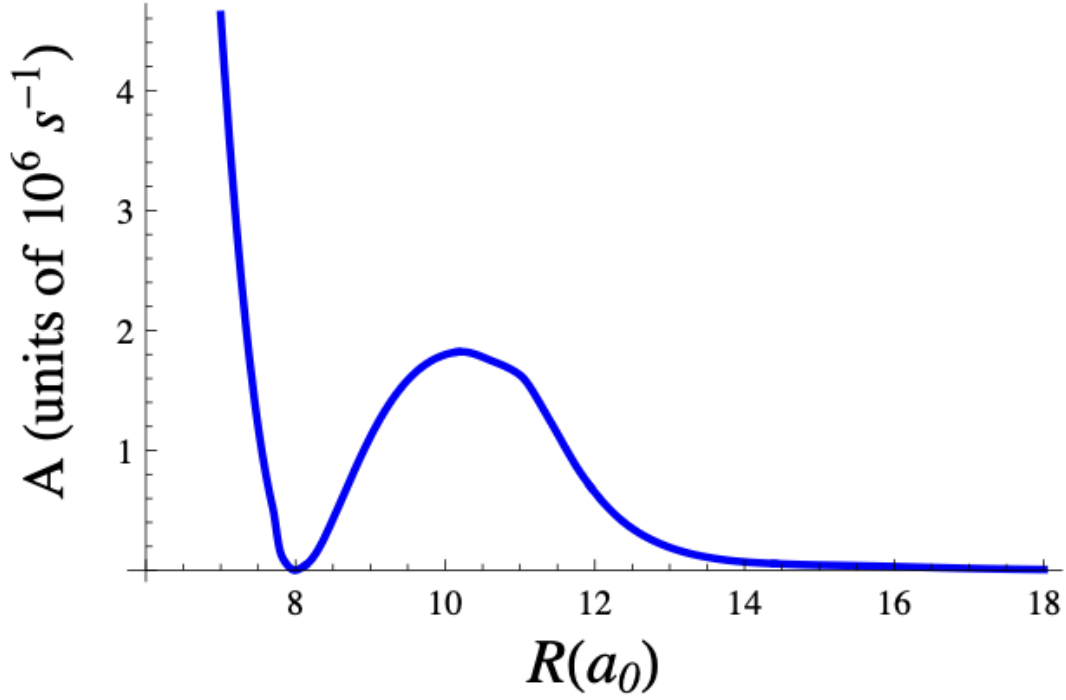


Figure 2.2: R-dependent Einstein A coefficient as a function of internuclear distance

## 2.5 Results

Figure 2.1 illustrates the mechanism for radiative association for the  $Yb^+$  ion and  $Ca$  atom that approach in the  $A^+\Sigma^2$  electronic BO state. The BO energies were taken from the data of the ab-initio calculations reported in [68]. At large internuclear distances, this potential has the form

$$V_A(R) \rightarrow \frac{C_4^A}{R^4}, \quad C_4^A = 78.5 \quad (2.24)$$

In the case of Radiative association, the incident  $A^2\Sigma^+$  channel can relax via photon emission. In that case, the final state is a bound rho-vibrational level of the  $X^2\Sigma^+$  channel. In radiative charge transfer, the collision partners can exit in that channel, as a re-arranged

$Yb - Ca^+$  pair. In the exit channel

$$V_X(R) \rightarrow \frac{C_4^X}{R^4} + \Delta E, \quad C_4^X = 71.5, \quad \Delta E = -0.0052 \quad (2.25)$$

as  $R \rightarrow \infty$

In calculating the radiative association cross sections given by (2.3), we need to itemize all bound states supported by the  $X^2\Sigma^+$  channel. The total number of bound states can be approximated using the JWKB expression

$$n = \text{Floor} \left[ \int_{R_c}^{\infty} dR \sqrt{-2\mu(V_X(R) - \Delta E)} - 1/2 \right] \approx 54,878 \quad (2.26)$$

where  $R_c$  is a classical turning point  $\text{Floor}[x]$  is the integer lower bound of  $x$ .

Here we have a large reduced mass  $\mu$ , i.e., the participant's mass is large, which in turn produces a very large number of bound states. For lighter participants, the number of rho-vibrational states is typically in the order of a few hundred [10].

Here we have another factor to consider. In the ultra-cold temperatures, the participants have lower energies, so the centrifugal repulsion in the entrance channel limits the number of partial waves that participate. Since the selection rules  $J \pm 1$  still apply, they constrain the possible rho-vibrational levels. For example, at a collision energy corresponding to a temperature of  $1mK$ , only levels with  $J$  up to the value  $\approx 15$  contribute to the association rate.

The following graph 2.3 shows the results of the calculation.

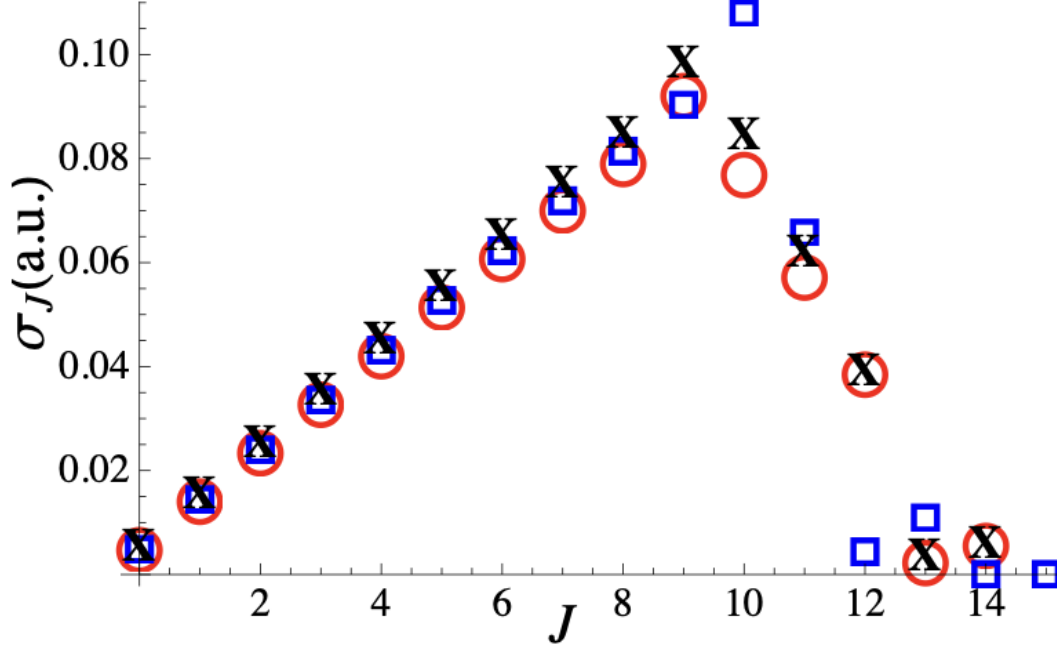


Figure 2.3: Plot of various cross sections as a function of incoming partial wave  $J$ . Circles represent data for the radiative association cross sections, Xs represent an upper bound for the total (association + radiative charge transfer), and squares represent the data for the total radiative relaxation obtained using the optical potential method.

We have made the calculations two ways. First, starting from the equation (2.4), we computed the cross section using equation (2.8). In the figure 2.3 symbol  $X$  represents the upper limit for total radiative relaxation, which is obtained by adding the association cross sections (2.8) with those given by expression (2.18).

The square icons represent the cross sections predicted by expression (2.20), i.e., by the optical potential method.

So from figure 2.3 for values  $J < 10$  we see that an optical potential method provides a very good approximation to the total cross section.

However, this approximation is good only for values  $J < J_{max}$  [92].

$$J_{max} = \sqrt[4]{8\mu k^3 C_4^A} = \sqrt[4]{24\mu^2 k_B T C_4^A} \approx 12 \quad (2.27)$$

The collision energy of the collision system is given as  $3/2k_B T$  where  $k_B$  is the Boltzman

constant and  $T$  is the temperature in Kelvin

Value  $J_{max}$  is the critical angular momentum for which the collision system has sufficient collision energy to overcome the centrifugal potential barrier [92]. For larger  $J$ , tunneling resonances can access the inner region where the transition dipole moment is non-negligible and induce radiative transitions.

In Table I of ref [93], we present the various cross sections at several representative collision temperatures. The second column contains the association cross-section obtained using the FGR method described above. For the radiative charge-transfer cross sections, as listed in the third column, we used (2.18). Thus, the upper bound for the total radiative relaxation cross sections is given in column 4. The last column gives the results obtained using the local optical potential method. The table shows that, over the temperature range considered, the local optical potential method predicts cross sections that are less than the upper bound itemized in column 4. Secondly, the differences between the two theories' predictions are small. The optical potential cross section differs by less than 4% from the upper limit values over the entire temperature range, including the ultra-cold region.

One can also note that the optical potential method predicts cross sections that are larger than the radiative association cross sections. At the same time the optical potential method provides a reliable upper bound for the total (RR) cross section.

In figure 2.4 we plot the total radiative relaxation cross section, obtained using the optical potential method, for the gas temperature range  $1mK < T < 1K$ .

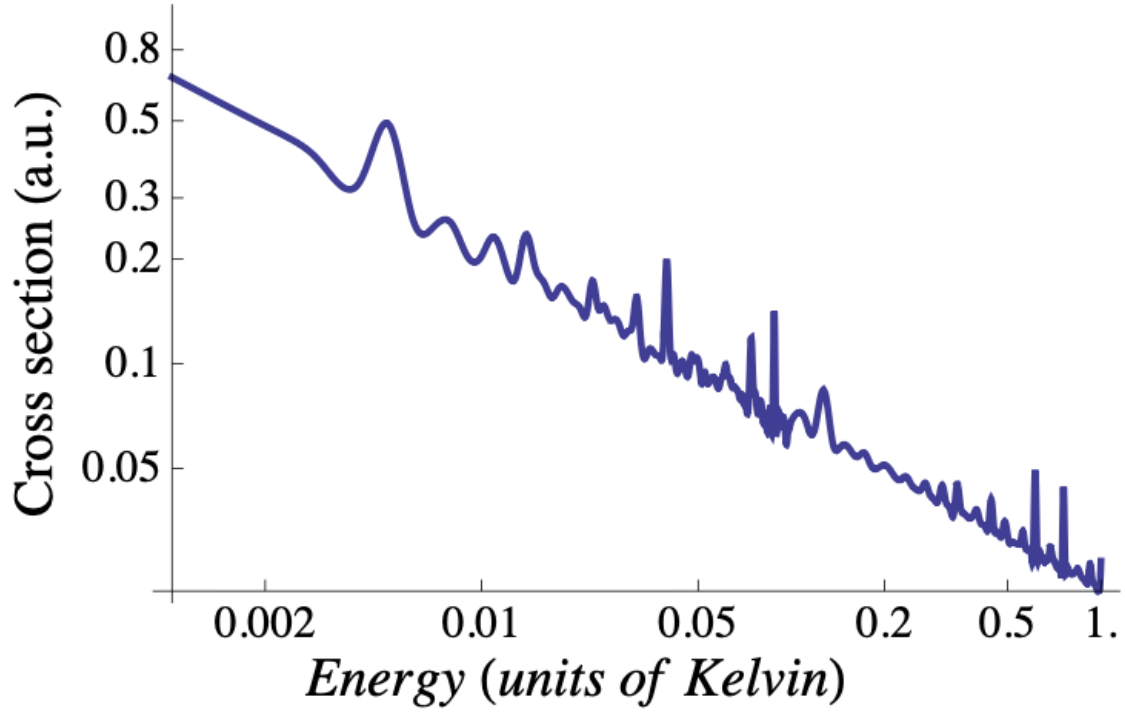


Figure 2.4: Total radiative relaxation cross section as a function of collision energy expressed as  $E = 3/2 k_B T$ , where  $T$  is the temperature in Kelvin

Although the optical potential method provides a good approximation for the total radiative relaxation rate, calculation of the photon emission spectrum requires the use of the FGR method. In 2.5 we illustrate the association cross sections  $\sigma_{nJ}$  at  $T = 1mK$  for the individual rho-vibrational levels as a function of the frequency of the emitted photon. The structure of the emission pattern is the result of the oscillation in the incoming wave as shown in 2.1

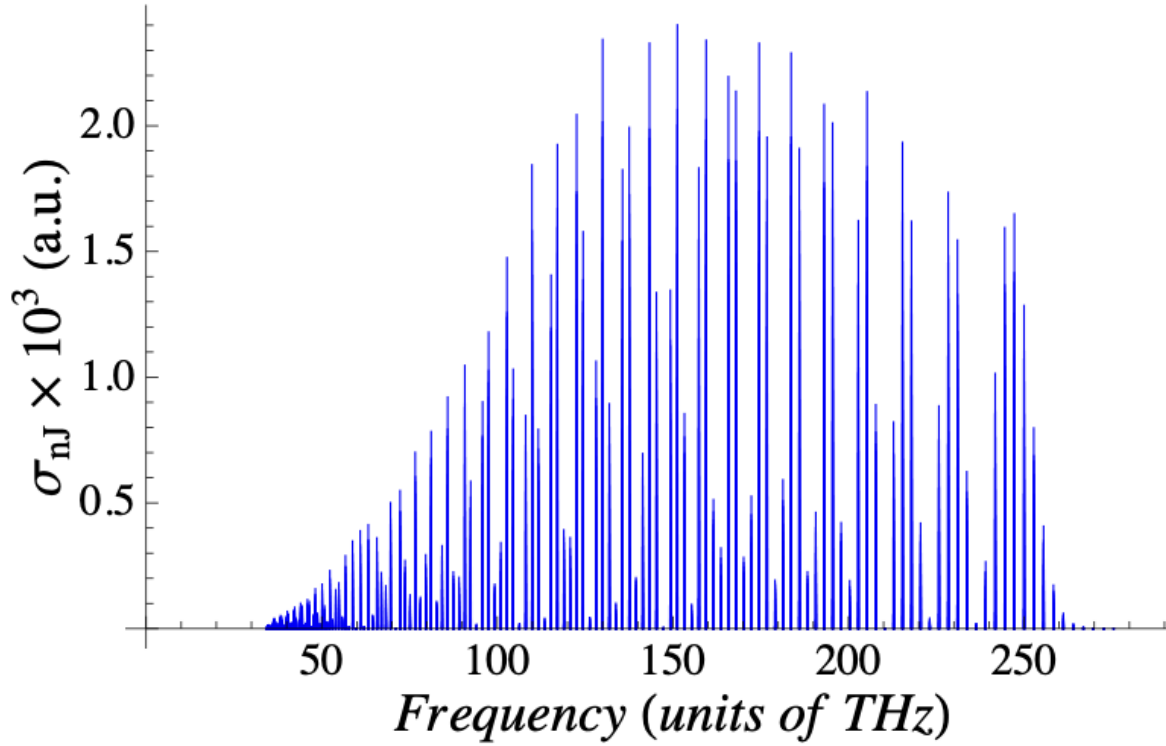


Figure 2.5: Emission spectrum for the radiative association process at a gas temperature of 1mK.

In the limit  $T \rightarrow 0$ , we define a complex scattering length for the s-wave solution to (2.21)

$$a \equiv -\frac{1}{k} \tan \delta(k) = 10785 - i0.008842; \text{ as } k \rightarrow 0 \quad (2.28)$$

Therefore, the total RR cross section, according to the optical potential method, has the limiting value

$$\sigma = \frac{4\pi}{k} 8.842 \times 10^{-3} \quad (2.29)$$

Defining the rate coefficient

$$k_{RR} \equiv \langle \nu \sigma \rangle \quad (2.30)$$

we use the results from [93] to show the values of  $k_{RR} \approx 1.15 \times 10^{-14}$ .

By looking at 2.4 we observe that  $k_{RR}$  varies slow, from  $1.6 \times 10^{-15} \text{ cm}^3 \text{ s}^{-1}$  at 1mK to  $2 \times$

$10^{-15} \text{cm}^3 \text{s}^{-1}$  at 1K.

We can summarize the current chapter as follows. Using both quantum and semi-classical methodologies, we calculated the rates for the radiative association and charge transfer in cold collisions of  $Yb^+$  with  $Ca$ . We have shown that the optical potential method provides very good results in predictions for the total radiative relaxation rates, in the ultra-cold s-wave scattering regime. In addition, we have computed and shown the cross section  $\sigma$  as a function of temperature  $K$  and frequency of the emitted photons.

### 2.5.1 Isotope Dependency

Different Yb isotopes ( $^{170}Yb$ ,  $^{174}Yb$ ,  $^{176}Yb$ ) differ only in nuclear mass, not in electronic structure. Thus, isotope dependence arises from nuclear motion. Following [94] we observe that radiative processes

$$Yb^+ + Ca \rightarrow Yb + Ca^+ + \hbar\omega \text{ and } Yb^+ + Ca \rightarrow YbCa^+ + \hbar\omega$$

are both controlled by the overlap integral

$$\int_0^{R_0} dR \phi(R) D(R) \phi_{J=1}^n(R) \quad (2.31)$$

Because isotope mass shifts the bound-state energies, it leads to order-of-magnitude differences in rate constants between isotopes.

At ultra-low temperatures ( $\mu K$  and below), only the s-wave is present, and the rates obey the Wigner threshold law. In that case, if a  $Yb$  isotope supports a bound state just below threshold, the scattering length becomes very large, and the radiative association rate is strongly enhanced. The enhancement becomes isotope-specific. This explains the pronounced isotope separation seen in calculated rate curves as shown in [94]. So, radiative association shows strong isotope dependence, while the radiative charge transfer isotope dependence is smaller. From [94], we copy, with permission, the diagram showing isotopic dependency of the radiative association.

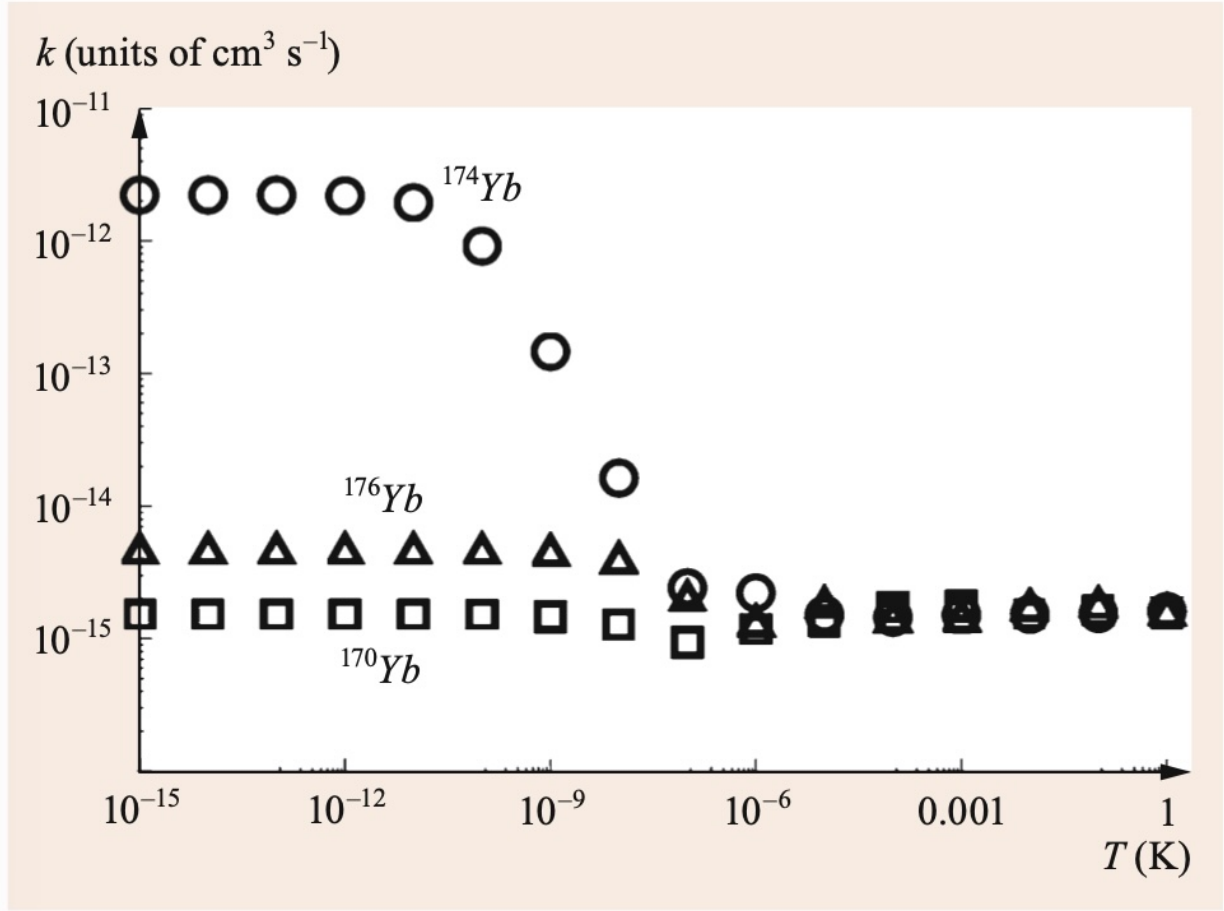


Figure 2.6: Isotopic dependence of radiative association into the  $\text{YbCa}^+$  molecular ion

In the following sections, we focus on 2D systems and describe Resonant Charge Transfer.

## Chapter 3

### Quantum Materials in 2 Dimensions

In this section , we will consider the following resonant charge transfer (RCT) process



i.e., the exchange of an electron in the collision of a proton with a hydrogen atom. This fundamental CT process has been studied extensively in three dimensions, but in this dissertation, we, for the first time, calculate the rate for (3.1) in a two-dimensional substrate, i.e., both the electron and nuclei are constrained to move in two dimensions. The motivation for this study stems from advances in laboratory studies of 2D systems and the fabrication of quantum materials in 2D. Thus, fundamental collision processes must be investigated theoretically in 2 dimensions. 2D materials hold great promise for innovative applications. To advance this emerging technology, it is essential to gain a fundamental understanding of the processes that govern the behavior of matter at the atomic and molecular levels. In this thesis, we will examine the interactions of ions with neutral matter in a 2D environment. Before we focus on the calculation of Eq. (3.1), we review, below, some advances in the physics of 2D systems.

#### 3.0.1 Overview of 2D Material Science

2D materials can exhibit unique topological properties that do not exist in the 3D case. An exciting and important feature in this context is the emergence of anyons [76]. Anyons are neither Fermions nor Bosons. The exchange of two anyons induces an arbitrary relative phase between the partners; an effect called fractional statistics. In 3D space, the

relative phase can only take the integer values of  $2\pi$  (Bosons) or  $\pi$  (Fermions).

### 3.0.2 On 2D Materials No-Go Theorems and Graphene

Landau and Piers[47][63] initially argued that 2D materials could not exist. That argument posited that the displacement of atoms at finite temperatures induces thermal fluctuations on the order of the atomic spacing, thereby melting the 2D material above 0 K.

There is an even stronger statement against 2D materials, in the form of the Mermin-Wagner-Hohenberg (MWH) theorem [38][57]. In these papers, the authors show that 2D systems with short-range interactions are unstable and cannot exist. Using thermodynamic arguments, one can show that a 3D system is stable if its free energy is bounded from below and convex (upwards). In this way, the system can minimize the ground-state energy. Because higher-order fluctuations are finite, the system is stable. The MWH argument posits that, in 2D systems, fluctuations around the ordered state (lattice in the 2D case) decorrelate over large distances, thus destroying the large-scale order. Therefore, it seemed that the possible existence and applications of the 2D materials were quite limited. The only 2D materials considered stable were thin molecular films, which could form on a solid's surface. Dash[41] predicted that 2D materials exist as a thin film either on the surface of the boundaries of 3D materials.

Despite those arguments, it has been speculated that stable 2D materials are possible[41]. The situation changed dramatically in 2004 when 2D carbon (graphene) was discovered and created in the laboratory[4, 5] at the University of Manchester, Great Britain.

2D materials have qualitatively different topological properties[16]. Moreover, those properties do not depend on the type of microscopic interactions between particles. While this subject is beyond the scope of this thesis, it does show that the field of 2D materials is a fertile ground for both fundamental and applied research. Interestingly, it has been shown

[61] that it takes 10 graphene layers for it to begin behaving like a 'regular' 3D material. The existence of graphene does not contradict the above-mentioned no-go theorem [57], but it does circumvent it [1]. In graphene, thermal oscillations produce ripples or bending of the graphene sheet. That way, graphene behaves like an elastic membrane that allows phonons to propagate in 2 dimensions and couple in a plane stretching along the transverse fluctuations out of the plane. These phonons mediate a long-range interaction, thereby circumventing the Mermin-Wagner no-go theorem.

Additionally [4] [59][15], 2D materials exhibit several other interesting properties. Its quantum excitations behave in accordance with relativistic dynamics. In condensed matter systems, the solutions of the Schrödinger equation describe the material's electronic properties. In graphene, the electrons obey the non-relativistic Schrodinger equation, but interactions with the graphene lattice's periodic potential give rise to solutions referred to as quasiparticle excitations, and they behave in the same manner as massless Dirac fermions. Even at low energies, these quasiparticles are described by a Dirac equation in 2+1 dimensions.

### 3.0.3 Excitons

These excitations represent a wide class of intrinsic electronic excitations (Excitons) [51, 35] in crystals of semiconductors and dielectrics. In one model, excitons are bound states of an electron and a hole, typically formed when an incident photon is absorbed, thereby exciting an electron from the valence to the conduction band. The attractive Coulomb interaction between the excited electron and the newly created hole binds them together to form a bound neutral compound system of the two charge carriers, similar to a hydrogen atom. The character of exciton motion depends on the strength of the exciton interaction with phonons. The electron-hole model is common in insulating and semiconducting crystals. Ya. I. Frenkel (1931) introduced excitons to explain the light absorption in crystals,

which does not lead to photoconductivity. In the Frenkel model, the exciton is considered as an electronic excitation of one crystal site with the energy close to, but a bit smaller than, that necessary for the excitation of a free electron. Due to the crystal's translational symmetry, the exciton can move along the lattice sites, transferring energy to electrically active or luminescent centers.

Similar excitations occur in 2D systems. [45, 22]. 2D excitons promise to serve as building blocks for 2D electronics. [74]. Although exciton-based transistor action has been demonstrated in bulk semiconductor coupled quantum wells, the low operating temperature required limits their practical application. The emergence of 2D semiconductors with large exciton binding energies may lead to excitonic devices and circuits that operate at room temperature. [74].

### 3.0.4 Graphene

Graphene exhibits similar properties[4], such as its Quantum Electrodynamics (QED) like electronic spectrum, electron tunneling. Another interesting consequence of the QED-like electron spectrum is the possibility of experimentally studying QED in curved space by controllable bending of a graphene sheet. Such a study may offer a possibility to address a certain class of cosmological problems.

At present, since the discovery of graphene, several new 2D materials have been identified[26]. As new 2D materials are discovered, researchers are investigating their potential applications. One of the most useful applications of exotic materials is electronics. These new 2D materials promise that they can be used to build existing electronic components, but with potentially better performances [8][9].

Nanotubes are rolled-up graphene sheets, and depending on the application, graphene (as a sheet) is sometimes superior, sometimes inferior, and sometimes completely different.

Another notable application is the emergence of high-temperature superconductivity in 2D materials[43].

## 2D Electronics

Driven by a demand for higher performance and lower consumption in electronic devices, 2D materials have found application in that arena[8],[9]. There is the hope that graphene can be used to build active electronic components, such as diodes, transistors (FET), and other electronic components. Alternatively, graphene can be used as a conductor in electronic components and batteries owing to its high electrical conductivity. Graphene itself is a zero-gap semiconductor. There is another unique opportunity in 2D semiconductors. In a 3D semiconductor, electronic states are buried deep inside the material. Unlike the surface state, electron states within the material are localized, i.e., their wave functions have exponentially decaying tails.

In 2D semiconductors, the electronic states exist on the surface. The surface states admit both localized and non-localized block-bulk states [23]. Another interesting application of 2D electronics is the LED elements in 2D [69]. Electronics research is closely related to research on new 2D materials. For example, graphene itself is not compatible with silicon. However, there is a new material, black phosphorus[34], which is both compatible with silicon and it holds promise for future electronic[40] devices. This fact relies on two key properties: the first is that black phosphorus has higher mobility, the speed at which it can carry an electrical charge, than silicon. The other is that it has a finite bandgap, whereas graphene does not. So, in essence, in the presence of an electric field, black phosphorus can act as a semiconductor[66].

## Quantum Computing

The concept of topological quantum computing [70] depends on the unique properties of 2D materials. It [70, 71, 67, 58, 30], relies on the unusual quantum properties of 2-dimensional (2D) matter. In principle, there are no theoretical objections to building a quantum computer; there are serious technical issues, such as noise and scaling limitations, for current qubit technologies that need to be overcome[65]. The topological approach promises a physically realizable quantum computer[71, 67].

The current computer architecture is based on the so-called 'von Neumann' architecture. The algorithms based on that architecture have inherent limitations, and, as it currently stands, the architecture cannot overcome them.

There are two main classes of computer algorithms, P and NP. There are subclasses within those classes, but those two classes capture the essential behavior of classical algorithms. The algorithms belonging to class P are solvable in polynomial time, with regard to the size of the input. Those algorithms belonging to class NP are those that require exponential time to complete[21].

Currently, the major unsolved problem in Computer Science is the P vs. NP problem [21]. While the hypothesis that  $P \neq NP$  is generally accepted, no proof has been found. Applying quantum-mechanical principles in engineering disciplines, such as computer science [29], promises to yield a qualitatively more powerful quantum computer.

Now, all problems in the NP class (more precisely, NP-complete subclass) can be transformed into each other in polynomial time [33], therefore finding a polynomial-time solution for one of them would amount to having a polynomial-time solution to all problems in the NP class. So the goal is rather worthy.

It is currently known that quantum computers cannot solve NP problems. They could solve the class of the bounded-error, quantum, polynomial time (BQP) problems [85].

Even if the NP problems remain intractable, just the possibility that, according to Feynman [29], a quantum computer can be used to simulate the quantum processes, which will in itself be a step forward.

For example, Schor's algorithm [73] can provably solve integer factorization in polynomial time, as opposed to the best classical algorithm (quadratic sieve), which works in sub-exponential time [64]. Unfortunately, the integer factorization problem, as well as other BQP problems, do not belong to the class of NP-complete problems.

Interestingly, according to Abrams and Lloyd [2], [3], if the small nonlinear term is added to Quantum Mechanics, quantum computers would be able to solve NP-complete problems in polynomial time.

## Anyons

Anyons are quasi-particles that can arise in 2D systems and whose quantum statistics in neither Fermionic nor Bosonic [52]. This fact has been proven [83], but arbitrary statistics are valid only in 2 dimensions [81]. After exchanging two identical particles, the wave function gains an arbitrary phase factor  $\Psi = e^{i\theta}\Psi$ , where  $\theta = 2\pi\nu^s$ . In contrast, for a 3D system, the exchange of two particles leads to either phase factor  $\pi$  or 0, for fermions and bosons, respectively.

In general, there are two kinds of Anyons: Abelian and non-Abelian. Abelian Anyons have been reliably detected in systems exhibiting the Fractional Quantum Hall Effect [49]. Whereas the computation with Abelian Anyons is theoretically possible [86], considerably more attention has been paid to non-Abelian Anyons. There are several possibilities for realizing non-Abelian statistics experimentally:

- A two-dimensional electron gas in a large magnetic field (the fractional quantum Hall effect)

- Rapidly rotating Bose-Einstein condensates[28]
- Frustrated magnets [19].
- One of the anyons candidates the Majorana fermion [78].

Majorana fermion (particle) is a particle that is its own antiparticle. A pair of localized Majorana states has been predicted to reside at the ends of a specially designed superconducting wire. Very recently (2020) [53] there is a possibility that Majorana particles has been observed for the first time. The other property of the Majorana fermion is that it is a non-Abelian anyon. This presents the possibility for the realization of fault-tolerant topological quantum computing[56].

## Excitons and Trions

Other interesting quasiparticle phenomena are called excitons [45]. It is a bound state of a hole and an electron in a solid-state material. Excitons enable energy transport without charge transport. There are two types of excitons:

1. Frenkel excitons are found in organic molecular crystals[31] and have binding energies on order of  $1eV$  and radii  $10\text{\AA}$ .
2. Wannier excitons are found in semiconductors[35] and have binding energies on the order of  $1meV$  with radii  $100\text{\AA}$ .

Excitons have been observed in 2D materials[74] as well as 2D excitons in 3D materials. The promise is that these quasi-particles can enable the development of new electronic components, particularly optoelectronic ones.

Since the exciton system resembles either the hydrogen atom or the hydrogen molecule, it seems conceivable that the dynamics of the  $H_2^+$  molecule is qualitatively similar and could be applied to the exciton systems as well.

A similar system is the trion, a bound state of three charged particles. It consists of either 2 electrons and a hole, or 2 holes and an electron, therefore resembling the hydrogen molecular ion.

## Chapter 4

### Overview of the Calculation of Molecular Potentials in 3D

:

## 4.1 Molecular Orbitals in 2D

In order to calculate the resonant charge transfer rate (RCT), we require the potential surfaces for the two lowest ground state Born-Oppenheimer (BO) that correlate to the degenerate asymptotic limits of the proton-hydrogen system. Many open-source and proprietary tools are available to calculate the latter in three dimensions; to the best of our knowledge, none of such tools are available for the 2D case. In this thesis, we use a first-principles approach to calculate accurate BO potential surfaces for application in our collision studies.

Prior to that discussion, we present a qualitative description and derive approximate solutions using the LCAO method to examine the gerade and ungerade potentials for the ground state of the  $H_2^+$  ion.

### 4.1.1 Gerade and Ungerade Molecular Potentials

In diatomic molecules, gerade (German for even) and ungerade (German for odd) describe how the electronic wavefunction behaves under inversion symmetry, i.e., symmetry of the molecular orbital with respect to the inversion center. If the internuclear axis is placed through the center of the molecule, the inversion manifests with every coordinate point  $r$

with  $-r$ . These terms are meaningful only for molecules or systems that possess a center of inversion, points such that  $r \rightarrow -r$  maps the molecule onto itself. The definition:

- Gerade(g): an orbital wavefunction  $\Psi(r)$  is gerade if it is unchanged by inversion:  $\Psi(-r) = \Psi(r)$ . It has even parity. Example:  $1s\sigma_g$
- Ungerade(u): an orbital wavefunction  $\Psi(r)$  is ungerade if it changes sign under inversion:  $\Psi(-r) = -\Psi(r)$ . It has odd parity. Example  $2p\sigma_u$ .

## 4.2 LCAO

The ground state expression gives amplitude for the 2-dimensional hydrogen atom [89]

$$\psi_{1s}(\mathbf{r}) = \frac{4}{\sqrt{2\pi}} e^{-2r} \quad (4.1)$$

where  $r = |\mathbf{r}|$ . In the LCAO method, an approximate description of the 2D gerade molecule  $H_2$ , for internuclear separation  $\mathbf{R}$  is given by

$$\psi_g(\mathbf{r}, \mathbf{R}) = C(R) \left( \psi_{1s}(\mathbf{r} - \mathbf{R}/2) + \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \right) \quad (4.2)$$

where  $C(R)$  is a normalization factor so that

$$\int d^2\mathbf{r} |\psi_g(\mathbf{r}, \mathbf{R})|^2 = C^2(R) \left( 2 + 2 \int d^2\mathbf{r} \psi_{1s}(\mathbf{r}) \psi_{1s}(\mathbf{r} + \mathbf{R}) \right) = 1 \quad (4.3)$$

or,

$$C(R) = \frac{1}{\sqrt{2(1 + S(R))}} \text{ and } S(R) \equiv \int d^2\mathbf{r} \psi_{1s}(\mathbf{r}) \psi_{1s}(\mathbf{r} + \mathbf{R}) \quad (4.4)$$

We now evaluate the expectation values  $\langle V \rangle$  of the potential

$$V(\mathbf{r}; \mathbf{R}) = -\frac{1}{|\mathbf{r} - \mathbf{R}/2|} - \frac{1}{|\mathbf{r} + \mathbf{R}/2|}. \quad (4.5)$$

Inserting the above wave amplitude to evaluate the expectation value, we get

$$\langle V \rangle_g = -2 \int \frac{d^2\mathbf{r}}{r} |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2. \quad (4.6)$$

We use the virial theorem to estimate the gerade ground state expectation value for the kinetic energy operator

$$\langle KE \rangle_g = \frac{1}{2} \langle \mathbf{r} \cdot \nabla V \rangle \quad (4.7)$$

and since

$$\mathbf{r} \cdot \nabla V = \mathbf{r} \cdot \frac{(\mathbf{r} - \mathbf{R}/2)}{|\mathbf{r} - \mathbf{R}/2|^3} + \mathbf{r} \cdot \frac{(\mathbf{r} + \mathbf{R}/2)}{|\mathbf{r} + \mathbf{R}/2|^3} \quad (4.8)$$

we find

$$\langle KE \rangle_g = \int \frac{d^2 \mathbf{r}}{r} |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2 + \int \frac{d^2 \mathbf{r}}{r^3} (\mathbf{r} \cdot \mathbf{R}/2) |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2. \quad (4.9)$$

In the same manner, we obtain the BO energy for the ungerade BO state,

$$\psi_u(\mathbf{r}, \mathbf{R}) = \frac{1}{\sqrt{2(1 - S(R))}} \int d^2 \mathbf{r} \left( \psi_{1s}(\mathbf{r} - \mathbf{R}/2) - \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \right). \quad (4.10)$$

In chapter 5 we shall compute the accurate solution of the 2D potential. Here for illustration, in Table 5.3, we present LCAO values for the gerade state, and compare it to the accurate results obtained in the discussions in chapter 5.

In Fig 4.1 we provide a graphical comparison of the accurate vs LCAO potential for the gerade state. The exact potential is shown in Red, and the approximation is shown in blue.

This is more readily observed in the enhanced plot.

The above graphs show the accurate gerade potentials (red dots) versus those obtained with the simple LCAO method described above. We observe that for  $R > 10$ , there is fair agreement; the agreement is poorer for  $R < 10$ . This is a reasonable result, as the LCAO, which includes only the ground-state orbital, is a poor description in the molecular region.

We now show the same comparison for the ungerade state in Table 5.4 we show the accurate vs LCAO values for the ungerade state.

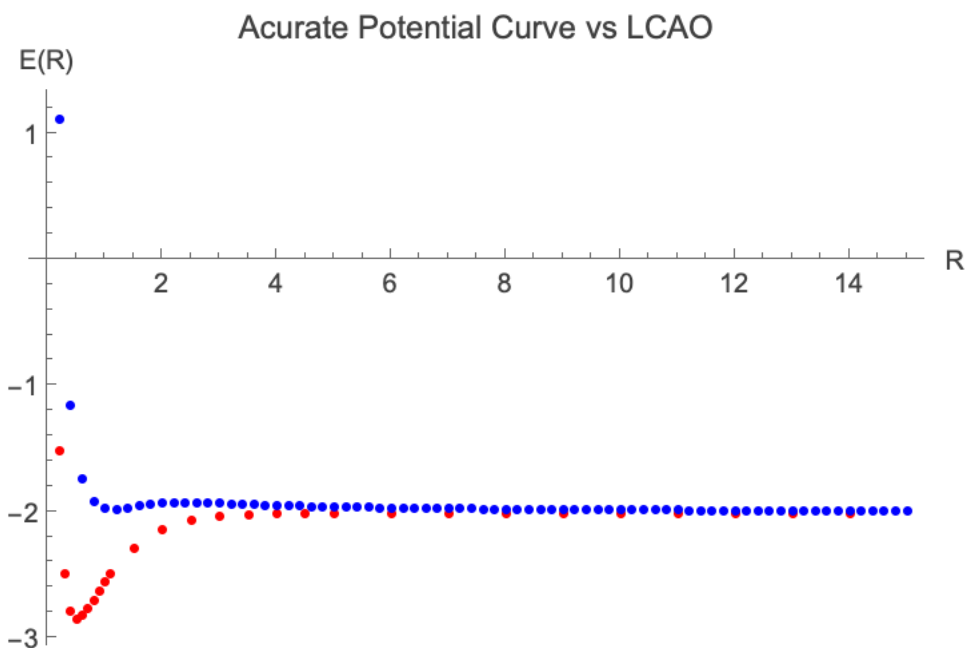


Figure 4.1: Accurate potential curve (red) vs LCAO approximation (blue) for the  $H_2^+$  ground gerade state

Graphical comparison of the accurate and LCAO potentials for the ungerade state is given in figure 4.3.

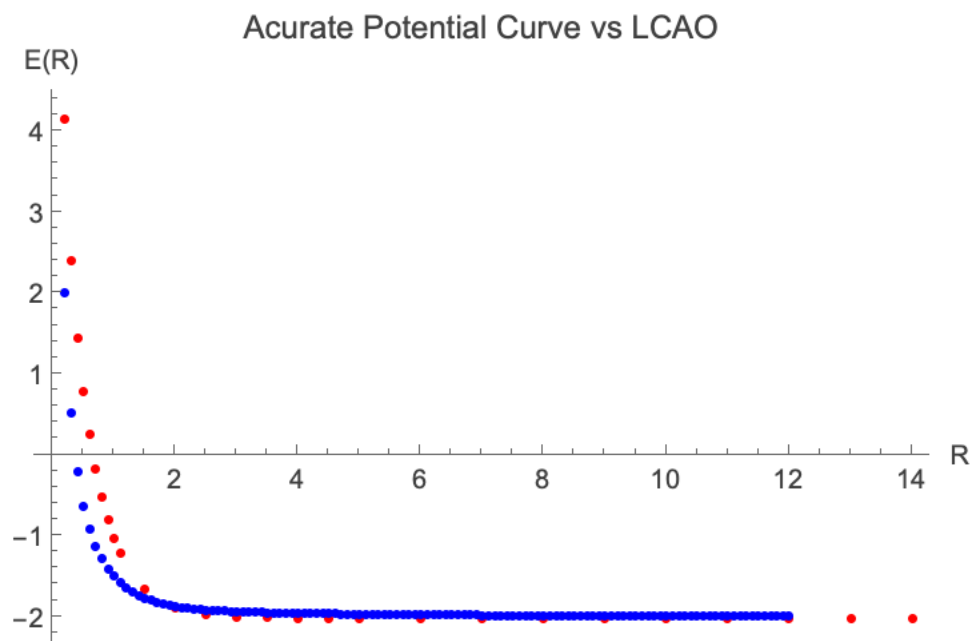


Figure 4.3: Accurate potential curve (red) vs LCAO approximation (blue) for the  $H_2^+$  1st excited ungerade state

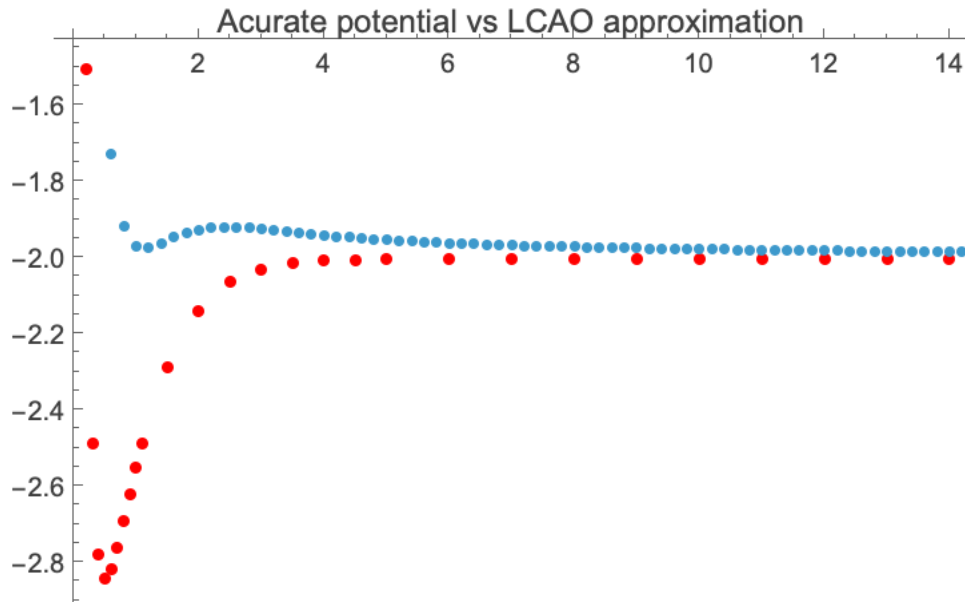


Figure 4.2: Accurate potential curve (red) vs LCAO approximation (blue) for the  $H_2^+$  ground gerade state, enhanced plot

Unlike for the gerade state, in the ungerade case the LCAO is quite a good approximation to the accurate potential.

The reason is the symmetries of the gerade and ungerade states and the overlap integral (4.4). In the ungerade case we have:

$$\psi_u(\mathbf{r}, \mathbf{R}) \approx \psi_{1s}(\mathbf{r} - \mathbf{R}/2) - \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \text{ which makes } 1 - S \approx 1.$$

It follows that the orbital are close to being orthogonal, which makes LCAO approximation more accurate.

In the gerade case, we have  $\psi_g(\mathbf{r}, \mathbf{R}) \approx \psi_{1s}(\mathbf{r} + \mathbf{R}/2) + \psi_{1s}(\mathbf{r} - \mathbf{R}/2)$ . In this case, we have  $1 + S > S$ . In this case, the LCAO orbits are much less orthogonal, so the overlap error becomes much greater.

In the panels of figure 4.2 we show the atomic orbitals for the gerade and ungerade cases as a function of the internuclear distance  $R$ .

Following our previous discussion, we have determined that our LCAO example provides only a rough qualitative description of the two-dimensional ground state potential surfaces

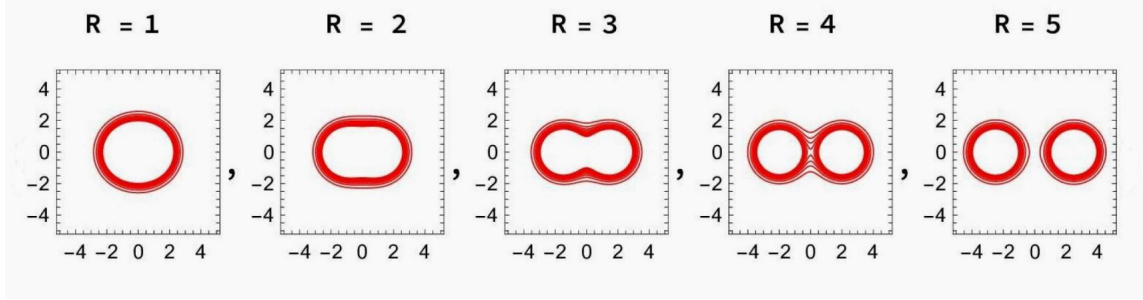


Figure 4.4: Gerade orbitals obtained from the LCAO approximation. The graph shows the overlap of the electronic orbitals for different values of the internuclear distance  $R$

for the  $H_2^+$  molecule. To obtain more accurate data, we will introduce a semi-analytic procedure initially developed by Bates and his colleagues for the treatment of the three-dimensional  $H_2^+$  molecular ion in the upcoming chapters.

In the next chapter, we will review this method as it applies to the three-dimensional system and provide a more precise description of the data presented in Ref. [13]. Subsequently, we will use a similar approach to generate the data for the two-dimensional case of  $H_2^+$ .

We will compare our results with those from a previous study [68] and apply our findings in the calculation of the RCT process given by Eq. (3.1) at cold temperatures.

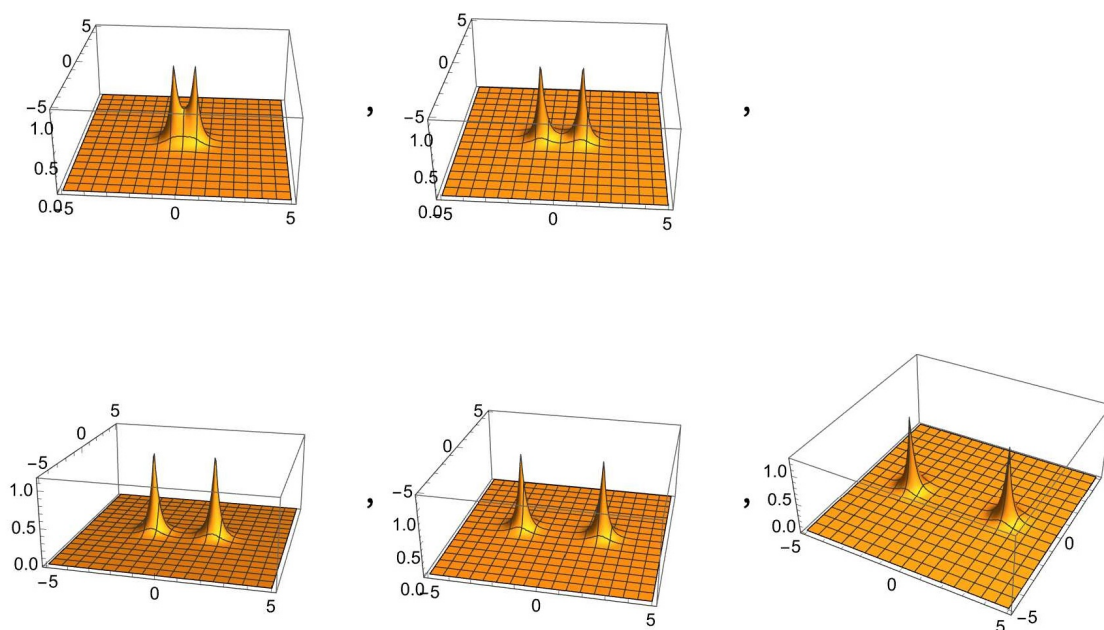


Figure 4.5: Gerade orbitals in 3D from the same LCAO approximation. The graph shows the magnitude of the potential function for different values of the internuclear distance  $R$

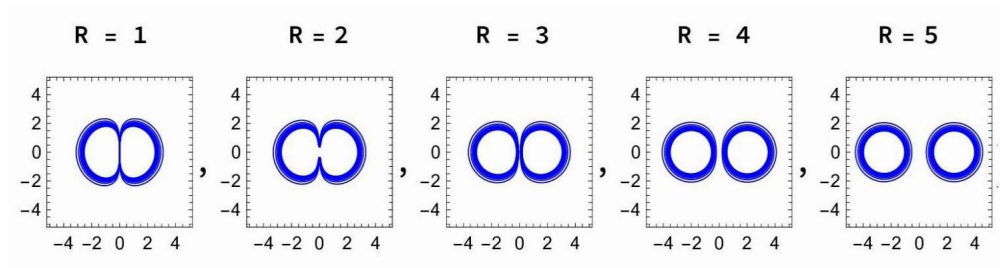


Figure 4.6: Ungerade orbitals obtained from the LCAO approximation. The graph shows the overlap of the electronic orbitals for different values of the internuclear distance  $R$

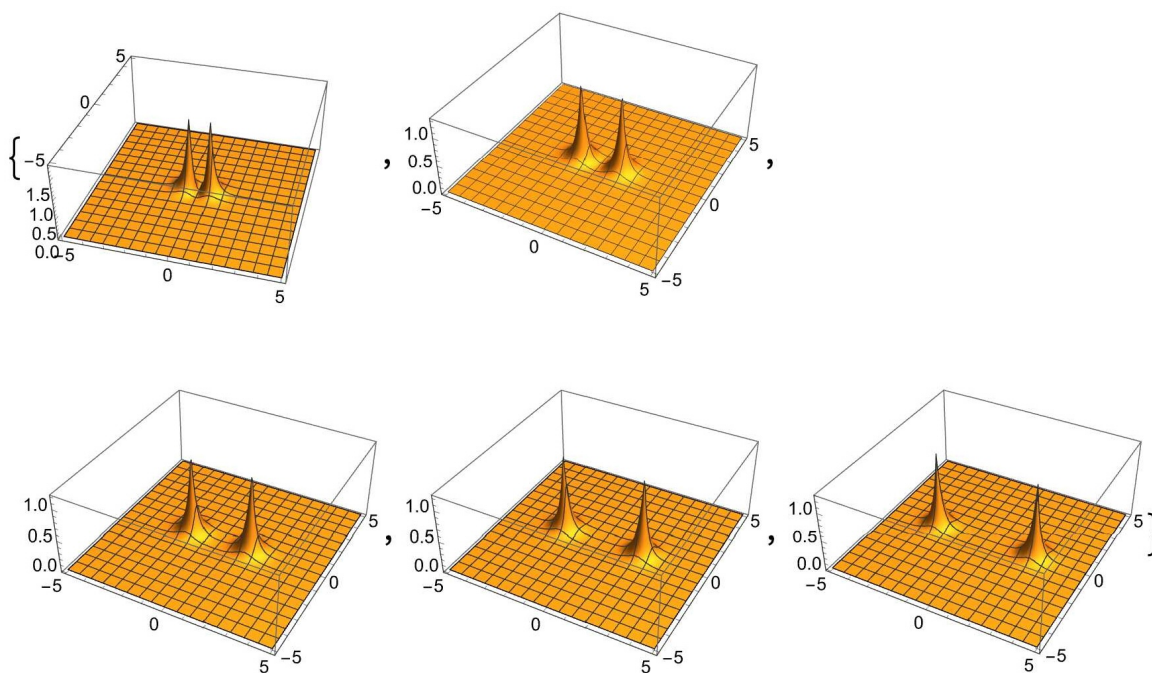


Figure 4.7: Ungerade orbitals in 3D from the same LCAO approximation. The graph shows the magnitude of the potential function for different values of the internuclear distance  $R$

## 4.3 Hydrogen Molecular Ion in 3 Dimensions, the Method of Bates

In this section, we describe first the work done by the Bates et al [13][12][75]. We review analytic methods[13][12][75] used to obtain solutions of the time-independent Schrodinger equation for the hydrogen molecular ion.

As we will apply a similar method to treat the hydrogen molecular ions in 2D, we first review the methodology of Bates [13][12] used to calculate the lowest electronic eigenstates and eigenvalues for the  $H_2^+$  system in 3D. \*

The hydrogen molecular ion, and similar systems (such as  $HeH^+$ ), are one of the few that allow a full analytic quantum mechanical description. Although its solutions do not exist in the closed form, they can be expressed as a convergent infinite series. Wasserman in his thesis [82] in 1963 reports that while the existence of  $H_2^+$  has been verified, no discrete spectrum of the  $H_2^+$  has been observed. In the same thesis, Wasserman reports that the ionization potential of  $H_2^+$  has been measured and shows good agreement with the theoretical calculations. However, in 1989 Carrington, McNab and Montgomerie report [18] that the radio frequency hyperfine transitions of the  $H_2^+$  have been measured. They note that the  $H_2^+$  is highly reactive and therefore hard to keep in the lab.

As a three-body problem, containing two protons and an electron, the Schrodinger PDE can be separated by expressing it in an elliptical coordinate system. Unfortunately, there does not exist a closed-form representation for the eigenvalues, and one must resort to numerical methods in their calculation. Bates et al. [13][12] did not provide numerical codes used in the evaluation of eigenvalues. Instead, they used an expanded fraction method for calculating the latter with the desired level of precision. The paper[12] does not report on the details of those calculations, and so in this thesis we provide an alternative description.

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\*In this section we use atomic (Hartree) units, unless otherwise stated.

We will compare the results obtained here with those reported in[12]. Harmony with those reported in[12] will provide confidence in the fidelity of the analogous calculations in 2D.

We use Hartree Atomic Units (4.11), in all equations. In this system, the numerical values of the following fundamental constants are 1:

$$\begin{aligned}
\text{Electron mass:} & \quad m_e = 1 \\
\text{Electron charge:} & \quad e = 1 \\
\text{Reduced Planck constant:} & \quad \hbar = 1 \\
\text{Coulomb's constant:} & \quad \frac{1}{4\pi\epsilon_0} = 1
\end{aligned} \tag{4.11}$$

In addition, in equations, variables that represent vectors are typeset either in **bold** or with an arrow above the symbol,  $\vec{r}$ . A caret ( $\wedge$ ) above a symbol indicates that the symbol is an operator.

The only instance in which we will avoid using atomic units is during the derivation that involves transforming the molecular Hamiltonian into a Jacobian coordinate system. Since certain terms will include the reduced mass, we aim to maintain the integrity of the units throughout this process.

## 4.4 Description

The starting point is the Schrodinger equation for the hydrogen molecular ion in 3D. We seek stationary states for non-relativistic motion, and neglect the spin degrees of freedom. With these assumptions the Schrodinger equation for such a system is given by:

$$\hat{H}\psi(\mathbf{R}, \mathbf{x}) = E \psi(\mathbf{R}, \mathbf{x}). \tag{4.12}$$

$\hat{H}$  represents total Hamiltonian and  $\psi(\mathbf{R}, \mathbf{x})$  is an eigenstate wavefunction. It is a function of both the collective nuclear coordinates  $\mathbf{R}$ , and electronic coordinates  $\mathbf{x}$ . Of course, since we are considering the hydrogen molecular ion,  $\mathbf{x}$  is the coordinate of a single electron.

The hydrogen molecular ion  $H_2^+$  is composed of 2 protons and a single electron. Thus, as we are neglecting relativistic effects and spin, its Hamiltonian is given as:

$$\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{en} + \hat{V}_{nn}. \quad (4.13)$$

The terms in the Hamiltonian are as follows:

The kinetic energy of the protons:

$$\hat{T}_n = -\frac{\hbar^2}{2M} \sum_{i=1}^2 \nabla_{R_i}^2 = \frac{1}{2M} \sum_{i=1}^2 \hat{P}_i^2 \quad (4.14)$$

The kinetic energy of the electron:

$$\hat{T}_e = -\frac{\hbar^2}{2m} \nabla_x^2 = \frac{1}{2m} \hat{p}^2 \quad (4.15)$$

The potential energy between the electron and the nuclei, i.e. the total Coulomb electron - nuclei attraction:

$$\hat{V}_{en} = -\sum_{i=1}^2 \frac{e^2}{|\mathbf{R}_i - \mathbf{x}|} \quad (4.16)$$

Potential energy (operators) between the nuclei - the total Coulomb proton - proton repulsion.

$$\hat{V}_{nn} = \frac{e^2}{|\mathbf{R}_1 - \mathbf{R}_2|} \quad (4.17)$$

The second term in the equation above represents the operator expression.

In the equations above,  $\mathbf{R}_i$ ,  $i = 1, 2$  and  $M$  represents the coordinates and mass of the nuclei (protons) 1, 2 respectively, and  $\mathbf{x}$  and  $m$  are the coordinates and mass of the electron, respectively. Coordinates are defined with respect to a common origin in a space fixed frame.

It is convenient to separate the motion of the center of mass from the motion of the point particles. A similar approach can be applied for the quantum case. We show in appendix A that the motion of the center of mass will be a plane wave, which possess a well defined

momentum and which is not square integrable. That makes the position of the center of mass equally probable in the all space, and confirms with the Heisenberg uncertainty principle.

In the appendix A we define  $\chi(\mathbf{R})$  as the wavefunction representing the motion of the nuclei. We then obtain:

$$\chi(\mathbf{R}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar}\sqrt{2ME_{CM}}\mathbf{R}} \quad (4.18)$$

where  $E_{CM}$  is the energy of the center of the mass.

Therefore, we proceed to transform the Hamiltonian into a Jacobian coordinate system, following reference [11]. The outcome of this procedure will be the transformation of the wave function as  $\psi(\mathbf{R}_i, \mathbf{r}) \rightarrow \psi(\mathbf{R}_{CM}, \mathbf{R}, \mathbf{r})$  where the  $\mathbf{R}_{CM}$  is the coordinate of the center of the mass,  $R$  is the distance between nuclei and  $\mathbf{r}$  is an electron coordinate. We choose the coordinates as:

$$\begin{aligned} \mathbf{R}_{CM} &= \frac{M \mathbf{R}_1 + M \mathbf{R}_2 + m \mathbf{x}}{2M + m} \\ \mathbf{R} &= \mathbf{R}_1 - \mathbf{R}_2 \\ \mathbf{r} &= \mathbf{x} - \frac{\mathbf{R}_1 + \mathbf{R}_2}{2} \end{aligned} \quad (4.19)$$

Now following [11],[20], we obtain the expression of the molecular Hamiltonian in the new

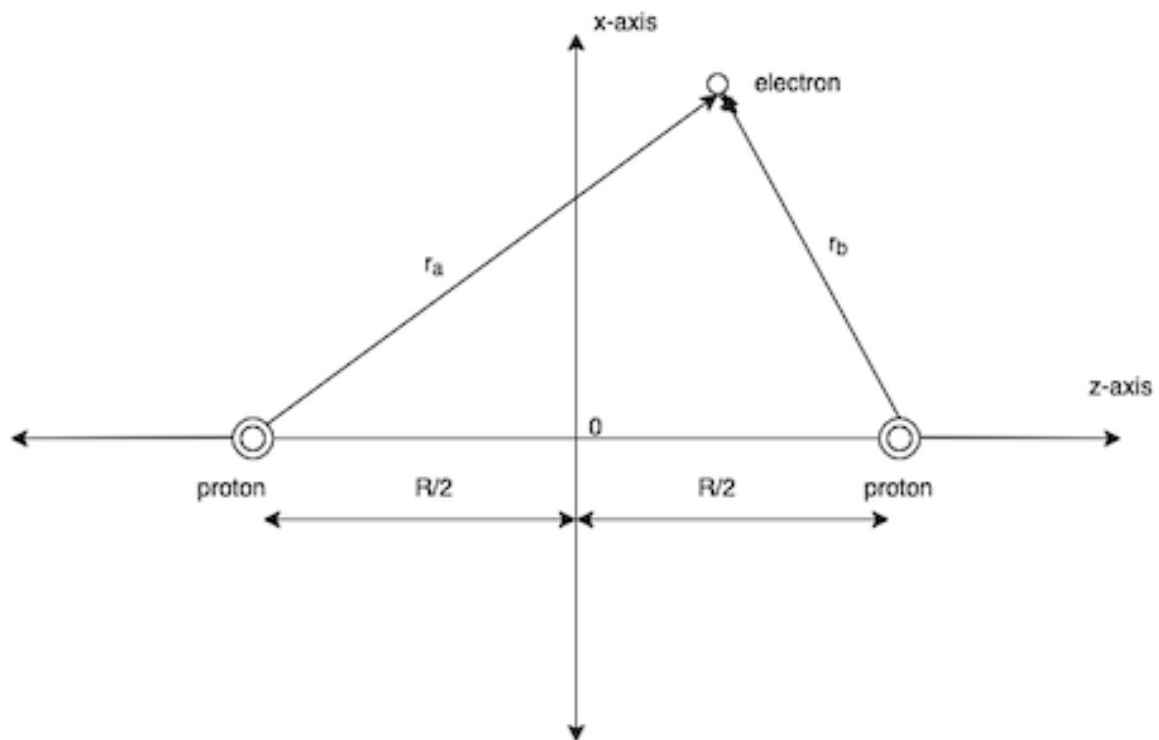


Figure 4.8:  $H_2^+$  ion, showing the position of the nuclei at distance  $R$ , and a coordinates of an electron

coordinates,

$$H' = T_{CM} + T_{KE} + H_{AD} \quad \text{where motion of the center}$$

of the mass that which be separated from the total energy of the nuclei:

$$T_{CM} = -\frac{1}{2(2M + m)} \nabla_{CM}^2$$

The relative motion of the nuclei (4.20)

$$T_{KE} = -\frac{1}{2\mu} \nabla_R^2$$

where

$$\mu = \frac{M}{2} \quad \text{is a reduced mass of the 'bare' nuclei}$$

and

The adiabatic Hamiltonian:

$$H_{AD} = -\frac{1}{2m} \nabla_r^2 - \frac{1}{4M} \nabla_r^2 + \frac{1}{|R|} - \frac{1}{|r + R/2|} - \frac{1}{|r - R/2|} \quad (4.21)$$

The second term  $\frac{1}{4M} \nabla_r^2$ , in the expression for the  $H_{AD}$  represents the mass polarization term, and because  $M \gg m$ , it can be neglected. This term represents the movement of the electron as the center of the mass is moving. In this case the mass polarization is the coupling of the two nuclei, since they are charged and interact with each other. The physical basis for the approximation is the fact that the  $m_p \approx 1836 m_e$ , where  $m_p$  and  $m_e$  are the proton's and electron's mass respectively. Because of the mass difference, the center of the mass (whose mass is mostly mass of the nuclei) is moving much slower than the electron, and thus justifies neglecting this term.

Now we get for the Schrodinger equation:

$$T_{KE}\psi(\mathbf{R}, \mathbf{r}) + H_{AD}\psi(\mathbf{R}, \mathbf{r}) = E \psi(\mathbf{R}, \mathbf{r}) \quad (4.22)$$

where  $E$  is the internal energy of the nuclei - electron system. To develop a working the-

ory we use Born-Oppenheimer (BO) approximation. Using BO approximation we solve for the electronic Hamiltonian in the Coulomb field of the nuclei, where the nuclei are assumed to be stationary with respect to inertial frame. In this case, the electronic Hamiltonian is the  $H_{AD}$  from (4.21). We neglect the mass polarization term and express the electronic Hamiltonian as:

$$H_{AD} = T_e + V_{NN} + V_{Ne} \quad \text{or} \quad (4.23)$$

$$H_{AD} = \frac{1}{2m} \nabla_r^2 + \frac{1}{|R|} - \frac{1}{|r + R/2|} - \frac{1}{|r - R/2|}$$

The BO approximation neglects the motion of the nuclei when describing the electron motion in a molecule. The physical basis for the approximation is the same one as above,  $m_p \approx 1836 m_e$ . In addition, due to their opposite charges, there is an attractive Coulomb force between proton and electron. But the magnitude of the acceleration is inversely proportional to the mass, so the acceleration of the electron is large and the acceleration of the proton is small, and can be neglected.

So at this step, we neglect the nuclear kinetic energy, effectively rendering the nuclei immobile. Following [75] we then solve for the motion of an electron in the Coulomb field of the nuclei, which leads to a following Schrodinger equation:

$$\left[ -\frac{1}{2} \nabla_r^2 + V(\mathbf{R}; \mathbf{r}) \right] u(\mathbf{R}; \mathbf{r}) = E(\mathbf{R}) u(\mathbf{R}; \mathbf{r}) \quad (4.24)$$

Now the equation (4.24) is the Schrodinger equation where  $u(\mathbf{R}; \mathbf{r})$  is the wavefunction for the electron with its coordinate  $\mathbf{r}$  and parameter  $\mathbf{R}$ . Electron is moving in the potential  $V(\mathbf{R}; \mathbf{r})$  with the coordinate  $\mathbf{r}$ . The electron's wavefunction  $u(\mathbf{R}, \mathbf{r})$  and the energy  $E(\mathbf{R})$  both depend on the position of the nuclei  $\mathbf{R}$  as a parameter. The last step in Born-Oppenheimer approximation is to solve the Schrodinger equation for the nuclei, using the above calculated  $E(\mathbf{R})$  as a potential energy, which leads to a following equation.

$$\left[-\frac{1}{2}\nabla_R^2 + E(\mathbf{R})\right]v(\mathbf{R}) = \epsilon v(\mathbf{R}) \quad (4.25)$$

where  $R$  is the internuclear distance.

Once these operations are carried, the Born-Oppenheimer approximation states that the energy  $\epsilon$  is a good approximation of the energy levels of the exact Schrodinger equation (4.12). As a result, the total molecular wave function is separated into the electronic and the nuclear wavefunction.

$$\psi(\mathbf{R}, \mathbf{r}) = u(\mathbf{R}; \mathbf{r})v(\mathbf{R}) \quad (4.26)$$

In addition, each electronic eigenvalue  $E_n(R)$  will give a rise to the electronic surface, known as Born-Oppenheimer surfaces.

One could proceed further for the complete treatment of the Born-Oppenheimer approximation, for example dealing with the Born-Oppenheimer off-diagonal correction [72], but it will go beyond the scope of this thesis .

At this moment, using BO approximation we have a Schrodinger equation for the electron, orbiting the two nuclei (each consisting of a single proton). Note that in the further text the  $\psi$  will denote the function  $u(\mathbf{R}; \mathbf{r})$ , since it is customary to denote the wave function as  $\psi$ .

## 4.5 Replicating the 3D Solution for the $H_2^+$ Molecular Ion

### 4.5.1 Solution Process

Following the seminal papers by Bates et. all. [13][12], we can re-express the equation as:

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r_a} - \frac{1}{r_b}\right)\psi = E\psi \quad (4.27)$$

where  $r_a = \left|\mathbf{r} - \frac{1}{2}\mathbf{R}\right|$  and  $r_b = \left|\mathbf{r} + \frac{1}{2}\mathbf{R}\right|$  are distances of the electron from the nuclei and  $\mathbf{R}$  is a distance between nuclei.

Due to the cylindrical symmetry of the problem, we choose the elliptical coordinates. We follow [13], where authors chose the confocal elliptic co-ordinates  $\lambda, \mu$ , which transforms the coordinates  $r_a$  and  $r_b$  from (4.27) into:

$$r_a = \frac{R}{2}(\lambda + \mu), \quad r_b = \frac{R}{2}(\lambda - \mu) \quad (4.28)$$

After applying the coordinate transformations (4.28) into the equation (4.27), we obtain another partial differential equations [13]:

$$\begin{aligned} &\frac{d}{d\lambda} \left\{ (\lambda^2 - 1) \frac{d\psi}{d\lambda} \right\} + \frac{d}{d\mu} \left\{ (1 - \mu^2) \frac{d\psi}{d\mu} \right\} + \left\{ \frac{1}{\lambda^2 - 1} + \frac{1}{1 - \mu^2} \right\} \frac{d^2\psi}{d\phi^2} + \\ &+ \left\{ \frac{1}{4} R^2 E (\lambda^2 - \mu^2) + 2R\lambda \right\} \psi = 0 \end{aligned} \quad (4.29)$$

We assume that the solution can be separated, in the product of the three functions of the single variable,  $\lambda, \mu, \phi$ . Under this assumption the solution can written as:

$$\psi(\lambda, \mu, \phi) = L(\lambda)M(\mu)e^{im\phi} \quad \text{where is } p^2 = -\frac{1}{4}R^2E \quad (4.30)$$

In this equation the  $L, M$  are the functions of their respective coordinates, and  $m$  is an integer, either positive or negative. The next step is to insert the ansatz (4.30) into the Eq (4.29) . After some algebra, one observes that the equation (4.29) separates into the system of two ordinary, coupled differential equations. an

$$\frac{d}{d\mu} \left\{ (1 - \mu^2) \frac{dM}{d\mu} \right\} + \left\{ -A + p^2\mu^2 - \frac{m^2}{1 - \mu^2} \right\} M = 0 \quad (4.31)$$

$$\frac{d}{d\lambda} \left\{ (\lambda^2 - 1) \frac{dL}{d\lambda} \right\} + \left\{ A + 2R\lambda - p^2\lambda^2 - \frac{m^2}{\lambda^2 - 1} \right\} L = 0 \quad (4.32)$$

In [13], Bates et al. state that they used the method of successive approximation to find the coefficients in the series. It was tacitly assumed that the successive approximation themselves converge. The approximation is terminated either when the desired precision is achieved or when the certain number of approximation is met. However the details of the numerical calculation are not provided in the paper.

### 4.5.2 Differential Operator Spectrum

We can view the equations for  $M$  and  $L$  as the eigenvalue operator equations  $D^1 M(\mu) = A M(\mu)$  and  $D^2 L(\lambda) = -A L(\lambda)$ . Since these equations model a real, physically realizable system, we assume that operators  $D^{1,2}$  are compact on  $l^2$  space, therefore each  $D^{1,2}$  operator satisfies:

$$\|D_n^{1,2} \rightarrow D^{1,2}\| \quad (4.33)$$

where  $D_n^{1,2} = P_n D^{1,2} P_n$  So with these assumptions, operators  $D^{1,2}$  can be approximated by the finite dimensional matrices.

### 4.5.3 Towards the Solution

Here we replicate the solution of Bates et. al. [13][12], by using a more of a brute force approach. The idea is to numerically calculate the spectrum of a differential operator to a desired precision. We expressed equations (4.31),(4.32) as a matrix eigenvalue problem. The system (4.31),(4.32) of the two ODEs then depend on the two parameters, total energy  $E$ , and the separation constant  $A$ . The problem is reduced to finding those values of  $E$  and  $A$  for which the solution to the differential equation satisfy the requisite boundary conditions. Since it is a Sturm-Liouville problem, or more precisely a system of two Sturm-Liouville problems, our approach is fundamentally the same approach as taken by Bates. but it is more practical, as it can better accommodate the existing numerical software. There exist excellent numerical packages (Lapack) [48] for computing the matrix eigenvalues, and they greatly simplify the actual computation. The Fortran code in appendix D uses Lapack library to compute eigenvalues.

We express both function  $M(\mu)$  and  $L(\lambda)$  using the set of orthogonal functions which satisfy the boundary conditions. We then follow the method of Robyn's. Once we replace the assumed solution into the equation and after some algebra, we obtain effectively two uncoupled, infinite systems of equations, where the unknowns are the coefficients in the solution series. The solution vector depends on the two parameters,  $p$ , which is the function of energy and  $A$  which is the separation constant.

The next step is rather simple. For functions  $M(\mu)$  and  $L(\lambda)$ , we need to find the functional dependency of the parameters  $p$  and  $A$ , so that solution exists. The intersections of the two curves  $p_1 = f(A_1)$  and  $p_2 = g(A_2)$  provides the value of  $p$  and  $A$  when the condition for the solutions exists. The desired precision is obtained by choosing the size number of terms in the series.

The remaining issue is that even with the modern fast computers and optimized numerical libraries [48], calculating eigenvalues is a slow process. The complexity of the QR de-

composition algorithms is  $O(n^3)$  [32], which make the QR factorization feasible by today's computer systems, but fairly slow for large matrices.

## 4.6 Series Expansions and Eigenvalue Determinant

Here we describe in details the method describe in the previous section.

### 4.6.1 L Equation

Start with:

$$\begin{aligned} \frac{d}{d\lambda} \left\{ (\lambda^2 - 1) \frac{d\Lambda}{d\lambda} \right\} + \{A - p^2\lambda^2 + 2R\lambda\} \Lambda = 0 &\Rightarrow \\ (\lambda^2 - 1) \frac{d^2\Lambda}{d\lambda^2} + 2\lambda \frac{d\Lambda}{d\lambda} + \{A - p^2\lambda^2 + 2R\lambda\} \Lambda = 0 \end{aligned} \quad (4.34)$$

For practical reasons we first shift the  $L$  equation b  $x = \lambda - 1$  to obtain:

$$\frac{d}{dx} \left\{ x(x+2) \frac{d\Lambda}{dx} \right\} + \{-p^2 x^2 + (-2p^2 + 2R)x - p^2 + 2R + A\} \Lambda = 0 \quad (4.35)$$

$$x(x+2) \frac{d^2\Lambda}{dx^2} + (2x+2) \frac{d\Lambda}{dx} + \{-p^2 x^2 + (-2p^2 + 2R)x - p^2 + 2R + A\} \Lambda = 0$$

Now we assume that the solution exists and that it can be represented as the sum of Laguerre polynomials:

$$\Lambda(x) = e^{-px} \sum_{n=0}^{\infty} c_n L_n(x) \quad (4.36)$$

Inserting the ansatz (4.36) into (4.35), applying the properties of Laguerre's polynomial some algebra we obtain.

$$\begin{aligned}
& -2p \sum_{n=0}^{\infty} c_n n(2n+1)L_n + 2p \sum_{n=0}^{\infty} c_{n+1} n(n+1)L_{n+1} + 2p \sum_{n=0}^{\infty} c_{n-1} n^2 L_{n-1} + \\
& + (2p-1) \sum_{n=0}^{\infty} c_{n-1} n(2n-1)L_{n-1} - (2p-1) \sum_{n=0}^{\infty} c_n n^2 L_n - \\
& - (2p-1) \sum_{n=0}^{\infty} c_{n-2} n(n-1)L_{n-2} + (-2p+2R) \sum_{n=0}^{\infty} c_n (2n+1)L_n - \\
& - (-2p+2R) \sum_{n=0}^{\infty} c_{n+1} (n+1)L_{n+1} - (-2p+2R) \sum_{n=0}^{\infty} c_{n-1} n L_{n-1} + \\
& + (-4p+1) \sum_{n=0}^{\infty} c_n n L_n + (4p+1) \sum_{n=0}^{\infty} c_{n-1} n L_{n-1} + \\
& + (-2p-p^2+2R+A) \sum_{n=0}^{\infty} c_n L_n = 0
\end{aligned} \tag{4.37}$$

The next step is to multiply the equation (4.37) with  $e^{-x}L_m(x)$  and integrate from 0 to  $\infty$  using the orthogonality property of the Laguerre's polynomials:  $\int_0^\infty dx e^{-x}L_n(x)L_m(x) = \delta_{mn}$  to obtain the recurrence equations for the unknown coefficients  $c_n$

$$\begin{aligned}
& -2p n(2n+1)c_n + 2p n(n+1)c_{n+1} + 2p n^2 c_{n-1} + (2p-1)n(2n-1)c_{n-1} - \\
& - (2p-1)n^2 c_n - (2p-1)n(n-1)c_{n-2} + (-2p+2R)(2n+1)c_n - \\
& - (-2p+2R)(n+1)c_{n+1} - (-2p+2R)nc_{n-1} + (-4p+1)nc_n + \\
& + (4p+1)nc_{n-1} + (-2p-p^2+2R+A)c_n = 0
\end{aligned} \tag{4.38}$$

This expression (4.38) is a system of linear equations, where the unknowns are the coefficients  $c_n$ . The coefficients multiplying the unknowns  $c_n$  in (4.38) form the band matrix.

### 4.6.2 M Equation

We use the same approach for the M equation.

$$\frac{d}{d\mu} \left\{ (1 - \mu^2) \frac{dM}{d\mu} \right\} + (-A + p^2 \mu^2) M(\mu) = 0 \quad (4.39)$$

We express the  $M(\mu)$  as the series of Legendre polynomials.

$$M(\mu) = \sum_{n=0}^{\infty} f_n P_n(\mu) \quad (4.40)$$

where

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2m+1} \delta_{m,n} \quad (4.41)$$

Substituting (4.40) into (4.39), multiplying by  $P_m(\mu)$ , and integrating from  $-1$  to  $1$  we obtain

$$\begin{aligned} \sum_{n=0}^{\infty} \left\{ - \int_1^1 P_m(\mu) n(n+1) P_n(\mu) d\mu - A \int_1^1 P_m(\mu) P_n(\mu) d\mu + \right. \\ \left. + p^2 \int_1^1 P_m(\mu) \mu^2 P_n(\mu) d\mu \right\} = 0 \quad \text{or} \end{aligned} \quad (4.42)$$

$$\sum_{n=0}^{\infty} \left\{ \frac{-2}{2n+1} n(n+1) \delta_{m,n} - A \frac{2}{2n+1} \delta_{m,n} + p^2 \int_1^1 P_m(\mu) \mu^2 P_n(\mu) d\mu \right\} = 0$$

Taking advantage of the orthogonality of Legendre Polynomials (4.41), we obtain a system of linear equations, where the coefficients form a band matrix.

$$\begin{aligned} \sum_{n=0}^{\infty} \left\{ \frac{-2n(n+1)}{2n+1} \delta_{m,n} - A \frac{2}{2n+1} \delta_{m,n} + \right. \\ \left. + p^2 \frac{2}{2n+1} \left[ \frac{(n+1)^2}{(2n+1)(2n+3)} + \frac{n^2}{(2n+1)(2n-1)} \right] \delta_{m,n} + \right. \\ \left. p^2 \frac{2(n+1)(n+2)}{(2n+1)(2n+3)(2n+5)} \delta_{m,n+2} + p^2 \frac{2n(n-1)}{(2n+1)(2n-1)(2n-3)} \delta_{m,n-2} \right\} = 0 \end{aligned} \quad (4.43)$$

In appendix B we present the Mathematica code for calculating electron energies of the molecular  $H_2^+$  ion in 3 dimensions using Mathematica eigenvalue solver.

### 4.6.3 Numerical Calculation

Equations (4.38) and (4.43) constitute the couple system of linear equation. Each series is parameterized by 2 parameters,  $p$  and  $A$ , as well as the parameter  $R$ , the distance between nuclei. We select the values of  $R$  as shown in table 4.1. We tacitly assume that the series (4.36) (4.39) converge, moreover, we assume that they converge uniformly. Therefore, we truncate the series at some index  $i = K$ , chosen to achieve the desired accuracy. Now the coefficient matrices are dimensions  $K \times K$  and each matrix has  $K$  eigenvalues, and the lowest eigenvalue corresponds to the lowest value of  $p$ , thus to the lowest energy.

In next step, for each value of  $R$ , we calculate the eigenvalues, sort them, and pick the lowest (or the  $i$ -th one from the bottom), and view them as functions  $A = A(p)$ . So for each of the equations (4.38) and (4.43), we choose the interval  $[0, 2 * R]$  and divide it into  $N$  points  $p_i, i = 1..N$ . For each point, we calculate, sort and pick desired the eigenvalue of the matrix  $M ( L )$ . That way we obtain two functions  $A_M = A_M(p)$  and  $A_L = A_L(p)$ , evaluated at points  $p_i$ . We interpolate these two functions, using Wolfram's Mathematica 3-rd order Spline interpolation. As a side note, for a small subset of values  $R$  we used the higher order Spline interpolation but we were unable to observe the difference in the results obtained. The intersection of these two curves in the value of  $p$  for which both  $M(\mu)$  and  $L(\lambda)$  equations have results satisfying the boundary conditions. Now we repeat the calculation for the next value of  $R$ .

After all this exercise, we get the function  $E = E(R)$ , from (4.24) which is the electron energy in the BO approximation. Adding the potential energy between nuclei is  $V(R) = 1/R$ , we obtain the potential energy curve. The paper by Bates does not explicitly use this curve to calculate the vibrational energies, but we will, in our 2D case.

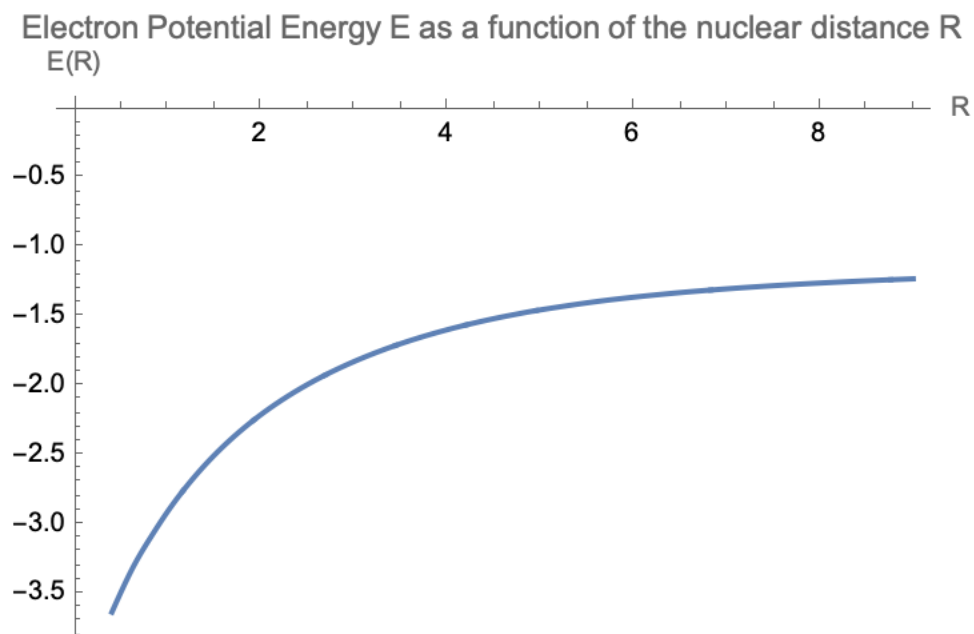


Figure 4.9: Electron potential energy as function of internuclear distance  $R$  for 3D  $H_2^+$

For illustration purposes we present some of the results below.

Plot of the ground state energies  $E(R)$  and  $E(R) + V(R)$  where  $V(R) = \frac{1}{R}$  represents the potential energy of the nuclei.

Electron energy as an eigenvalue of the equations (4.38) and (4.43).

Total energy, calculated by adding the potential energy of the nuclei  $V(R) = 1/R$ .

Electron energy as an eigenvalue of the equations (4.38) and (4.43).

Table comparing the electron energies obtain using our method vs method applied by Bates et al. We chose a subset of values to indicate the overall trend. The  $1s\sigma_g$  and  $2p\sigma_u$  represent the ground Gerade and Ungerade state respectively. In the table below we used atomic units for both  $R$  and  $E(R)$ .

For reference we use the values obtained by Bates [13][12]. Valued obtained by our method are in good alignment with the referent values.

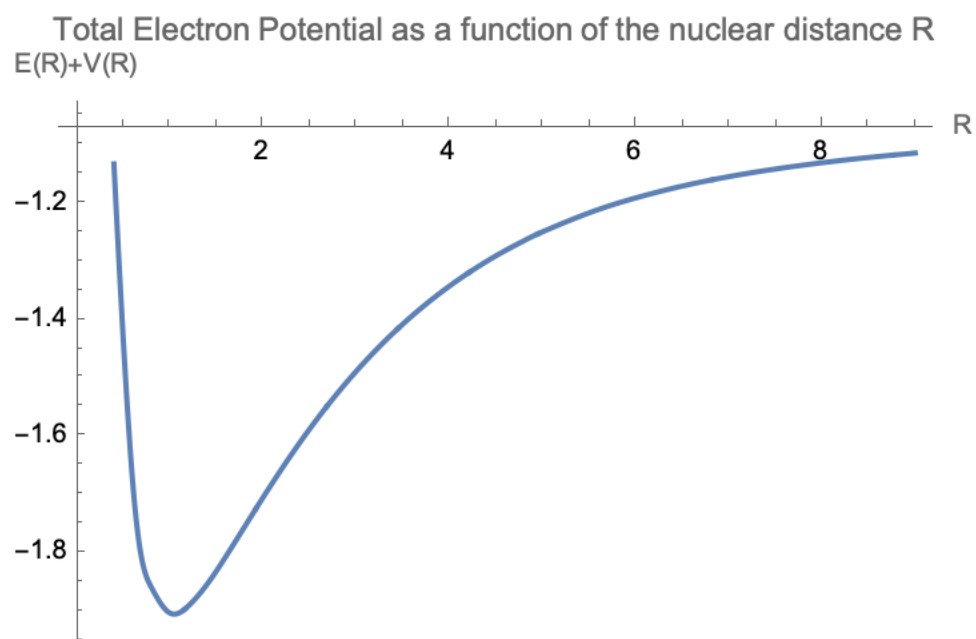


Figure 4.10: Total electron potential energy as function of internuclear distance  $R$  for 3D  $H_2^+$ , given as  $E(R) + V(R)$

Table 4.1: Electron energy,  $1s\sigma_g$  state,

| R   | -E(R)<br>(Our<br>Method) | -E(R)<br>(Bates) |
|-----|--------------------------|------------------|
| 0.4 | 3.63220                  | 3.601570         |
| 0.6 | 3.34296                  | 3.34301          |
| 0.8 | 3.10895                  | 3.10895          |
| 1.0 | 2.90357                  | 2.90356          |
| 1.2 | 2.7246154                | 2.72461          |
| 1.4 | 2.5685373                | 2.56853          |
| 1.6 | 2.4318740                | 2.43186          |
| 1.8 | 2.3116176                | 2.31162          |
| 2.0 | 2.2052681                | 2.20525          |
| 2.2 | 2.1107695                | 2.11076          |
| 2.4 | 2.0264405                | 2.02642          |
| 2.6 | 1.9508966                | 1.95090          |
| 2.8 | 1.88299                  | 1.88299          |
| 3.0 | 1.82179                  | 1.82178          |
| 3.2 | 1.76648                  | 1.76647          |
| 3.4 | 1.71640                  | 1.71639          |
| 3.6 | 1.67097                  | 1.67097          |
| 3.8 | 1.62970                  | 1.62971          |
| 4.0 | 1.59216                  | 1.59216          |
| 4.2 | 1.55799                  | 1.557998         |
| 4.4 | 1.52685                  | 1.52685          |
| 4.6 | 1.49844                  | 1.49844          |
| 4.8 | 1.47252                  | 1.47252          |
| 5.0 | 1.44884                  | 1.44884          |

Table 4.2: Electron energy,  $2p\sigma_u$  state

| R   | -E(R)<br>(Our<br>Method) | -E(R)<br>(Bates) |
|-----|--------------------------|------------------|
| 0.4 | 1.3456                   | 1.02158          |
| 0.6 | 1.0487                   | 1.04861          |
| 0.8 | 1.08552                  | 1.08548          |
| 1.0 | 1.12961                  | 1.12962          |
| 1.2 | 1.17719                  | 1.1772           |
| 1.4 | 1.22415                  | 1.22415          |
| 1.6 | 1.26723                  | 1.26721          |
| 1.8 | 1.30446                  | 1.30446          |
| 2.0 | 1.33507                  | 1.13507          |
| 2.2 | 1.35912                  | 1.35912          |
| 2.4 | 1.37715                  | 1.37715          |
| 2.6 | 1.38992                  | 1.38991          |
| 2.8 | 1.39823                  | 1.39822          |
| 3.0 | 1.40284                  | 1.40285          |
| 3.2 | 1.40443                  | 1.40443          |
| 3.4 | 1.40359                  | 1.40358          |
| 3.6 | 1.40082                  | 1.40082          |
| 3.8 | 1.39655                  | 1.39654          |
| 4.0 | 1.3911                   | 1.39110          |
| 4.2 | 1.38477                  | 1.38478          |
| 4.4 | 1.37778                  | 1.37779          |
| 4.6 | 1.37033                  | 1.36255          |
| 4.8 | 1.36255                  | 1.36255          |
| 5.0 | 1.35458                  | 1.35458          |

## Chapter 5

### Non-Relativistic Solution of the Hydrogen Molecular Ion in 2 Spatial Dimensions

#### 5.1 Existing Solutions

We apply similar method used in the 3D case of the  $H_2^+$  ion to find the solution in 2 dimensions. The problem itself is analogous to the 3D problem, but leads to the Mathieu's function as the solution for the radial problem.

There are not too many papers dealing with solving the  $H_2^+$  but here is some relevant, and recent work in this problem. In addition to the Bates paper [13], there are other, more recent papers [62] [42], [79] which deal with the solution of the Schrodinger equation and spectrum of the  $H_2^+$  molecule in 3 dimensions. As with other work in this problem, the paper relies on the work done by Bates et al. [13]. The solution in [62] agrees well with our solution. One can also apply approximate methods, but they should be considered inferior to the analytical solution, we provide here.

Analogous to the 3D case, we rely on Born-Oppenheimer (BO) approximation, in order to provide an analytical solution. The same justification for using the BO approximation in 3D molecular systems applies to the 2D problems as well, since the masses of the nuclei and electron(s) remain unchanged. Following the same BO approximation as in Chapter 1, and using atomic units, the Schrodinger equation for the  $H_2^+$  molecules is given by equation (5.1) below.

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r_a} - \frac{1}{r_b}\right)\psi = E\psi \quad (5.1)$$

The equation (5.1) is superficially similar to the equation (4.27) and the diagram 4.8, but in this case, the  $r_a$  and  $r_b$  represent the nuclei coordinates in 2 dimensions.

Following the [60], The electron wavefunction depends on only 3 quantum numbers, principal quantum number  $n$ , angular quantum number  $l$  and spin  $s$ . Again we ignore the spin degrees of freedom, considering the electron to be the regular 3 dimensional particles, where only its orbit is restricted to 2 dimensions. Therefore the spin magnetic moment of the electron is not affected by the 2D restriction and the spin vector can point in any direction in 3D space. Also the orbit can have different shapes, thus there remains the need for an orbital quantum number. However, there can be one orientation of the orbit, thus we do not consider the magnetic quantum number. This reasoning agrees with the solution of the electron wavefunction to the hydrogen atom in 2 dimensions [88] where electron wavefunction solution only depends on the principal and orbital quantum numbers.

The geometry of the 2D problem leads to choosing the elliptical coordinates (as in 3D problem), with two coordinates,  $\mu$  and  $\lambda$  denoting the position of the electron in 2D plane. [6]

## 5.2 Exact Solution of the Electron Energies of the $H_2^+$ Molecule

The goal is to provide the exact solution to the wavefunction of the  $H_2^+$  electron, for a given definition of exact. However even for this, relatively simple problem, it is impossible to find a closed form solution. So the solution is exact, in the sense that it can be done to an arbitrary precision.

As illustrated by the figure 5.1, we express equation (5.1) in the elliptical coordinates, and by setting the  $x$  axis to be perpendicular to the inter-nuclear axis, we have the nuclei at:  $z = \pm \frac{R}{2}$ ,  $R$  being the distance between nuclei. So in 2D elliptic coordinates,  $\lambda, \mu$  we have:

$$r_a = \frac{R}{2} (\lambda + \mu) \quad r_b = \frac{R}{2} (\lambda - \mu) \quad (5.2)$$

so

$$\lambda = (r_a + r_b) / R; \quad \mu = (r_a - r_b) / R \quad (5.3)$$

$$\text{where } \lambda \in [1, \infty], \quad \mu \in [-1, 1] \quad (5.4)$$

Now if we plug in the (5.2) into (5.1) we get by using  $\nabla$  in 2D elliptic cylindrical coordinates [55] .

$$\begin{aligned} \nabla^2 = \frac{4}{R^2(\lambda^2 - \mu^2)} \left[ \sqrt{\lambda^2 - 1} \frac{\partial}{\partial \lambda} \left( \sqrt{\lambda^2 - 1} \frac{\partial \psi}{\partial \lambda} \right) + \sqrt{1 - \mu^2} \frac{\partial}{\partial \mu} \left( \sqrt{1 - \mu^2} \frac{\partial \psi}{\partial \mu} \right) \right] = \\ \frac{4}{R^2(\lambda^2 - \mu^2)} \left[ (\lambda^2 - 1) \frac{\partial^2 \psi}{\partial \lambda^2} + \lambda \frac{\partial \psi}{\partial \lambda} + (1 - \mu^2) \frac{\partial^2 \psi}{\partial \mu^2} - \mu \frac{\partial \psi}{\partial \mu} \right] \end{aligned} \quad (5.5)$$

So the equation transforms to (with  $\psi = \psi(\lambda, \mu)$ :

$$-\frac{1}{2} \frac{4}{R^2(\lambda^2 - \mu^2)} \left[ (\lambda^2 - 1) \frac{\partial^2 \psi}{\partial \lambda^2} + \lambda \frac{\partial \psi}{\partial \lambda} + (1 - \mu^2) \frac{\partial^2 \psi}{\partial \mu^2} - \mu \frac{\partial \psi}{\partial \mu} \right] - \frac{4}{R} \frac{\lambda}{\lambda^2 - \mu^2} \psi = E \psi \quad (5.6)$$

We assume that the total electronic wavefunction can be written as the product of two functions:

$$\psi(\lambda, \mu) = F(\mu) L(\lambda) \quad (5.7)$$

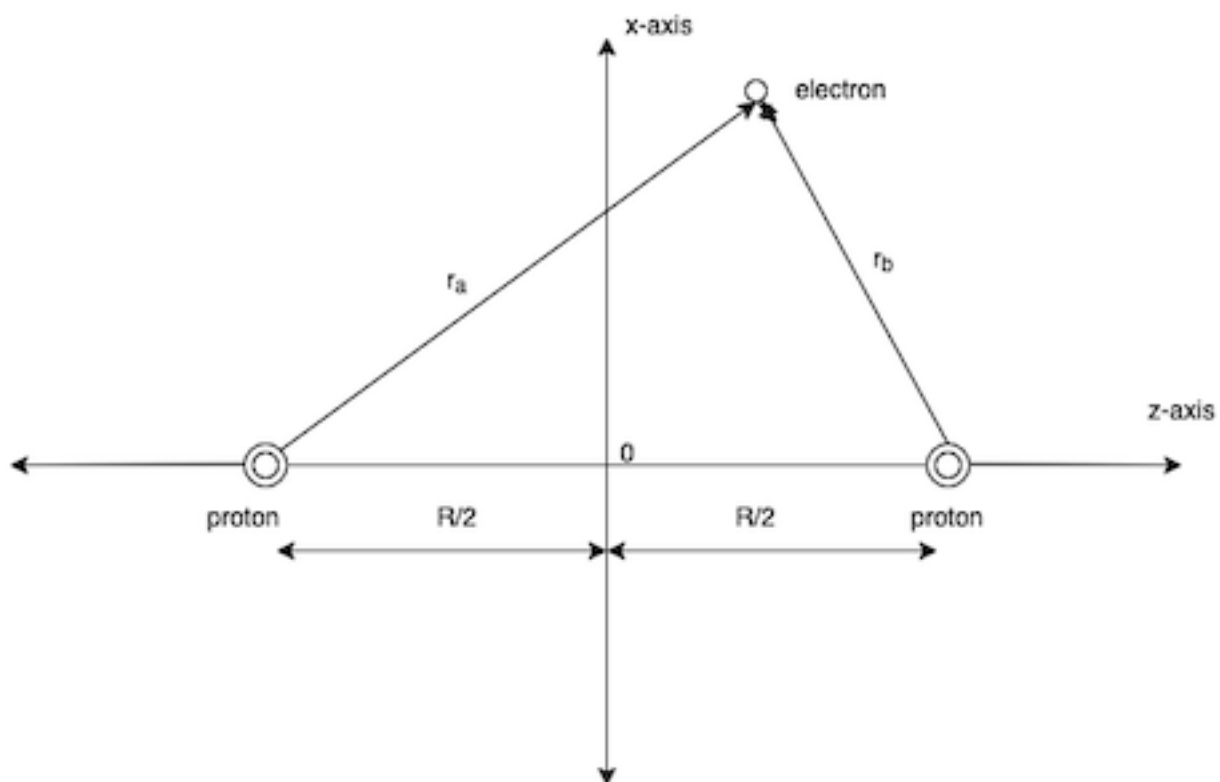


Figure 5.1: Hydrogen molecular ion in 2 dimensions

it follows:

$$\begin{aligned} & \frac{2}{R^2(\lambda^2 - \mu^2)} \left[ (\lambda^2 - 1)F \frac{\partial^2 L}{\partial \lambda^2} + \lambda F \frac{\partial L}{\partial \lambda} + (1 - \mu^2)L \frac{\partial^2 F}{\partial \mu^2} - \mu L \frac{\partial F}{\partial \mu} \right] \\ & + \frac{4}{R} \frac{\lambda}{\lambda^2 - \mu^2} FL + EFL = 0 \Rightarrow \end{aligned} \quad (5.8)$$

$$(\lambda^2 - 1) \frac{1}{L} \frac{\partial^2 L}{\partial \lambda^2} + \frac{\lambda}{L} \frac{\partial L}{\partial \lambda} + (1 - \mu^2) \frac{1}{F} \frac{\partial^2 F}{\partial \mu^2} - \frac{\mu}{F} \frac{\partial F}{\partial \mu} + 2R\lambda + \frac{R^2}{2} E(\lambda^2 - \mu^2) = 0$$

We can conclude that both equations for  $\lambda$  and  $\mu$  must be equal to the separation constant  $A$ .

$$(\lambda^2 - 1) \frac{\partial^2 L}{\partial \lambda^2} + \lambda \frac{\partial L}{\partial \lambda} + \left( 2R\lambda - \frac{R^2}{2} E\lambda^2 \right) L = AL \quad (5.9)$$

$$(1 - \mu^2) \frac{\partial^2 F}{\partial \mu^2} - \mu \frac{\partial F}{\partial \mu} - \frac{R^2}{2} E\mu^2 F = -AF$$

where  $A$  is the separation constant.

## 5.2.1 Algebraic Solution

### M Equation

Using the substitution  $\mu = \cos x$  and  $F(\mu) = M(x)$  we get a new equation:

$$\frac{d^2 M}{dx^2} + \left[ -A + \frac{p^2}{2} + \frac{p^2}{2} \cos(2x) \right] M = 0 \quad (5.10)$$

The equation (5.10) can be written as:

$$\frac{d^2 M}{dx^2} + [w - 2q \cos(2x)] M = 0 \quad (5.11)$$

where

$$\begin{aligned} w &= -A + \frac{p^2}{2} \\ q &= -\frac{p^2}{2} \end{aligned} \tag{5.12}$$

The equation (5.11) is a form of a Mathieu's equation [54].

From the geometry of the problem we conclude that  $M(\mu)$  must be an even function. Following [36] we look for the solution as class I and class II, which has even and odd eigenvalue functions as:

$$V_0 = \frac{2}{V_2 - \frac{1}{V_4 - \frac{1}{V_6 - \dots}}} \tag{5.13}$$

$$V_1 - 1 = \frac{1}{V_3 - \frac{1}{V_5 - \frac{1}{V_7 - \dots}}} \tag{5.14}$$

where

$$V_m = \frac{w - m^2}{q} \tag{5.15}$$

The equations 5.13 and 5.14 provide the first gen equation for the even and odd solution respectively.

## F Equation

$$(\lambda^2 - 1) \frac{\partial^2 F}{\partial \lambda^2} + \lambda \frac{\partial F}{\partial \lambda} + \left( 2R\lambda - \frac{R^2}{2} E\lambda^2 \right) F = -AF \tag{5.16}$$

We observe that both equations for the functions  $M(\mu)$  and  $F(\lambda)$  are either exact match

or related to the Mathieu's equations. In general the Mathieu's equation represents the standing wave on an elliptical drum, (2D space), and the solution for the time independent Schrodinger equation is in general a standing wave, in 2D in this case. So it is plausible that these types of solutions are similar.

There are actually two ways to try to solve this equation. One would be a brute force, by using some orthogonal polynomial (Laguerre) and looking for the eigenvalues.

The Laguerre approach seems to be numerically unstable. The algorithm is fast but the results are only good to the order of magnitude.

The second is a bit more algebraic, and likely more accurate.

Following [13] and [90] we could look for the solution in the form of:

$$L(\lambda) = (\lambda + 1)^\sigma e^{-p\lambda} \sum_{n=0}^{\infty} a_n x^n \quad (5.17)$$

where we use the substitution

$$\sigma = \frac{R}{p} - \frac{1}{2} \quad \text{and} \quad x = \frac{\lambda - 1}{\lambda + 1} \quad (5.18)$$

Substituting (5.18) into (5.17) and after some formidable algebra, we get a recurrence relation:

$$\alpha_n a_{n+1} - \beta_n a_n + \gamma_n a_{n-1} = 0 \quad n \geq 0 \quad (5.19)$$

with

$$\alpha_n = (n + 1) \left( n + \frac{1}{2} \right)$$

$$\beta_n = \left[ 2n^2 + (4p - 2\sigma)n - A + p^2 - 2p\sigma - \frac{\sigma}{2} \right] \quad (5.20)$$

$$\gamma_n = (n - 1) \left( n - 2\sigma - \frac{1}{2} \right) + \sigma \left( \sigma - \frac{1}{2} \right)$$

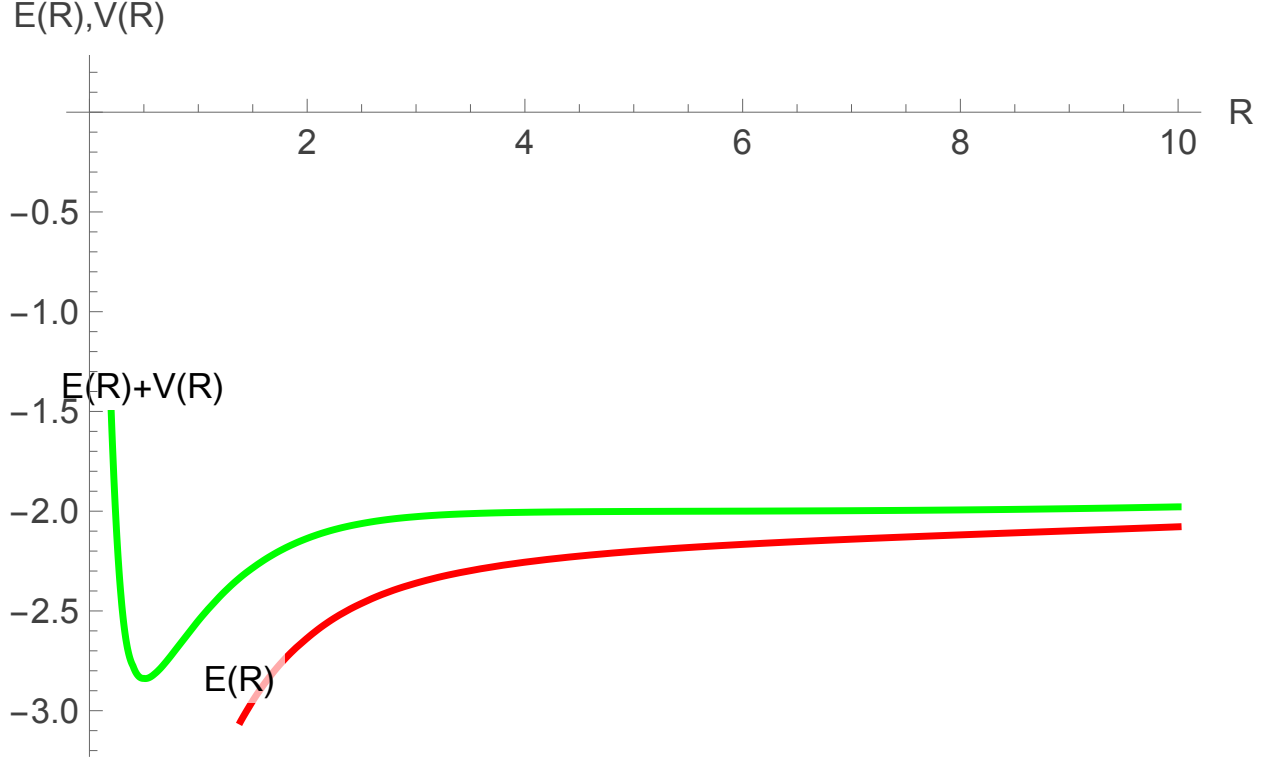


Figure 5.2:  $H_2^+$  energy and potential curves for the  $1s_g^+$  state

and it follows that

$$\frac{a_n}{a_{n-1}} = F_n \quad \text{where} \quad F_n = \frac{\gamma_n}{\beta_n - \frac{\alpha_n \gamma_{n+1}}{\beta_{n+1} - \frac{\alpha_{n+1} \gamma_{n+1}}{\beta_{n+1} - \dots}}} \quad (5.21)$$

Since  $a_{-1} = 0$  we have

$$\frac{\beta_0}{\alpha_0} = F_1 \quad (5.22)$$

The Mathematica code is in appendix C. The energies calculated and the plot of the potential for the states  $1s_g^+$  and  $1s_u^+$  are in the tables 5.1 and 5.2

The states  $1s_g$  and  $1s_u^+$  represent the Gerade ground state and the Ungerade state respectively.

For comparison we used paper by Patil [62]. Our results are very close to the results ob-

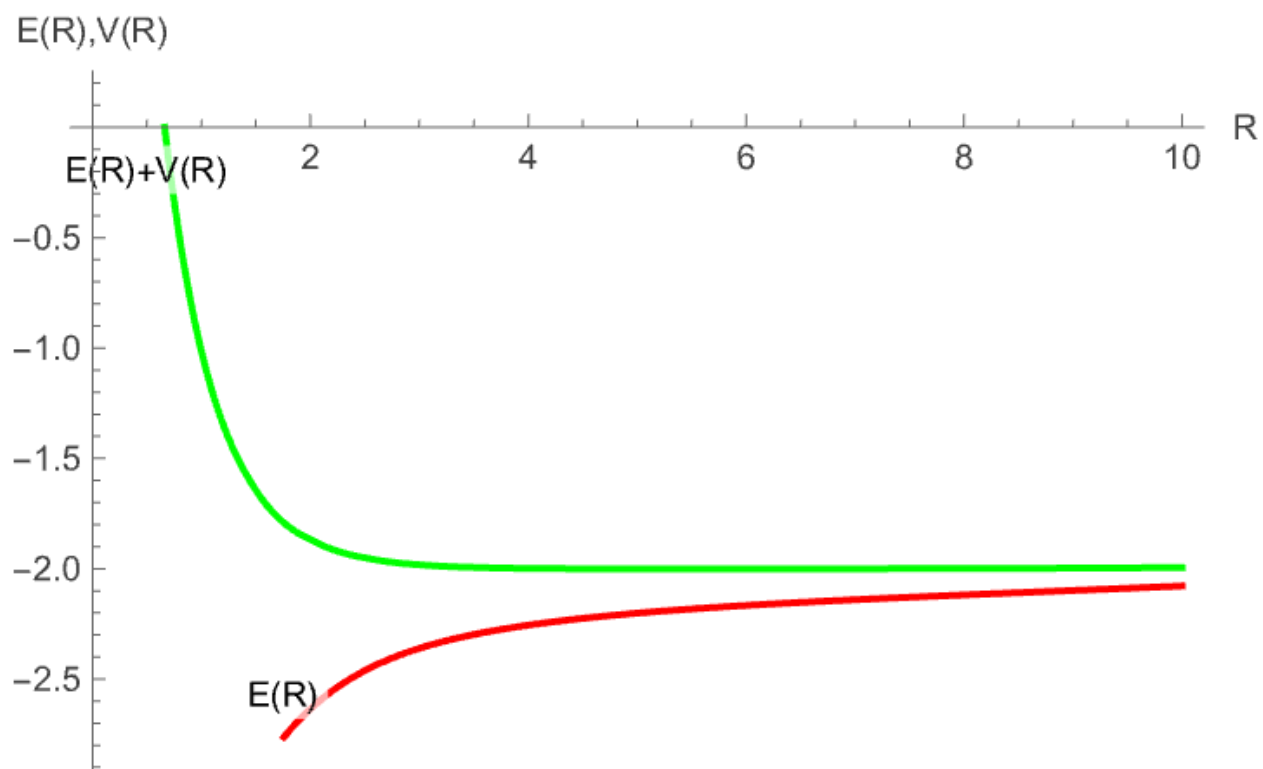


Figure 5.3:  $H_2^+$  energy and potential curves for the  $1s_u^+$  state

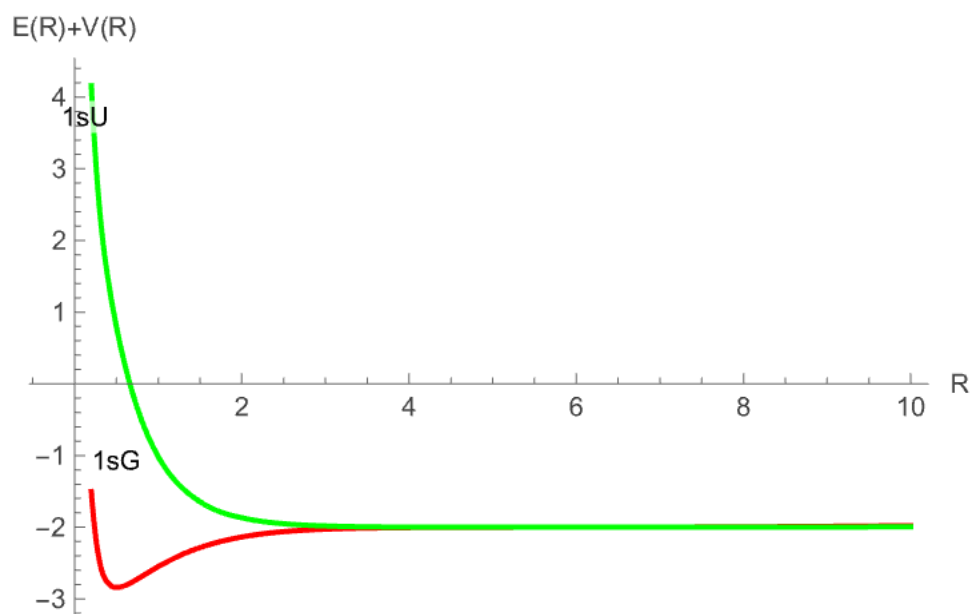


Figure 5.4:  $H_2^+$  potential curves for the first gerade (red) and first ungerade (green) states

Table 5.1: The separation constant A, electronic energy E, and total energy  $E + 1/R$  for the  $1s_g$  state at selected internuclear separations R

| R   | E        | $E + \frac{1}{R}$ | A       | p        |
|-----|----------|-------------------|---------|----------|
| 0.2 | -6.50361 | -1.50361          | 19.2757 | 0.360655 |
| 0.3 | -5.81983 | -2.48649          | 175.774 | 0.511754 |
| 0.4 | -5.27552 | -2.77552          | 198.466 | 0.649647 |
| 0.5 | -4.83822 | -2.83822          | 218.834 | 0.777674 |
| 0.6 | -4.48148 | -2.81482          | 30.6031 | 0.898146 |
| 0.7 | -4.18616 | -2.75759          | 255.124 | 1.01272  |
| 0.8 | -3.93852 | -2.68852          | 271.721 | 1.12264  |
| 0.9 | -3.72861 | -2.6175           | 287.554 | 1.22886  |
| 1.  | -3.54905 | -2.54905          | 302.754 | 1.33211  |
| 1.1 | -3.39429 | -2.4852           | 317.417 | 1.43302  |
| 1.5 | -2.95083 | -2.28416          | 371.99  | 1.822    |
| 2.  | -2.63638 | -2.13638          | 433.864 | 2.29625  |
| 2.5 | -2.4621  | -2.0621           | 491.12  | 2.77382  |
| 3.  | -2.36086 | -2.02752          | 545.139 | 3.25943  |
| 3.5 | -2.29797 | -2.01225          | 596.728 | 3.75167  |
| 4.  | -2.25565 | -2.00565          | 646.407 | 4.24796  |
| 4.5 | -2.22496 | -2.00274          | 694.536 | 4.74634  |
| 5.  | -2.20134 | -2.00134          | 741.373 | 5.24564  |
| 6   | -2.16637 | -1.9997           | 16608.  | 6.24457  |
| 7   | -2.14043 | -1.99757          | 919.107 | 7.24158  |
| 8   | -2.11874 | -1.99374          | 1003.68 | 8.23405  |
| 9   | -2.09856 | -1.98745          | 1086.12 | 9.21909  |
| 10  | -2.07822 | -1.97822          | 1166.79 | 10.1937  |

tained by Patil. Also one can observe that in 2 dimensions the max depth of the potential well of the  $H_2^+$  ground state is approx  $V(R = 0.5) = -2.83822$  a.u. compared to the  $V(R = 0.4) = -3.63220$  in the 3D case, which we showed in table 4.1 Also the shape of the potential well is more compact in the 2D case, for the ground  $1s_g^+$  gerade state. For the ungerade  $1s_u^+$  state the situation is even more different. The potential curve of the  $1s_u^+$  state does not really have a minimum for and does not allow a bound state.

Table 5.2: The separation constant A, electronic energy E, and total energy  $E + 1/R$  for the  $1s_u^+$  state at selected internuclear separations R

| R   | E         | $E + \frac{1}{R}$ | A       | p        |
|-----|-----------|-------------------|---------|----------|
| 0.2 | -0.841481 | 4.15852           | 33.4451 | 0.129729 |
| 0.3 | -0.911818 | 2.42152           | 39.6245 | 0.202563 |
| 0.4 | -1.03317  | 1.46683           | 44.6816 | 0.287495 |
| 0.5 | -1.20111  | 0.798885          | 49.0907 | 0.387478 |
| 0.6 | -1.39266  | 0.274004          | 53.0632 | 0.500679 |
| 0.7 | -1.58212  | -0.153551         | 56.7159 | 0.622591 |
| 0.8 | -1.75267  | -0.502667         | 395.73  | 0.748901 |
| 0.9 | -1.89725  | -0.786142         | 417.783 | 0.876577 |
| 1.  | -2.01521  | -1.01521          | 66.3746 | 1.0038   |
| 1.1 | -2.10899  | -1.1999           | 459.254 | 1.12957  |
| 1.5 | -2.31161  | -1.64494          | 534.677 | 1.61262  |
| 2.  | -2.36744  | -1.86744          | 619.719 | 2.17598  |
| 2.5 | -2.35029  | -1.95029          | 698.046 | 2.7101   |
| 3.  | -2.31526  | -1.98193          | 111.724 | 3.2278   |
| 3.5 | -2.2797   | -1.99399          | 841.77  | 3.73673  |
| 4.  | -2.24845  | -1.99845          | 909.103 | 4.24118  |
| 4.5 | -2.22222  | -2.               | 974.188 | 4.74341  |
| 5.  | -2.20044  | -2.00044          | 1037.4  | 5.24457  |
| 6   | -2.16704  | -2.00038          | 1159.29 | 6.24554  |
| 7   | -2.14278  | -1.99992          | 1276.36 | 7.24556  |
| 8   | -2.12404  | -1.99904          | 1389.63 | 8.24435  |
| 9   | -2.10851  | -1.9974           | 1499.83 | 9.24093  |
| 10  | -2.09461  | -1.99461          | 1607.45 | 10.2338  |

### 5.2.2 Long-Range Behavior of $E(R)$

In the calculation for the phase shifts, or  $b_m$ , we use the accurate  $g, u$  potential curves calculated using the method outlined in Chap. 5.

However, for large internuclear separations  $R \approx > 13a_0$ , that numerical method becomes unstable. Instead, we match those calculated curves with the analytic asymptotic form described below. Although the LCAO approximation given above correctly predicts the asymptotic energy  $E(R) \rightarrow -2$  as  $R \rightarrow \infty$ , it does not accurately predict the long range behavior of  $E(R)$ . To that end we note that as  $R \rightarrow \infty$ ,

$$V(\mathbf{r}, \mathbf{R}) + 1/R \rightarrow -\frac{1}{r} - \frac{z}{R^2} + \frac{x^2 - 2z^2}{2R^3} + \dots \quad (5.23)$$

where  $z$  is the electronic coordinate along the direction  $\mathbf{R}$ , and we ignored terms that contain higher inverse powers of  $R$ . Thus the Hamiltonian

$$H \approx H_0 - \frac{z}{R^2} + \frac{x^2 - 2z^2}{2R^3} + \dots \quad (5.24)$$

where  $H_0$  is the Hamiltonian for the 2D hydrogen atom. Making use of perturbation theory,

$$E(R) \approx E_0 + E^{(1)}(R) + E^{(2)}(R) + \dots \quad (5.25)$$

and

$$E^{(1)} = -\frac{\langle \psi_{1s} | z | \psi_{1s} \rangle}{R^2} + \frac{\langle \psi_{1s} | (x^2 - 2z^2) | \psi_{1s} \rangle}{2R^3} = 0 \quad (5.26)$$

The lowest non-vanishing contribution arises from second-order perturbation theory

$$E^{(2)} = -\frac{C_4}{R^4}$$

$$C_4 = \sum_n \frac{|\langle \psi_{1s} | z | \psi_n \rangle|^2}{E_{1s} - E_n}$$

The polarization constant  $C_4 = 0.082$  was given in [89].

In the tables below we show the accurate and LCAO values for the potential of the gerade  $1sG$  table 5.3 and ungerade  $1sU$  table 5.4 states.

Table 5.3: Accurate and LCAO energy as function of R (internuclear distance) for the gerade state

| R   | E(R) +<br>V(R)<br>accurate | E(R) +<br>V(R)<br>LCAO |
|-----|----------------------------|------------------------|
| 0.2 | -1.50328                   | 1.11882                |
| 0.3 | -2.48639                   | -<br>0.436367          |
| 0.4 | -2.77548                   | -1.14571               |
| 0.5 | -2.8382                    | -1.51849               |
| 0.6 | -2.81481                   | -1.72704               |
| 0.7 | -2.75759                   | -1.84613               |
| 0.8 | -2.68852                   | -1.91333               |
| 0.9 | -2.61749                   | -1.94936               |
| 1.  | -2.54904                   | -1.96642               |
| 1.1 | -2.4852                    | -1.97198               |
| 1.5 | -2.28416                   | -1.95159               |
| 2.  | -2.13638                   | -1.9237                |
| 2.5 | -2.0621                    | -1.91727               |
| 3.  | -2.02753                   | -1.9224                |
| 3.5 | -2.01226                   | -1.93066               |
| 4.  | -2.00568                   | -1.93856               |
| 4.5 | -2.00284                   | -1.94523               |
| 5.  | -2.00156                   | -1.95066               |
| 6   | -2.00065                   | -1.95882               |
| 7   | -2.00036                   | -1.96462               |
| 8   | -2.00023                   | -1.96898               |
| 9   | -2.00016                   | -1.97238               |
| 10  | -2.00012                   | -1.97512               |
| 11  | -2.00009                   | -1.97736               |
| 12  | -2.00006                   | -1.97924               |

Table 5.4: Accurate and LCAO energy as function of R (internuclear distance) for the ungerade state

| R   | E(R) +<br>V(R)<br>accurate | E(R) +<br>V(R)<br>LCAO |
|-----|----------------------------|------------------------|
| 0.2 | 4.15868                    | 2.01244                |
| 0.3 | 2.42148                    | 0.534181               |
| 0.4 | 1.46676                    | -<br>0.186921          |
| 0.5 | 0.798837                   | -<br>0.617186          |
| 0.6 | 0.273983                   | -<br>0.905274          |
| 0.7 | -<br>0.153558              | -1.11241               |
| 0.8 | -<br>0.502669              | -1.26834               |
| 0.9 | -<br>0.786143              | -1.38934               |
| 1.  | -1.01521                   | -1.48521               |
| 1.1 | -1.1999                    | -1.56225               |
| 1.5 | -1.64494                   | -1.75286               |
| 2.  | -1.86744                   | -1.85654               |
| 2.5 | -1.95029                   | -1.90027               |
| 3.  | -1.98193                   | -1.92094               |
| 3.5 | -1.99399                   | -1.93255               |
| 4.  | -1.99845                   | -1.94038               |
| 4.5 | -2.                        | -1.94638               |
| 5.  | -2.00046                   | -1.95129               |
| 6   | -2.00049                   | -1.95897               |
| 7   | -2.00034                   | -1.96465               |
| 8   | -2.00023                   | -1.96899               |
| 9   | -2.00016                   | -1.97239               |
| 10  | -2.00012                   | -1.97512               |
| 11  | -2.00009                   | -1.97736               |
| 12  | -2.00007                   | -1.97923               |

## Chapter 6

### Scattering in Two Dimensions

In the PSS (perturbed stationary state) approximation, and after factoring out the center of mass coordinates, the system wave amplitude is given by the expression

$$\psi_g(\mathbf{R})\phi_g(\mathbf{R}, \mathbf{r}) + \psi_u(\mathbf{R})\phi_u(\mathbf{R}, \mathbf{r}) \quad (6.1)$$

where  $\phi_{u,g}(\mathbf{R}, \mathbf{r})$  are, respectively, the gerade and ungerade electronic ground BO eigenstates and  $\psi_g(\mathbf{R}), \psi_u(\mathbf{R})$  are the gerade and un-gerade amplitudes for the relative motion of the atom and projectile. The amplitudes satisfy the following

$$-\frac{1}{2\mu}\nabla^2\psi_{u,g}(\mathbf{R}) + V_{u,g}(R)\psi_{u,g}(\mathbf{R}) = E\psi_{u,g}(\mathbf{R}) \quad (6.2)$$

where  $E = k^2/2\mu$  is the collision energy, and  $V_{u,g}$  are the Born-Oppenheimer potentials for the gerade and ungerade electronic states respectively. Here  $\nabla^2$  is the two-dimensional Laplacian, and because  $V_{g,u}(R)$  are rotationally invariant, we can express the amplitudes as a partial wave expansion

$$\psi_{g,u}(\mathbf{R}) = \sum_{m=-\infty}^{m=\infty} e^{im\phi} F_m^{g,u}(R) \quad (6.3)$$

where the radial functions  $F_m(R)$  obey

$$F_m''(R) + \frac{F_m'}{R} - \frac{m^2}{R^2}F_m(R) + (k^2 - 2\mu V_{g,u}(R))F_m(R) = 0 \quad (6.4)$$

where we suppressed the  $g, u$  superscripts in the radial functions, and the prime notation denotes derivatives with respect to  $R$ .

In the asymptotic region as  $R \rightarrow \infty$  and  $V_{g,u} \rightarrow 0$ , Eq. (6.4) tends to the Bessel equation

for integer order, and the radial solutions tend to

$$F_m(R) \rightarrow a_m J_{|m|}(kR) + b_m H_{|m|}^{(1)}(kR) \quad (6.5)$$

where  $J_m(kR)$  is the regular Bessel function and  $H_m^1(kR)$  is a Hankel function of the first kind, and  $a_m, b_m$  are constants. We impose the boundary condition

$$\psi(\mathbf{R}) \rightarrow e^{ikR \cos \phi} + f(\phi) \frac{e^{ikR}}{\sqrt{R}} \quad (6.6)$$

Using the identity

$$e^{ikR \cos \phi} = \sum_{m=-\infty}^{m=\infty} -i^{-|m|} J_{|m|}(kR) e^{im\phi} \quad (6.7)$$

we set  $a_m = -i^{-|m|}$  and inserting into (6.6) we get another equation

$$\psi(\mathbf{R}) \rightarrow e^{ikR \cos \phi} + \frac{e^{ikR}}{\sqrt{R}} \sqrt{\frac{2}{\pi k}} \sum_m (-1)^{m+1} e^{-i\frac{\pi}{4}} e^{im\phi} b_m \quad (6.8)$$

where we used the asymptotic form

$$H_m^{(1)} \rightarrow \sqrt{\frac{2}{\pi kR}} e^{[i(kR - \frac{m\pi}{2} - \frac{\pi}{4})]}. \quad (6.9)$$

We can solve for the coefficients  $b_{|m|}$  by matching the radial logarithmic derivative

$$y_m \equiv \frac{F'_m}{F_m} \quad (6.10)$$

evaluated at a cutoff distance  $R_c$  so that the ratio

$$b_{|m|} \rightarrow i^{-|m|} \frac{J'_{|m|} - y_m J_{|m|}}{H'^{(1)}_{|m|} - y_m H^{(1)}_{|m|}} \quad (6.11)$$

tends to a constant value  $b_{|m|}$  at  $R > R_c$ .

Using the expression

$$\mathbf{j} = \text{Re} \left[ \frac{-i}{\mu} \psi^*(\mathbf{R}) \nabla \psi(\mathbf{R}) \right] \quad (6.12)$$

where  $\nabla$  is the two-dimensional gradient operator for the probability current, we obtain

for the incident current along the  $z$ - direction

$$j_{inc} = \frac{k}{\mu} \quad (6.13)$$

and for the leading order term of outgoing radial current

$$j_r = \frac{k}{R\mu} f^*(\phi) f(\phi) \quad (6.14)$$

Defining the cross section as the ratio of particle flux scattered into the region  $R d\phi$  centered at angle  $\phi$  with the incident current  $j_{inc}$ , we get

$$d\sigma \equiv |f(\phi)|^2 d\phi \quad (6.15)$$

Comparing equation (6.6) with the equation (6.8) we find that

$$f(\phi) = \sqrt{\frac{2}{\pi k}} \sum_m (-1)^{(m+1)} e^{-i\frac{\pi}{4}} e^{im\phi} b_{|m|} \quad (6.16)$$

Thus

$$\sigma = \int_0^{2\pi} d\phi \frac{d\sigma}{d\phi} = \int_0^{2\pi} d\phi |f(\phi)|^2 = \frac{4}{k} \sum_m |b_{|m|}|^2 \quad (6.17)$$

Alternatively, the constant  $b_m$  is sometimes expressed as a phase shift.

## 6.1 Resonant Charge Transfer in Two Dimensions

### 6.1.1 Resonant Charge Transfer Cross Sections

Consider the amplitude given by expression (6.5)

$$F(\mathbf{R}, \mathbf{r}) = \psi_g(\mathbf{R})\phi_g(\mathbf{R}, \mathbf{r}) + \psi_u(\mathbf{R})\phi_u(\mathbf{R}, \mathbf{r}) \quad (6.18)$$

In the limit as  $R \rightarrow \infty$ ,

$$F \rightarrow \psi_g(\mathbf{R})\frac{1}{\sqrt{2}}(\phi_a + \phi_b) + \psi_u(\mathbf{R})\frac{1}{\sqrt{2}}(\phi_a - \phi_b) \quad (6.19)$$

where

$$\phi_a \equiv \psi_{1s}(\mathbf{r} - \mathbf{R}/2) \quad \phi_b \equiv \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \quad (6.20)$$

Now imposing the boundary conditions

$$\psi_{g,u}(\mathbf{R}) \rightarrow e^{ikR \cos \phi} + f_{g,u}(\phi) \frac{e^{ikR}}{\sqrt{R}} \quad (6.21)$$

we get

$$F \rightarrow \sqrt{2}e^{ikR \cos \phi} \phi_a + \frac{1}{\sqrt{2}}(f_g(\phi) + f_u(\phi))\phi_a + \frac{1}{\sqrt{2}}(f_g(\phi) - f_u(\phi))\phi_b \quad (6.22)$$

Due to the  $H$  nuclei being equal, the charge-transfer probability is governed by the phase difference. So the ratio of flux coming into the  $\phi_a$  channel to that going into the  $\phi_b$  channel (i.e. resonant charge transfer) is given by

$$\sigma_{RCT} = \frac{1}{4} \int_0^{2\pi} |f_g(\phi) - f_u(\phi)|^2 = \frac{1}{k} \sum_m \left| b_{|m|}^{(g)} - b_{|m|}^{(u)} \right|^2 \quad (6.23)$$

where  $b_{|m|}^{(g,u)}$  are the elastic scattering coefficients for the gerade and ungerade potentials, respectively.

We compute them in the Wolfram Mathematica code listed in appendix E.

## 6.2 Role of Symmetry and Anyonic Flux Tube

The nuclei of the H molecule are identical particles, and so we must enforce symmetry constraints for the latter. Symmetrization requires the inclusion of the spin degree of freedom, and here, for now, we assume that both nuclei are spin-polarized. Therefore, under traditional interpretations in which the nuclei are fermions, we take the spatial component of the wave amplitude to be antisymmetric under interchange

$$F(\mathbf{R}, \mathbf{r}) = \psi_g(\mathbf{R})\phi_g(\mathbf{R}, \mathbf{r}) + \psi_u(\mathbf{R})\phi_u(\mathbf{R}, \mathbf{r}) \quad (6.24)$$

where  $P_{12}$  is an interchange operator. Now since

$$F = \psi_g(\mathbf{R})\phi_g + \psi_u(\mathbf{R})\phi_u$$

$$P_{12}F = \psi_g(-\mathbf{R})\phi_g + \psi_u(-\mathbf{R})\phi_u$$

We symmetrize by requiring the amplitude

$$F \rightarrow (\psi_g(\mathbf{R}) - \psi_g(-\mathbf{R}))\phi_g + (\psi_u(\mathbf{R}) - \psi_u(-\mathbf{R}))\phi_u \quad (6.25)$$

or

$$\begin{aligned} F &\rightarrow \sqrt{2} \left( e^{ikR \cos \phi} + e^{-ikR \cos \phi} \right) \phi_a \\ &+ \frac{1}{\sqrt{2}} (f_g(\phi) - f_g(\phi + \pi) + f_u(\phi) - f_u(\phi + \pi)) \phi_a \\ &+ \frac{1}{\sqrt{2}} (f_g(\phi) - f_g(\phi + \pi) - f_u(\phi) + f_u(\phi + \pi)) \phi_b \end{aligned}$$

Thus

$$\begin{aligned} \sigma_{RCT} &= \frac{1}{8} \int_0^{2\pi} |f_g(\phi) - f_g(\phi + \pi) - f_u(\phi) + f_u(\phi + \pi)|^2 = \\ &\frac{2}{k} \sum_{m=\pm 1, \pm 3, \dots} |b_m^g - b_m^u|^2. \end{aligned} \quad (6.26)$$

In a pioneering paper, J. M. Leinass and J. Myrheim [50] argued that in two dimensions, identical particles can exhibit interchange statistics that differ from bosons or fermions. As emphasized in that paper, the topology of the configuration space of identical particles in a 2D spatial manifold differs from that of a 3-dimensional manifold. The former allows a generalized amplitude  $\psi(\mathbf{r}_1, \mathbf{r}_2)$  for a pair of identical particles to obey

$P_{12}\psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\pi\alpha}\psi(\mathbf{r}_2, \mathbf{r}_1)$  where  $\alpha$  is an arbitrary constant.

For fermions  $\alpha = 1$ , and for bosons  $\alpha = 0$ . For fractional values of  $\alpha$ , the amplitudes are elements of the braid group [44]

Particles that differ from fermion/boson statistics are called anyons (Wilzcek [84]).

Whether such particles exist in nature is of topical and intense investigation. Several studies have examined the role of the latter in the bulk properties of an anyon gas [7]. There are two popular methods used to enforce anyonic statistics among identical particles. The first involves positing a Chern-Simons [39] term that couples each anyon to a pure gauge field. Alternatively, one posits the presence of an Aharonov-Bohm infinitesimal magnetic flux tube piercing the anyon at an angle perpendicular to the 2D plane (see Fig 6.1).

We shall assume the latter for the nuclei of our collision partners. Hence, the Hamiltonian for a 2D Hydrogen atom/ion pair, whose nuclei are anyons, takes the form

$$H = \frac{1}{2m} (\mathbf{p}_1 - \mathbf{A})^2 + \frac{1}{2m} (\mathbf{p}_2 - \mathbf{A})^2 + H'(\mathbf{r}_1, \mathbf{r}_2) \quad (6.27)$$

where  $H'(\mathbf{r}_1, \mathbf{r}_2)$  includes the electronic kinetic energy and all Coulombic interactions discussed in the previous chapters and

$$\mathbf{A} \equiv \alpha \frac{\mathbf{n} \times (\mathbf{r}_1 - \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^2} \quad (6.28)$$

is a gauge potential [87]. Here  $\mathbf{n}$  is a unit normal vector to the 2D scattering plane and  $0 < \alpha < 1$ . Introducing the relative coordinates

$$\begin{aligned} \mathbf{R} &= \mathbf{r}_2 - \mathbf{r}_1 \\ \mathbf{R}_{cm} &= \frac{1}{2} (\mathbf{r}_2 + \mathbf{r}_1) \end{aligned} \quad (6.29)$$

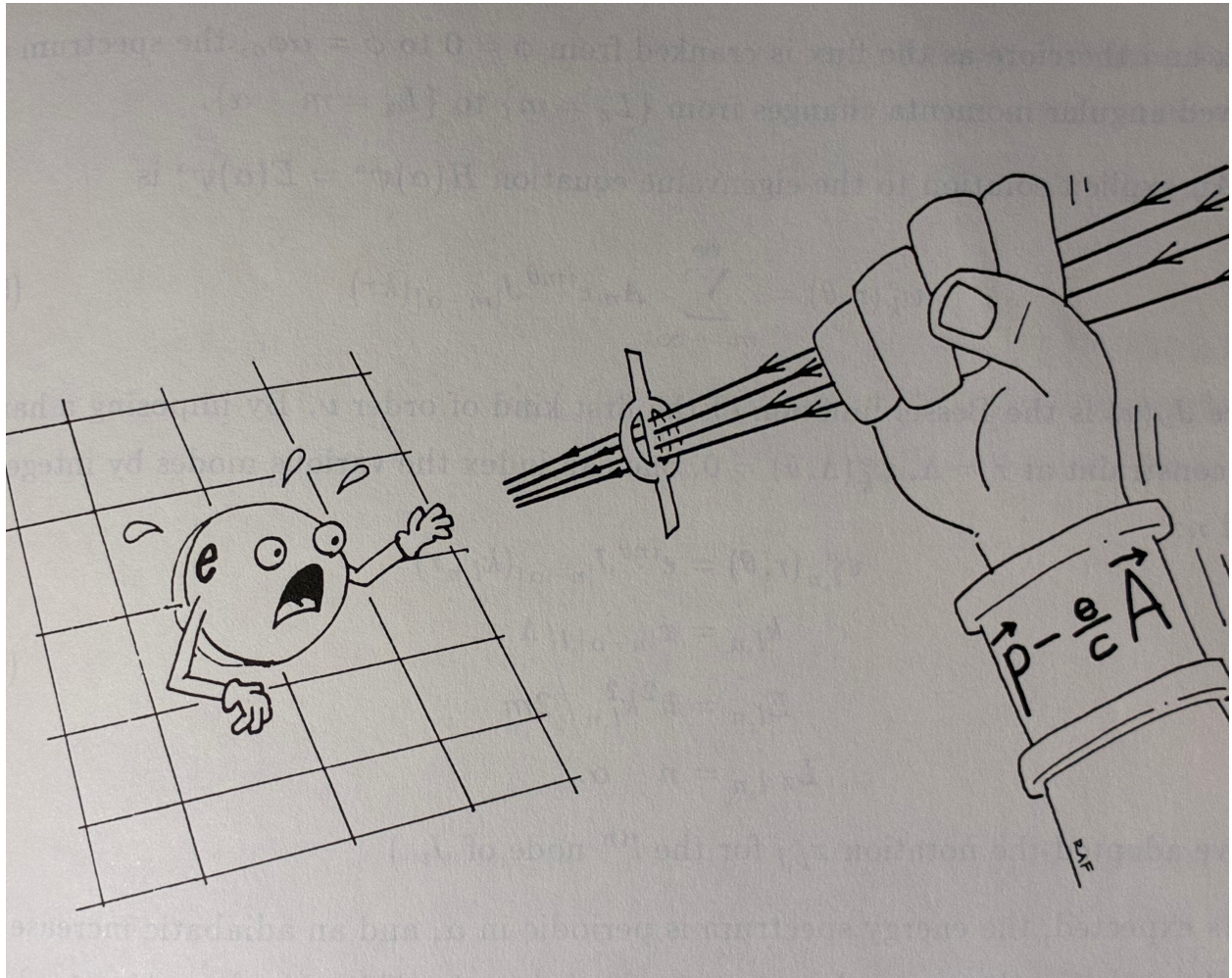


Figure 6.1: Illustration of anyon as a particle-flux tube composite. Illustration taken from D. P. Arovas Geometric Phases in Physics, pg. 287, ed. A. Shapere and F. Wilczek, World Scientific (1989)

The Schrodinger equation for the relative motion  $F(\mathbf{R})$  becomes

$$-\frac{1}{2\mu}(\nabla - i\mathbf{A})^2 F(\mathbf{R}) + V_{g,u}(R)F(\mathbf{R}) - EF(\mathbf{R}) = 0 \quad (6.30)$$

where  $\nabla$  is the 2D gradient operator and, in a cylindrical coordinate system,

$$\mathbf{A} = \hat{\phi} \frac{\alpha}{R} \quad (6.31)$$

Or

$$\begin{aligned} \frac{d^2 F(\mathbf{R})}{dR^2} + \frac{dF(\mathbf{R})}{dR} \frac{1}{R} + \frac{1}{R^2} \left( \frac{\partial}{\partial \phi} - i\alpha \right)^2 F(\mathbf{R}) + \\ (k^2 - 2\mu V_{g,u}(R))F(\mathbf{R}) = 0 \end{aligned} \quad (6.32)$$

Note that if we impose a singular gauge transformation  $F(\mathbf{R}) \rightarrow e^{ri\alpha\phi} \tilde{F}(\mathbf{R})$  then  $\tilde{F}(\mathbf{R})$  obeys the standard Schrodinger without the presence of a flux tube, however, the pre-factor  $e^{i\alpha}$  introduces multi-valuedness in the total amplitude and special methods [14] must be invoked to cancel that singularity. Under particle interchange  $\mathbf{R} \rightarrow -\mathbf{R}$ , which in this coordinate system is equivalent to the action of the parity operator  $\Pi$ .

Here, we follow the procedure of Aharonov-Bohm and expand  $F(\mathbf{R})$  in terms of single-valued function  $e^{im\phi}$ ,  $m = 0, \pm 1, \dots$ , resulting in the radial equations (6.33)

$$F_m''(R) + \frac{F_m'}{R} - \frac{(m - \alpha)^2}{R^2} F_m(R) + (k^2 - 2\mu V_{g,u}(R))F_m(R) = 0 \quad (6.33)$$

from which we construct the amplitude in the asymptotic region

$$\sum_{m=0,\pm 1,\dots} e^{im\phi} \left( a_m J_{|m-\alpha|}(kR) + b_m H^{(1)}_{|m-\alpha|}(kR) \right) \quad (6.34)$$

If we choose (see (6.34))  $a_m = (-i)^{|m-\alpha|}$ , the above expression reduces to

$$F_{g,u} \rightarrow e^{i\alpha\phi} e^{-ikz} + \frac{e^{ikR}}{\sqrt{2\pi kR}} f_{AB}(\phi) + f_{g,u}(\phi) \frac{e^{ikR}}{\sqrt{R}} \quad (6.35)$$

where  $f_{AB}$  is the Aharonov-Bohm scattering amplitude, and  $f_{g,u}(\phi)$  is given by the expression

$$f^{g,u}(\phi) = \sqrt{\frac{2}{\pi k}} \sum_{m=0,\pm 1,\dots} (-i)^{|m-\alpha|} e^{im\phi} b_m^{g,u} \quad (6.36)$$

$$b_m^{g,u} = (-i)^{|m-\alpha|} \frac{J'_{|m-\alpha|} - y_m^{g,u} J_{|m-\alpha|}}{H'_{|m-\alpha|} - y_m^{g,u} H_{|m-\alpha|}}$$

and  $J', H'$  represent derivatives with respect to  $R$ , and  $y^{g,u}$  the log-derivates corresponding to gerade and ungerade potentials respectively. Note that the wave amplitude, though single valued, the incident wave in equation (6.35) is multi-valued. As Schrodinger equation (6.30) is invariant to the parity operation  $\Pi$ , we seek the eigensolutions  $F(\mathbf{R})$  that are either even or odd with respect to a parity transformation, i.e.  $\Pi F(\mathbf{R}) = F(\mathbf{R})$  or  $\Pi F(\mathbf{R}) = -F(\mathbf{R})$  for even/odd functions respectively. We shall call the former the bosonic/fermionic bases respectively.

Following the discussion leading to equation (6.35) we obtain the outgoing wave of the charge transfer amplitude given by Eq. (6.36) but where  $b_{|m|}^{g,u}$  satisfies relation (6.36) instead of (6.11), and we get

$$\sigma_{CT} = \frac{2}{k} \left( \sum_m |b_m^{(g)} - b_m^{(u)}|^2 \right) \quad (6.37)$$

$$m = 0, \pm 2, \dots \quad \text{for bosonic base/}$$

$$m = \pm 1, \pm 3 \quad \text{for fermionic base}$$

Not/e that the terms proportional to  $f_{AB}(\phi)$  cancel as they are independent of the gerade/ungerade potentials. The restriction to odd values of  $m$  is determined by the fermionic symmetry requirement of the base ( $\alpha = 0$ ) system.

### 6.3 Calculations and Discussion

Below we illustrate the result of the calculations corresponding to the analysis given above.

In Figures 2-7 we present values for the calculation of the partial wave contributions to  $|b_m^{(g)} - b_m^{(u)}|^2$  for the values of temperatures from  $T = 1, 10, 100 K$ . The abscissa correspond to the partial wave index  $m$ , and the blue/red icons refer to bosonic/fermionic basis respectively. The parameter  $\alpha$  gives the strength of the flux tube, so  $\alpha = 0$  corresponds to the absence of a flux tube, and the partial wave contributions reflect the standard symmetry identification for the 3D case. For  $\alpha = 1$ , the Wilson loop integral

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = 2\pi n \quad (6.38)$$

where  $C$  is any circuit enclosing the flux with winding number  $n$ . In that case we notice, from these figures, that the partial wave spectra are identical to the  $\alpha = 0$  spectra, with the important difference that the bosonic  $\alpha = 0$ , contributions now correspond to the fermionic  $\alpha = 1$  contributions, and vice/versa. One can interpret this phenomenon as a transmutation of bosons into fermions, and vice/versa.

The case  $\alpha = 1/2$  exhibits fractional statistics, and in order to distinguish it from bosonic/fermionic statistics, particles that exhibit this feature are commonly called semions in the literature (cite papers). In the figures below, we also illustrate the partial wave contributions for semions ( $\alpha = 1/2$ ), at  $T = 1K, 10K, 100K$ .

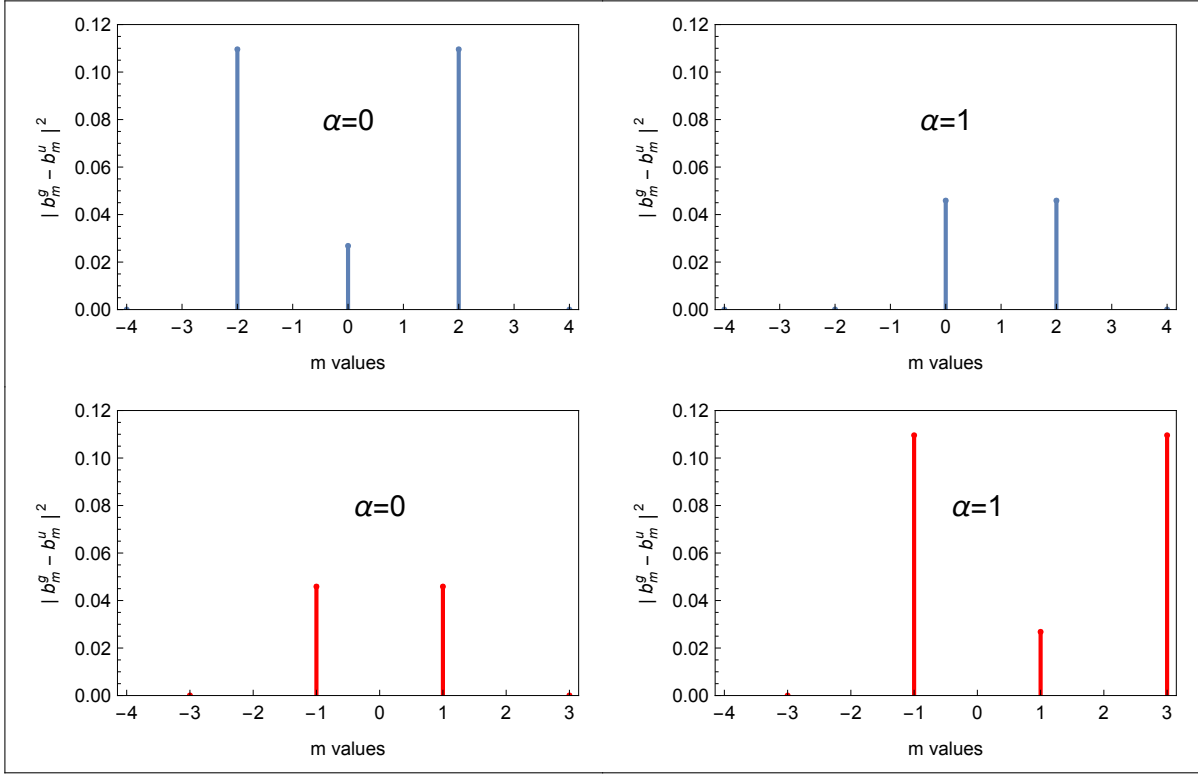


Figure 6.2: Collision temperature  $T = 1 K$ . Blue icons refer to a bosonic base, and red icons to a fermionic base.

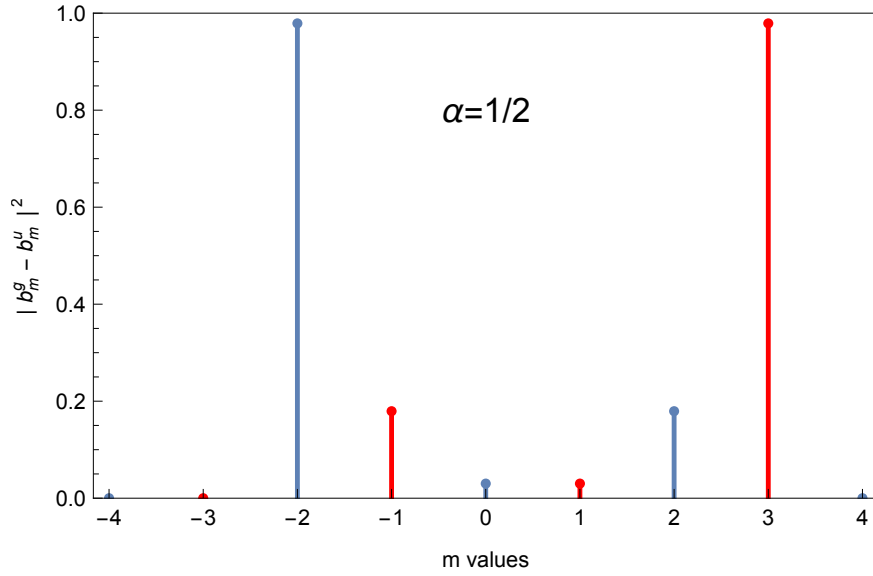


Figure 6.3: Collision temperature  $T = 1 K$ . Blue icons refer to a bosonic base, and red icons to a fermionic base.

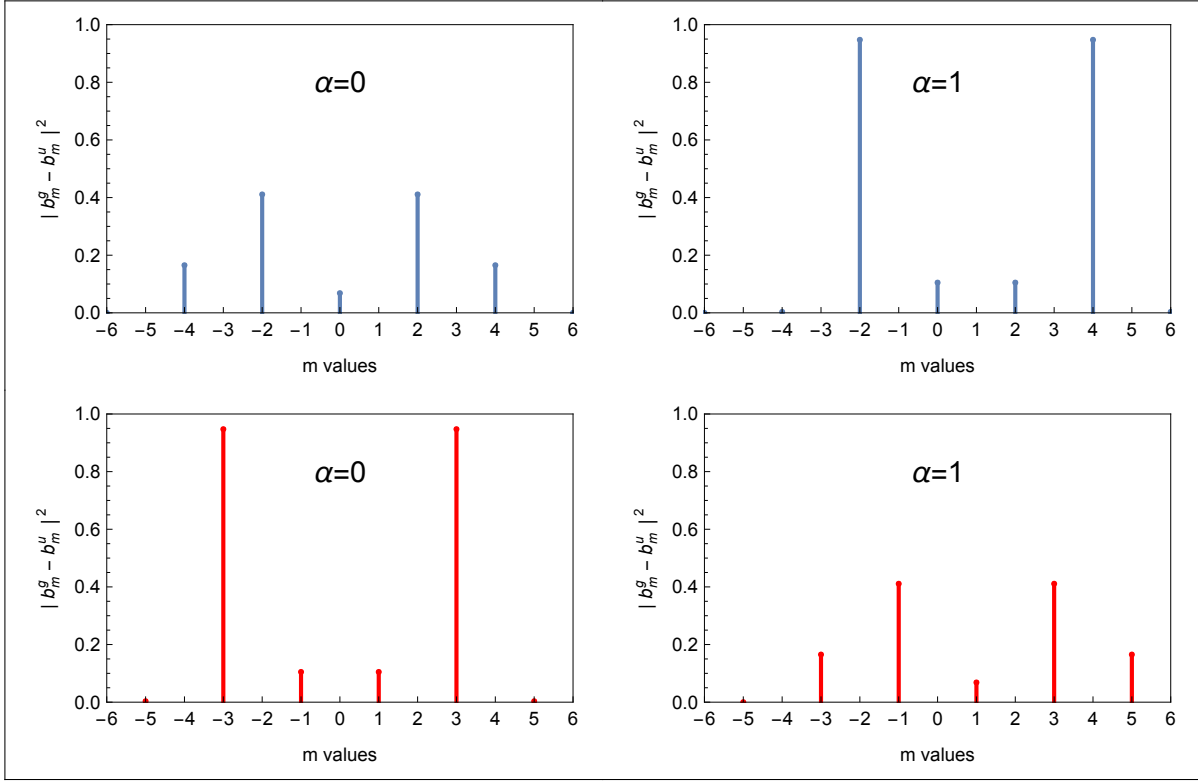


Figure 6.4: Collision temperature  $T = 10$  K. Blue icons refer to a bosonic base, and red icons to a fermionic base.

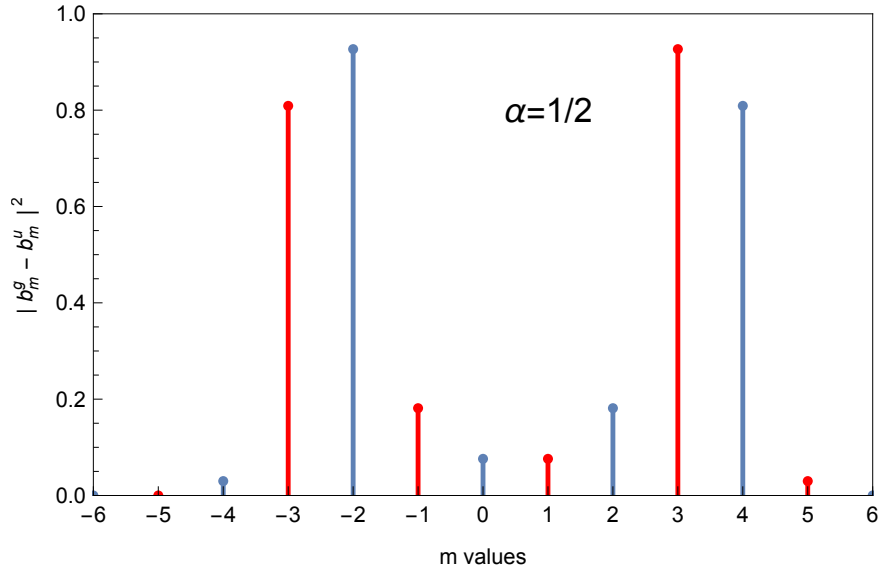


Figure 6.5: Collision temperature  $T = 10$  K. Blue icons refer to a bosonic base, and red icons to a fermionic base.

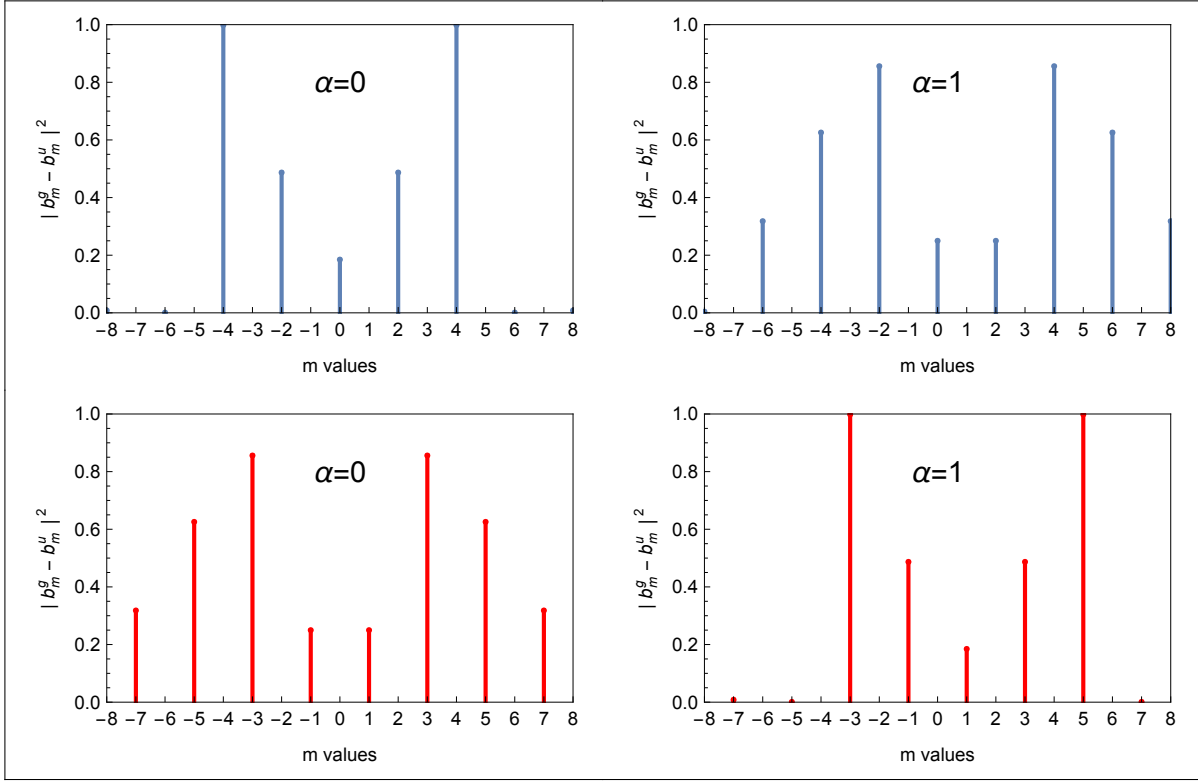


Figure 6.6: Collision temperature  $T = 100 K$ . Blue icons refer to a bosonic base, and red icons to a fermionic base.

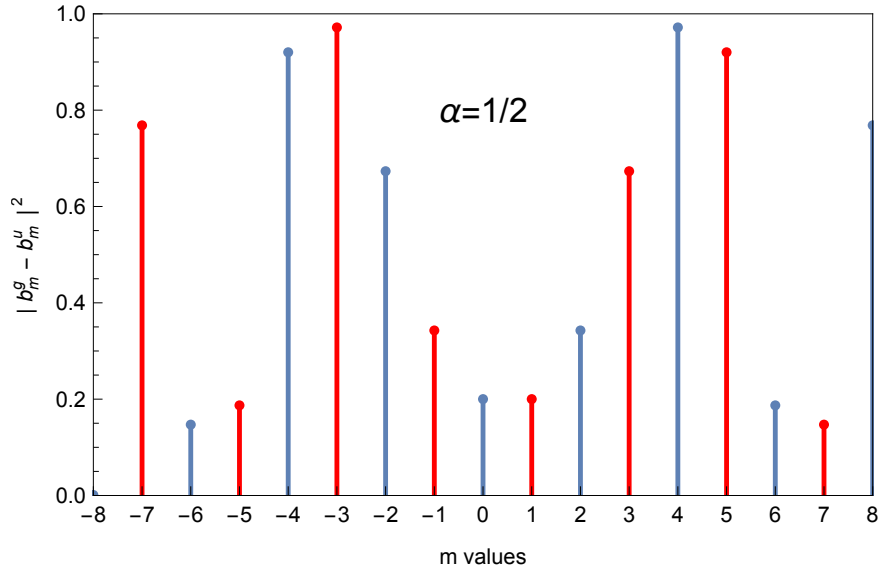


Figure 6.7: Collision temperature  $T = 100 K$ . Blue icons refer to a bosonic base, and red icons to a fermionic base. Flux tube charge  $\alpha = 0$ , and  $T = 1 K$

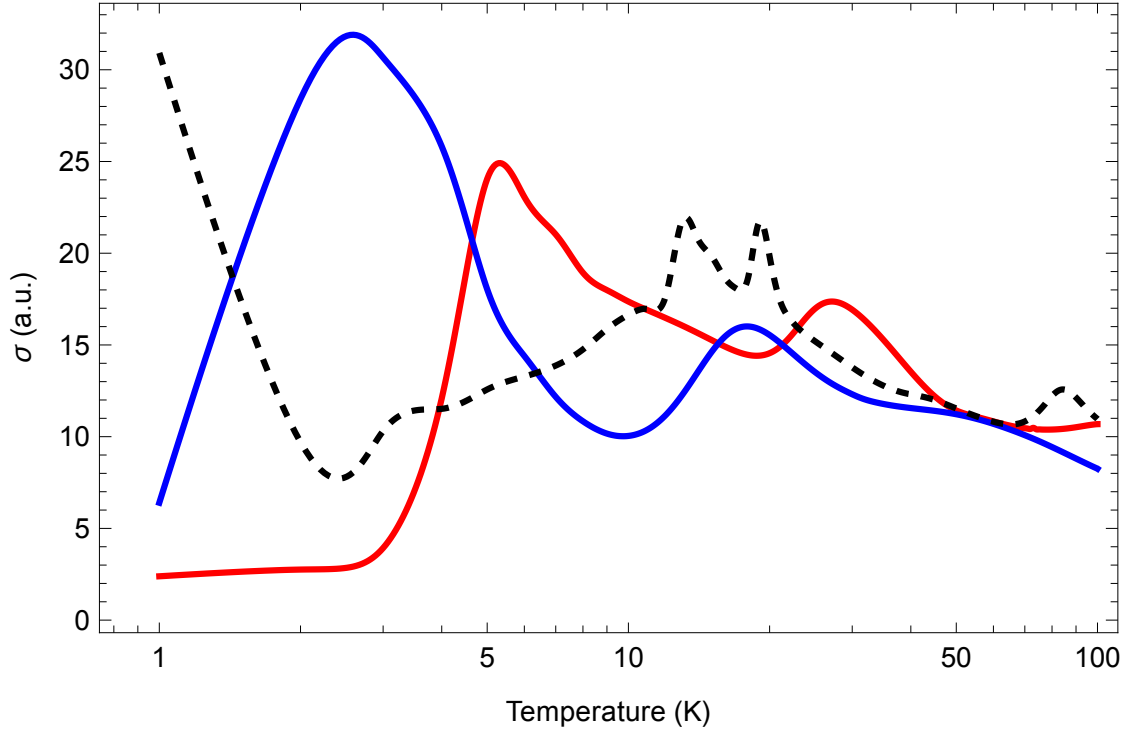


Figure 6.8: Total RCT cross sections. Solid lines refer to  $\alpha = 0$  (i.e., absence of flux tube), for the bosonic base (blue line) and the fermionic base (red line) respectively. The dashed line refers to a semion ( $\alpha = 1/2$ ), and the cross section is identical for both fermionic and bosonic bases.

## Chapter 7

### Conclusion

This dissertation has presented a comprehensive theoretical investigation of charge transfer (CT) processes in both ultracold ion-atom systems and reduced-dimensional (2D) quantum environments. The central objective has been to deepen our understanding of fundamental collision dynamics in regimes where quantum effects dominate, while also extending existing theoretical frameworks into previously unexplored domains.

A major contribution of this work lies in its re-examination of radiative charge transfer and association (RadCT) in the  $Yb^+ - Ca$  system. By employing both semi-classical and fully quantum approaches including the local optical potential method and a Fermi Golden Rule formalism we have demonstrated that previously reported reaction rates are significantly overestimated. Our calculations establish that Langevin-type behavior persists into the single-Kelvin regime and provide rigorous upper bounds on radiative relaxation rates. These results not only resolve discrepancies with earlier experimental interpretations but also underscore the importance of accurate molecular data and careful treatment of long-range interactions in cold collision theory.

In addition, this work has revealed pronounced isotope-dependent effects in radiative association processes in the ultra-cold regime, arising from subtle shifts in bound-state structure and scattering lengths. The identification of such strong isotope sensitivity, particularly in the ultracold s-wave regime, constitutes a novel contribution to the field and opens new avenues for experimentally controllable chemistry at the quantum level.

The second major advance of this dissertation is the extension of charge transfer theory

to two-dimensional systems. Motivated by rapid developments in 2D materials and quantum technologies, we have formulated and implemented a framework for analyzing resonant charge transfer in a strictly planar geometry. This required the development of new methodologies, including the construction of 2D molecular potential surfaces and adaptations of scattering theory to reduced dimensionality. Most notably, this work represents the first investigation into how anyonic exchange statistics influence charge-transfer processes. By incorporating the possibility of fractional statistics, this study connects atomic collision physics with topological phases of matter, thereby bridging AMO physics and condensed matter theory in a fundamentally new way.

More broadly, the research presented here highlights the importance of charge transfer as a unifying process across multiple domains: from astrophysical environments and plasma physics to quantum simulation, precision measurement, and emerging quantum technologies. In ultracold systems, CT serves as a sensitive probe of quantum scattering and molecular formation, while in 2D materials, it provides insight into the interplay between topology, dimensionality, and interaction dynamics. In summary, this thesis advances the field in three key ways:

- Quantitative refinement of cold and ultracold charge transfer rates, resolving discrepancies with prior work and establishing reliable theoretical benchmarks.
- Discovery of strong isotope-dependent effects in ultracold radiative processes, with implications for controlled quantum chemistry.
- Investigate the charge transfer process in 2 dimensions

These contributions not only deepen our understanding of fundamental quantum collision processes but also lay the groundwork for future experimental and theoretical studies at the intersection of AMO physics, quantum materials, and quantum information science.

## Appendix A

### Center of Mass as a Plane Wave

Take the isolated system of 2 interacting spinless particles. The particles are of masses  $m_1$  and  $m_2$  and are located at positions  $\mathbf{r}_1$  and  $\mathbf{r}_2$ . The potential energy of the particles depends only on their relative position  $\mathbf{r}_1 - \mathbf{r}_2$ . The Langrangian of this system can be written as:

$$L = T - V = \frac{1}{2}m_1 \cdot \dot{r}_1^2 + \frac{1}{2}m_2 \cdot \dot{r}_2^2 + V(\mathbf{r}_1 - \mathbf{r}_2) = \frac{1}{2}M \cdot \dot{R}^2 + \frac{1}{2}\mu \cdot \dot{r}^2 + V(\mathbf{r}) \quad (\text{A.1})$$

where

$M = m_1 + m_2$  is a total mass of the system

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \text{ is a reduced mass of the system} \quad (\text{A.2})$$

$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}$  are the coordinates of the center of the mass of the system, and

$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$  are the relative coordinates.

with:

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \text{ we define}$$

$$\mathbf{P} = M \cdot \dot{\mathbf{R}} \text{ and } \mathbf{p} = \mu \cdot \dot{\mathbf{r}} \text{ and we obtain} \quad (\text{A.3})$$

$$\dot{\mathbf{P}} = 0 \quad \dot{\mathbf{p}} = -\nabla V \mathbf{r}$$

Now using the variables  $\mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{r}$  we get for the Hamiltonian:

$$H = \frac{P^2}{2M} + \frac{p^2}{2\mu} + V(\mathbf{R}) = H_{CM} + H_r \quad (\text{A.4})$$

The first term represents the motion of the center of the mass, which is a uniform inertial motion, since  $\dot{\mathbf{P}} = 0$ . The other terms represent the energy associated with the motion relative to the center of the mass. Now if we choose the inertial frame in which the center of the mass is at rest, we get the Hamiltonian to reduce to  $H_r$ , representing the motion of a single, fictitious particle in an external potential.

Now transitioning to a quantum case. The variables  $H, \mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{r}$  become operators, with the usual commutation relations:

$$[R_i, P_j] = [r_i, p_j] = i\delta_{ij} \text{ and } [R_i, p_j] = [r_i, P_j] = 0 \implies \quad (\text{A.5})$$

$$[H_{cm}, H_r] = 0, [H_r, H] = 0 [H_{CM}, H] = 0$$

Since all the Hamiltonians commute, there exists a common eigenbasis in some state space  $E$  of  $H, H_{CM}, H_r$ . We can express the space  $E$  as a tensor product of spaces  $E_R$  and  $E_{cm}$  with  $E = E_{CM} \otimes E_r$ . The operators  $\mathbf{P}, \mathbf{R}$  operate in space  $E_{CM}$  and operators  $\mathbf{p}, \mathbf{r}$  operate in  $E_r$  space. From the Lagrangian and the Hamiltonian above we observe that the motion of the center of mass and the relative motion are completely independent of each other.

$$H_{CM} |\chi\rangle = E_{CM} |\chi\rangle \quad H_r |\omega\rangle = E_r |\omega\rangle$$

$$H |\phi\rangle = (H_{CM} + H_r)(|\chi\rangle \otimes |\omega\rangle) = \quad (\text{A.6})$$

$$H_{CM}(|\chi\rangle \otimes |\omega\rangle) + H_r(|\chi\rangle \otimes |\omega\rangle) = (E_{CM} + E_r)(|\chi\rangle \otimes |\omega\rangle)$$

Expressing the expressions above in coordinate representation, we have:

$$\phi(\mathbf{R}, \mathbf{r}) = \chi(\mathbf{R})\omega(\mathbf{r})$$

$$H_{CM}\chi(\mathbf{R}) = E_{CM}\chi(\mathbf{R}) \quad H_r\omega(\mathbf{r}) = E_r\omega(\mathbf{r}) \quad (\text{A.7})$$

$$-\frac{\hbar_R^2}{2M}\nabla^2\chi(\mathbf{R}) = E_{CM}\chi(\mathbf{R}) \quad -\frac{\hbar^2}{2M}\nabla_r^2\omega(\mathbf{r}) = E_r\omega(\mathbf{r})$$

So we can solve for  $\chi(\mathbf{R})$  to obtain:

$$\chi(\mathbf{R}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar}\sqrt{2ME_{CM}}\mathbf{R}} \quad (\text{A.8})$$

## Appendix B

### Mathematica Code for Calculating the Eigenvalues for the H2 Ion in 3 Dimensions

This is a fairly simple program which illustrates the power of Mathematica (and similar) mathematical applications. Following the derivation in chapter 1, we are left with two differential equations  $M(\mu)$  and  $L(\lambda)$ . Each equation depends on two parameters  $p^2 = -\frac{1}{4}R^2E$  where  $E$  is the electron energy and separation constant  $A$ .

We then create a array of values of  $R$  (nuclear distance), and for each value of  $F$  calculate eigenvalues of the equations (4.38) and (4.43), as a functions of  $p$  and  $A$ . Interpolate and the intersection of the functions of  $p$  and  $A$  provides an eigenvalue.

The last step is to add nuclear potential energy  $1/R$ , tabulate and plot the function.

```

(* Matrix dimensions *)
numCoefficients := 10;

numInterPoints := 20;

d[m_, n_] := KroneckerDelta[m, n];

lP[n_, x_] := LegendreP[n, x];

mSum[n_, x_] := (-n (n + 1) lP[n, x] + p^2 x^2 lP[n, x]);

mSumExpanded[m_, n_] :=  $\left( \frac{-2 n (n + 1)}{2 n + 1} + p^2 \frac{2 (2 n^2 + 2 n - 1)}{(2 n + 3) (2 n + 1) (2 n - 1)} \right) d[m, n] +$ 
 $\left( p^2 \frac{2 (n + 1) (n + 2)}{(2 n + 1) (2 n + 3) (2 n + 5)} \right) d[m, n + 2] + \left( p^2 \frac{2 n (n - 1)}{(2 n + 1) (2 n - 1) (2 n - 3)} \right) d[m, n - 2];$ 

lSumExpanded[m_, n_] :=
(-2 p n (2 n + 1) - (2 p - 1) n^2 + (-2 p + 2 R) (2 n + 1) + (-4 p + 1) n + (-2 p - p^2 + 2 R))
d[m, n] + (2 p n (n + 1) - (-2 p + 2 R) (n + 1)) d[m, n + 1] +
(2 p n^2 + (2 p - 1) n (2 n - 1) - (-2 p + 2 R) n + (4 p - 3) n) d[m, n - 1] +
(- (2 p - 1) n (n - 1)) d[m, n - 2];

tableEofR = {};

allRs = Table[i, {i, 0.2, 5, 0.2}];
allRs = Join[allRs, Table[i, {i, 5.5, 9, 0.5}]];

CalcE[radius_, max_, np_] := Module[{matrixM, matrixL, ee, q,
pe, rm, pMax, pMin, pStep, eigenM, eigenL, pgrid, mFunc, lFunc},

(* m = rows, n = columns.
Sums go from 0 (zero), matrix indices go from 1
*)

(* M matrix *)
matrixM = Table[0, {i, 1, max}];
Do[
matrixM[[m + 1]] = Table[mSumExpanded[m, n]  $\left( \frac{2 m + 1}{2} \right)$ , {n, 0, max - 1}],
{m, 0, max - 1}
];

```

```

(* L Matrix For Power series solution *)

matrixL = Table[0, {i, 1, max}];

Do[
  matrixL[[m+1]] = Table[lSumExpanded[m, n], {n, 0, max-1}],
  {m, 0, max-1}
];

q = 2.;
pMax =  $\frac{2. q \text{ radius}}{2.} + 1.$ ;
pMin =  $\frac{\text{radius}}{2.}$ ;
pStep =  $\frac{pMax - pMin}{np}$ ;

(*Output *)

(* {p, eigenvalue @ p } *)
eigenM = Table[0, {i, 1, np}, {j, 1, 2}];
eigenL = Table[0, {i, 1, np}, {j, 1, 2}];
pgrid = Table[pMin + (i-1) * pStep, {i, 1, np}];

rm[x_] := If[Im[x]  $\neq$  0,  $10^{-99}$ , x];

Do[
  eigenM[[i]][[2]] =
    Sort[Map[rm, Eigenvalues[N[matrixM /. p  $\rightarrow$  pgrid[[i]], 10.]], Greater][[1]];
  eigenM[[i]][[1]] = pgrid[[i]];
  eigenL[[i]][[2]] = Sort[Map[rm,
    (Eigenvalues[N[matrixL /. {R  $\rightarrow$  radius, p  $\rightarrow$  pgrid[[i]]}, 10.]] (-1))][[1]];
  eigenL[[i]][[1]] = pgrid[[i]],
  {i, 1, np}
];

(* Now Interpolation *)

mFunc = (eigenM // Interpolation);
lFunc = (eigenL // Interpolation);

pe = x /. FindRoot[mFunc[x] == lFunc[x], {x, radius}];

```

```

ee = -4  $\left(\frac{pe}{radius}\right)^2$ ;
ee
]

```

```

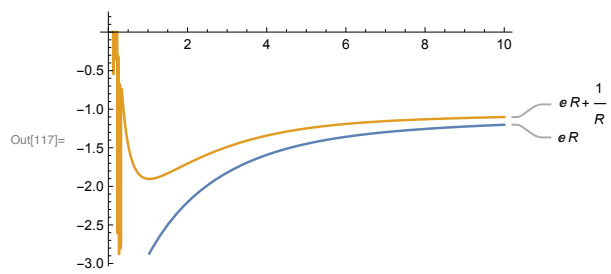
(* Syntax:
  [E(R), E(R)+1/R] = CalcE[1,numCoefficients, numInterPoints]
*)

```

```

Energies := CalcE[#, numCoefficients, numInterPoints] &;
Plot[{Energies[r], Energies[r] +  $\frac{1}{r}$ }, {r, 0.1, 10},
  AxesOrigin -> {0, 0}, PlotLabels -> {E(R), E(R) +  $\frac{1}{R}$ }]

```



## Appendix C

### Hydrogen Ion in 2 Dimension Equation, Mathematica Code

```
(*UsingZhu, Xiongpaper*)
(* Distance between nuclei*)

(**)

(* L equation and coefficients *)
σ[r_]:=r/p - 1/2;
α[n_]:= (n + 1) (n + 1/2);
β[n_, r_]:= 2n² + (4p - 2σ[r])n - A + p² - 2pσ[r] - σ[r]/2;
γ[n_, r_]:= (n - 1) (n - 2σ[r] - 1/2) + σ[r] (σ[r] - 1/2);

(* Create a continued fraction equation for the coefficients of L equation *)

lCoeff2[n_, r_]:=β[0,r]/α[0] + Simplify[ContinuedFractionK[-α[i - 1]γ[i, r], β[i, r], {i, 1, n}]];

(* M equation and coefficients *)

w = -A + p²/2; q = -p²/4;
v[n_]:= w - n²/q;
(* Evaluate as a continued fraction *)
mCoeffV0[n_]:=v[0] + 2Simplify[ContinuedFractionK[-1, v[2i], {i, 1, n}]];

mCoeffV1[n_]:=v[1] - 1 + Simplify[ContinuedFractionK[-1, v[2i + 1], {i, 1, n}]];

(*Parameters :
n : Depthofthecontinuedfraction,
R : Internucleardistance,
gerade : flagindicatingwhetherwearecalculatinggeradeorungeradestates
output :
R, electronenergy, totalenergy, separationconstant, andpvalue
*)
energySingleR[n_, R_, gerade_]:=
Module[{asAndps, pe, energy, aa},
(*Now find p and A *)
If[gerade==1,
asAndps = NSolve[{lCoeff2[n, R] == 0,
mCoeffV0[n] == 0}, {A, p}, Reals],
asAndps = NSolve[{lCoeff2[n, R] == 0,
mCoeffV1[n] == 0}, {A, p}, Reals]];
aa = Sort[asAndps[[All, 1, 2]], Greater][[1]];
pe = Sort[asAndps[[All, 2, 2]], Greater][[1]];
energy = (-2)/R² pe²;
{R, energy, energy + 1/R, aa, pe}
]
```

```

allRs = Table[0.2 + i * 0.1, {i, 0, 9}];
allRs = Join[allRs, Table[1 + .5 * i, {i, 1, 8}]];
allRs = Join[allRs, Table[5 + i, {i, 1, 5}]];

energiesGerade = Map[energySingleR[3, #1, 1] &, allRs];
energiesUngerade = Map[energySingleR[3, #1, 2] &, allRs];

Prepend [energiesGerade, {R, E, "E +  $\frac{1}{r}$ ", A, p}] // MatrixForm
Prepend [energiesUngerade, {R, E, "E +  $\frac{1}{r}$ ", A, p}] // MatrixForm

eG = Interpolation[Transpose[{energiesGerade[[All, 1]], energiesGerade[[All, 2]]}]];
eGTotal = Interpolation[Transpose[{energiesGerade[[All, 1]], energiesGerade[[All, 3]]}]];
Plot[{eG[x], eGTotal[x]}, {x, 0.2, 10}, PlotStyle -> {Red, Green}, AxesLabel -> {R, "(E), V(R)"},
AxesOrigin -> {0, 0}, PlotLabel -> {"Electron vs Total potential for 1sG (Gerade) state"},
PlotLabels -> Placed[{ "E(R)", "E(R)+V(R)" }, {Left}], PlotRange -> Automatic]

eU = Interpolation[Transpose[{energiesUngerade[[All, 1]], energiesGerade[[All, 2]]}]];
eUTotal = Interpolation[Transpose[{energiesUngerade[[All, 1]], energiesUngerade[[All, 3]]}]];
Plot[{eU[x], eUTotal[x]}, {x, 0.2, 10}, PlotStyle -> {Red, Green}, AxesLabel -> {R, "(E), V(R)"},
AxesOrigin -> {0, 0}, PlotLabel -> {"Electron vs Total potential for 2pU (Ungerade) state"},
PlotLabels -> Placed[{ "E(R)", "E(R)+V(R)" }, {Left}], PlotRange -> Automatic]

Plot[{eGTotal[x], eUTotal[x]}, {x, 0.2, 10}, PlotStyle -> {Red, Green}, AxesLabel -> {R, "E(R)+V(R)"},
AxesOrigin -> {0, 0}, PlotLabel -> {"Gerade 1sg and Ungerade 2pu total potential curves"},
PlotLabels -> Placed[{ "1sG", "2pU" }, {Above, Left}], PlotRange -> Full]

```

## Appendix D

### Fortran Code for Calculating Energies for the 2D Problem

To run the numerical calculations I used gfortan 5.2.0, running on 2.2 GHz, Intel Code 7, Macbook Pro 15" Laptop (mid 2014). The numerical library used was Lapack, version 3.6.0, compiled on the same machine using the gfortran above.

This is Fortran 90 code for calculating even and odd eigenvalues and consequently the energies.

```
! To Compile
! gfortran -O4 -o h2Plus h2plus.f90 hpsort.f90 -L ./ -l lapacke -l lapack -l blas -l install
program main

implicit none

interface

subroutine M_Matrix(mMatrix,max,p)
  integer :: max
  double precision :: p
  double precision,dimension(:,:) :: mMatrix
end subroutine

subroutine L_Matrix(lMatrix,max,R,p)
  integer :: max
  double precision :: R,p
  double precision,dimension(:,:) :: lMatrix
end subroutine

subroutine CalculateEigenvalues(c,matrix,max,wr,wi)
  character*1 :: c
  double precision,dimension(:,:) :: matrix
  integer :: max,np,count
```

```

    double precision,dimension(:) :: wr,wi
end subroutine

subroutine hpsort(n,rra)
    integer :: n
    double precision, dimension(:) :: rra
end subroutine

end interface

! Declare local variables
integer,parameter :: max = 500    ! matrix dimesions
integer,parameter :: np = 1000    ! number of p points
integer,parameter :: state = 0    ! 0=ground state, 1=1st excited, etc..
integer,parameter :: numRadii = 1
double precision,dimension(numRadii) :: Radii
double precision,dimension(max,max) :: mMatrix,lMatrix
double precision,dimension(np) :: pGrid
double precision :: R    ! Nuclei distance
double precision,dimension(max,np) :: mEig, lEig
double precision,dimension(max) :: wr,wi
double precision :: pStep, pMin, pMax
integer :: count,i,j, index
double precision :: p,rr
character(len=6) :: fmt
character(len=16) :: rOut

i = 1;
rr = 20.0d0
do while( (rr <= 5.0d0) .and. (i <= numRadii))
    Radii(i) = rr
    rr = rr + 0.2d0
    i = i + 1
end do

do while(i <= numRadii)
    Radii(i) = rr
    rr = rr + 0.5d0
    i = i + 1
end do

```

```

do j = 1, numRadii
  R = Radii(j)

  print *, "R = ", R

  pMin = 0
  pMax = 2.d0*R + 1.d0
  pStep = (pMax - pMin) / dble(np)

  count = 1;
  p = pMin
  do while( count <= np )
    ! Initialize M matrix
    call M_Matrix(mMatrix,max,p)

    !Calculate eigenvalues for M matrix
    call CalculateEigenvalues('m',mMatrix,max,wr,wi)
    ! sort
    do i = 1,max
      if( wi(i) /= 0) then
        wr(i) = -1.0d99
      end if
    end do

    call hpsort(max,wr)
    do i = 1,max
      mEig(i,count) = wr(max-i+1)
    end do

    ! And L Matrix
    call L_Matrix(lMatrix,max,R,p)
    call CalculateEigenvalues('l',lMatrix,max,wr,wi)
    do i = 1,max
      if( wi(i) /= 0) then
        wr(i) = 1.0d99
      end if
    end do
    call hpsort(max,wr)
    ! wr contains eigenvalues from largest to smallest
    do i = 1,max
      lEig(i,count) = wr(i)
    end do
  end do
end do

```

```

end do

pGrid(count) = p
p = p + pStep
count = count + 1
end do

! Now write the eigenvalues to file
if( R < 10.0d0 ) then
    fmt = "(F3.1)"
else
    fmt = "(F4.1)"
end if
write(rOut, fmt) R
open(unit=11,file="Eigenvalues-Even-M-R="//trim(rOut)//".dat")
open(unit=12,file="Eigenvalues-Even-L-R="//trim(rOut)//".dat")
! Todo: Excited states
index = 1
do i=1,np
    write(11,*) pGrid(i), mEig(1,i),mEig(2,i), mEig(3,i), mEig(4,i), mEig(5,i)
    write(12,*) pGrid(i), lEig(1,i),lEig(2,i), lEig(3,i), lEig(4,i), lEig(5,i)
end do
close(11)
close(12)
end do

end ! main

subroutine CalculateEigenvalues(id,matrix,max,wr,wi)
! Calculate and return the eigenvalues
implicit none

character*1 ::id
double precision,dimension(:,:) :: matrix
integer :: max
double precision,dimension(:) :: wr,wi
!local
double precision,dimension(max,max) :: z
double precision,dimension(:), allocatable :: work
integer :: info,i
double precision,dimension(1,1) :: dummy

```

```

integer :: lda,lwork
double precision :: temp

lda = max
lwork = max * 11

allocate( work(lwork) )

call DGEEV('N','N',max,matrix,lda,wr,wi,dummy,1,dummy,1,work,lwork,info)

deallocate(work)

if( info /= 0 ) then
  print *, "Error in info , info = ", info
  call exit(1)
end if

end subroutine

subroutine M_Matrix(mMatrix,max,p)
! create the M matrix
implicit none
interface
  function Delta(m,k)
    double precision :: Delta
    integer, intent(in) :: m,k
  end function Delta
end interface

integer :: max
double precision :: p, val
double precision,dimension(:,:) :: mMatrix
integer :: m,k

do m = 0,max-1
  do k = 0,max-1
!    Even state
    val = ( (-1)*((2.0d0 * k)**2) + ( (p**2.0)/2.0d0) ) * Delta(m,k) + &
!    For the Odd state use
    val = ( (-1)*((2.0d0 * k + 1)**2) + ( (p**2.0)/4.0d0) ) * Delta(m,k) + &
    ( (p**2)/4.0d0 ) * ( Delta(m,k+1) + Delta(m,k-1) )

```

```

    mMatrix(m+1,k+1) = val
end do
end do

!      Even state
mMatrix(1,2) = 2 * mMatrix(1,2)
!      For the Odd state use
! mMatrix(1,1) = -1 + (p**2.)/4.0d0

end subroutine

subroutine L_Matrix(lMatrix,max,R,p)
! create the L matrix
implicit none
interface
    function Delta(m,k)
        double precision :: Delta
        integer, intent(in) :: m,k
    end function Delta

    function SubSum(m,k)
        double precision :: SubSum
        integer, intent(in) :: m,k
    end function SubSum
end interface

integer :: max
double precision :: p,R,val
double precision, parameter :: PI = 3.141592653589793238462643d0
double precision,dimension(:,:) :: lMatrix
integer :: m,k

do m = 0,max-1
do k = 0,max-1
    val = ((-2.0d0)*p*k*(2*k+1) - (2*p-1)*(k**2) - 4*p*k + &
(2*R-p)*(2*k+1)-p -(p**2)+2*R) * Delta(m,k) + &
(2*p*k*(k+1) - (2*R-p)*(k+1))*Delta(m,k+1) + &
(2.0d0*p*(k**2) + (2.0d0*p - 1)*k*(2.0d0*k - 1) +(4.0d0*p - 2)*k-&
(2*R - p)*k)*Delta(m,k-1) - &
(2.0d0*p - 1)*k*(k - 1)*Delta(m,k-2) + SubSum(m,k)
lMatrix(m+1,k+1) = (-1)*val

```

```

        end do
    end do

end subroutine

function SubSum(m,k)
implicit none
interface
    function Delta(m,k)
        double precision :: Delta
        integer, intent(in) :: m,k
    end function Delta
end interface

double precision :: SubSum
integer, intent(in) :: m,k
integer :: i

SubSum = 0.0d0

do i = 0,k-1
    SubSum = SubSum + Delta(m,i)
end do

end function SubSum

function Delta(m,n)
implicit none
double precision :: Delta
integer, intent(in) :: m,n

if (m == n) then
    Delta = 1.d0
else
    Delta = 0.d0
end if

end function Delta

```

## Appendix E

### Scattering Cross Section Code

```
In[1]:= (* this routine needs to run Vpots.nb *)

geradeData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
    MathematicaCodeFinal/gerade1sV3.mx"];
geradePotential = Table[{geradeData?i, 1?, geradeData?i, 3?}, {i, 1, 27}];
ungeradeData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
    MathematicaCodeFinal/ungerade1sV3.mx"];
ungeradePotential = Table[{ungeradeData?i, 1?, ungeradeData?i, 3?}, {i, 1, 27}];
long[R_] := -2.0 - 0.082 / R ^ 4;
geradefun = Interpolation[geradePotential];
ungeradefun = Interpolation[ungeradePotential];
(* Fit to a model *)
gdata = Drop[geradePotential, {5, 27}];
udata = Drop[ungeradePotential, {5, 27}];
model[R_, C1_, C2_, C3_] := C1 Exp[-C2 R] / R + C3;
gnlm = NonlinearModelFit[gdata, model[R, C1, C2, C3], {C1, C2, C3}, R];
unlm = NonlinearModelFit[udata, model[R, C1, C2, C3], {C1, C2, C3}, R];
gshort[R_] := model[R, C1, C2, C3] /. gnlm["BestFitParameters"];
ushort[R_] := model[R, C1, C2, C3] /. unlm["BestFitParameters"];
gpot[R_] = Piecewise[
  {{gshort[R], R <= 0.3}, {geradefun[R], 0.3 < R < 13.0}, {long[R], R >= 13.0}}];
upot[R_] = Piecewise[
  {{ushort[R], R <= 0.3}, {ungeradefun[R], 0.3 < R < 13.0}, {long[R], R >= 13.0}}];
Plot[{gpot[R], upot[R]}, {R, 0.2, 5}, PlotRange ? All]

(*e=0.1/27.21;*)
u = 1866.0 / 2.0;
(*k= 2 u e ;*)
kB = 0.695; (* Boltzman in AU, cm-1 ?K *)
(*?=2?/k*)
a0 = 5.29188 ^ -11;

r0 = 0.01;
rend = 100;

(*For a single partial wave m, energy e, and range r0 ? rend*)
computePhaseShifts[m_, e_, r0_, rend_] :=
  Module[{k, eqG, eqU, solsG, solsU, fg, fu},
```

```

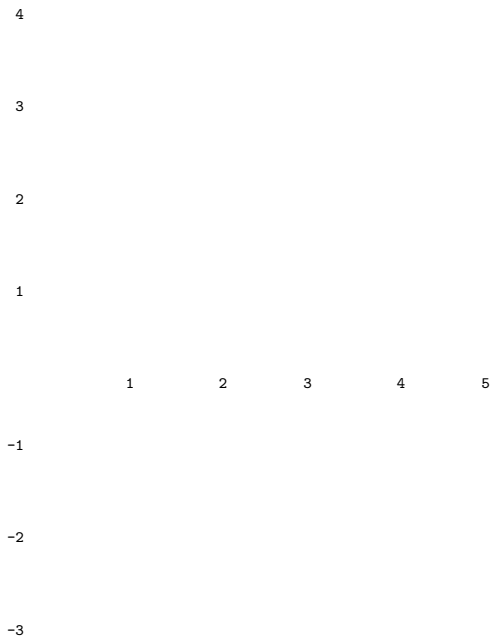
k=      2ue;
equG := yg ''[r] + yg '[r] / r - 2 u (gpot[r] + 2) yg[r] - m ^ 2 / r ^ 2 yg[r] + 2 u e yg[r] ;
equU := yu ''[r] + yu '[r] / r - 2 u (upot[r] + 2) yu[r] - m ^ 2 / r ^ 2 yu[r] + 2 u e yu[r];

2      Z_u_g_first-5-v1.nb

solsG = Flatten[NDSolve[
  {equG ? 0, yg[r0] ? 0, yg'[r0] ? 1}, yg, {r, r0, rend}, MaxSteps ? 200 000];
solsU = Flatten[NDSolve[
  {equU ? 0, yu[r0] ? 0, yu'[r0] ? 1}, yu, {r, r0, rend}, MaxSteps ? 200 000];
fg[r_] = yg[r] /. solsG;
fu[r_] = yu[r] /. solsU;
ylogG[r_] := fg'[r] / fg[r];
ylogU[r_] := fu'[r] / fu[r];
sl[n_, r_] := BesselJ[n, k r];
cl[n_, r_] := HankelH1[n, k r];
slp[n_, r_] := Derivative[0, 1][sl][n, r];
clp[n_, r_] := Derivative[0, 1][cl][n, r];
bG[n_, r_] :=
  (-I) ^ (Abs[n]) ( slp[n, r] - ylogG[r] x sl[n, r]) / ( clp[n, r] - ylogG[r] x cl[n, r]);
bU[n_, r_] :=
  (-I) ^ (Abs[n]) ( slp[n, r] - ylogU[r] x sl[n, r]) / ( clp[n, r] - ylogU[r] x cl[n, r]);
If?m ? 0,
  Return??Abs[N[bG[m, rend]]], Abs[N[bG[m, rend] - bU[m, rend]]]2 ??,
  Return??2 Abs[N[bG[m, rend]]], 2 Abs[N[bG[m, rend] - bU[m, rend]]]2 ???
?;
(*computePhaseShifts[10,0.003,0.1,100];*)

```

Out[18]=



```

(* e = collision energy in eV, r0, rend - R range *)
computeCrossSect[eV_, r0_, rend_, mMin_, mMax_] :=
Module[{allBs, eH, kk, crossSection},
  eH = eV / 27.21; (* energy Hartree *)
  kk = Sqrt[2 u eH];
  (* m - partial waves *)
  allBs = Table[{m, computePhaseShifts[m, eH, r0, rend]}, {m, mMin, mMax}];
  crossSection = Total[allBs, {1}]^2;
  {eV, crossSection}
];

(* Up to this partial wave *)
mMax = 5;

crossSection =
Table[computeCrossSect[e, r0, rend, 0, mMax], {e, 0.00001, .001, 0.00005}];

4      Z_u_g_first-5-v1.nb

In[29]:=

crossSectionGraphSC =
  Interpolation[{{crossSection?All, 1?, crossSection?All, 2??All, 1?} // Transpose];
crossSectionGraphCT =
  Interpolation[{{crossSection?All, 1?, crossSection?All, 2??All, 2?} // Transpose];

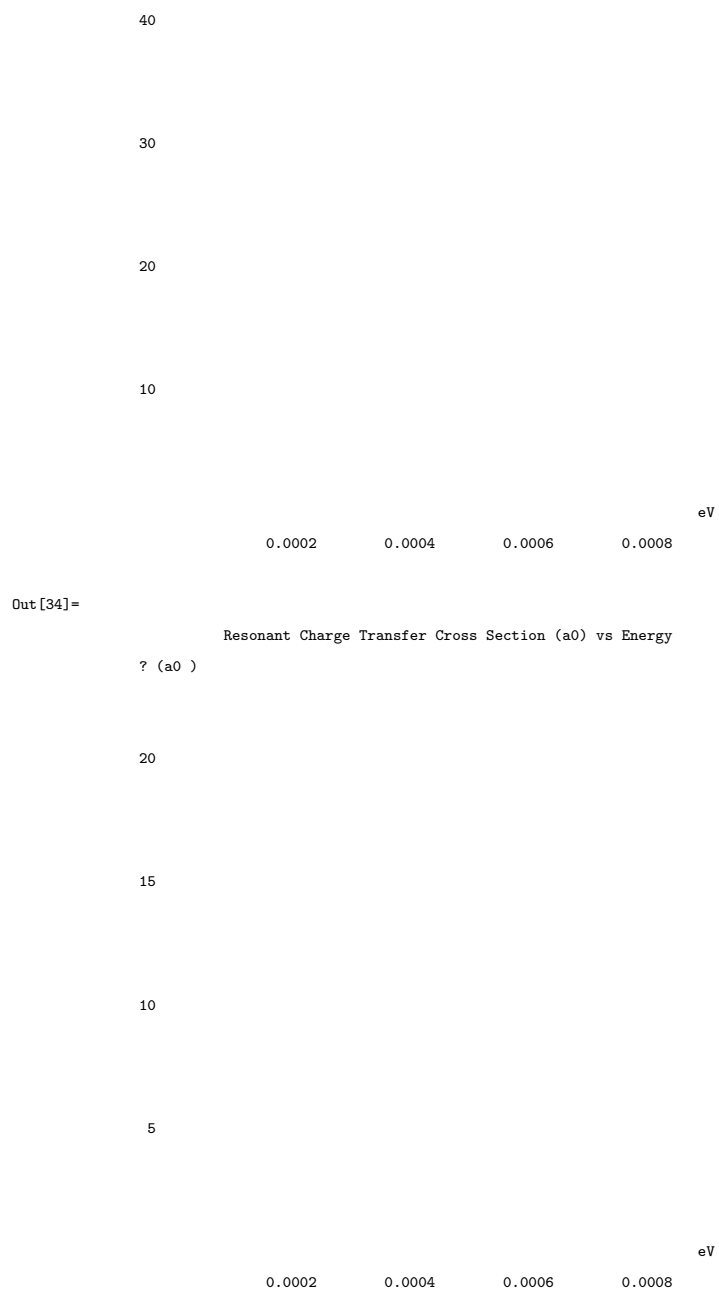
start = crossSection?1??1?
end = crossSection?Length[crossSection]??1?
Plot[crossSectionGraphSC[e], {e, start, end},
  AxesOrigin ? {0, 0}, AxesLabel ? {"eV", "?" (a0)"},
  PlotLabel ? "Scattering Cross Section (a0 ) vs Energy (eV) " ]
Plot[crossSectionGraphCT[e], {e, start, end},
  AxesOrigin ? {0, 0}, AxesLabel ? {"eV", "?" (a0)"},
  PlotLabel ? "Resonant Charge Transfer Cross Section (a0) vs Energy"]

Out[33]=

      Scattering Cross Section (a0 ) vs Energy (eV)

      ? (a0)

```



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